

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
```

```

        df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()

```

```

        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)

    return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : column

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : column

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv

- r_{train} : $\text{train_cv}/\text{test_cv}$

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

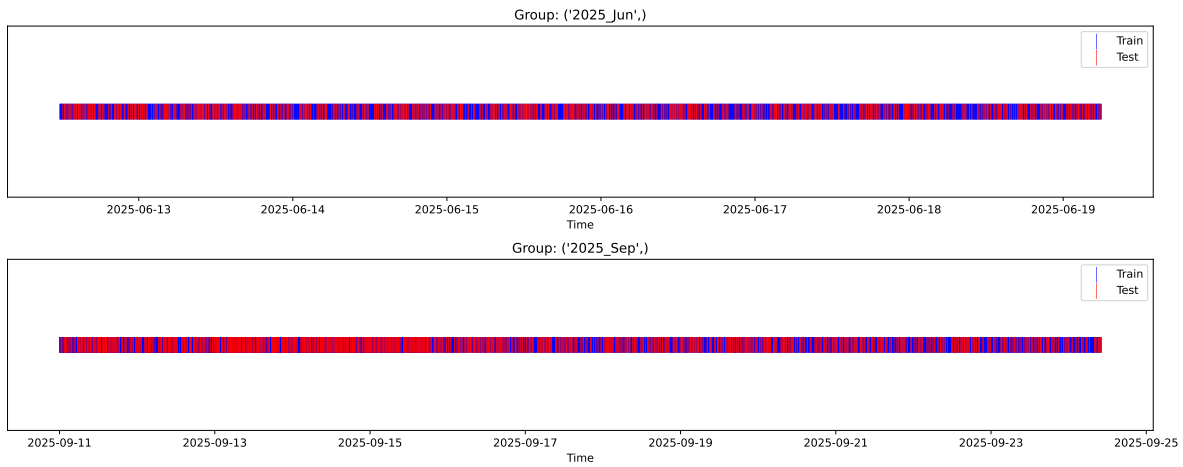
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

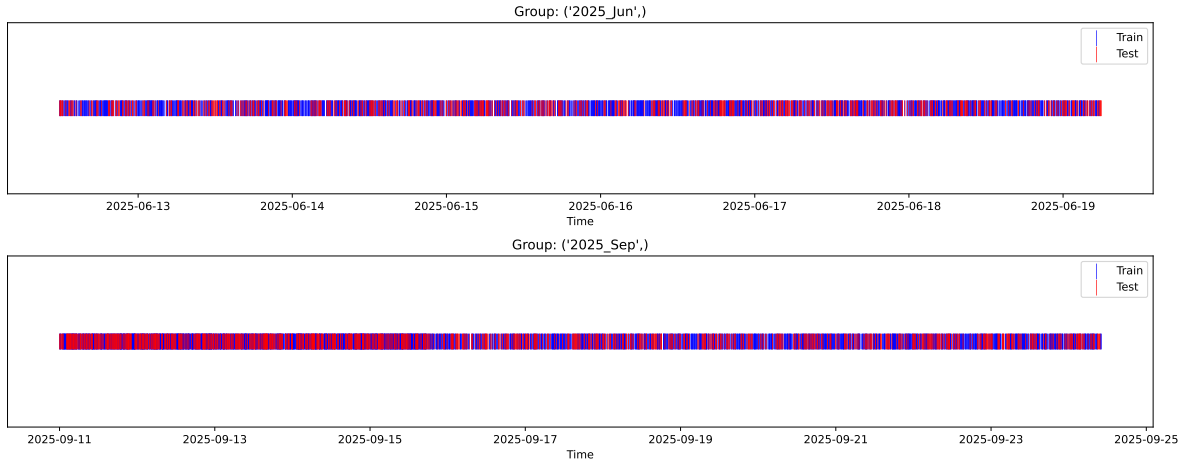
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

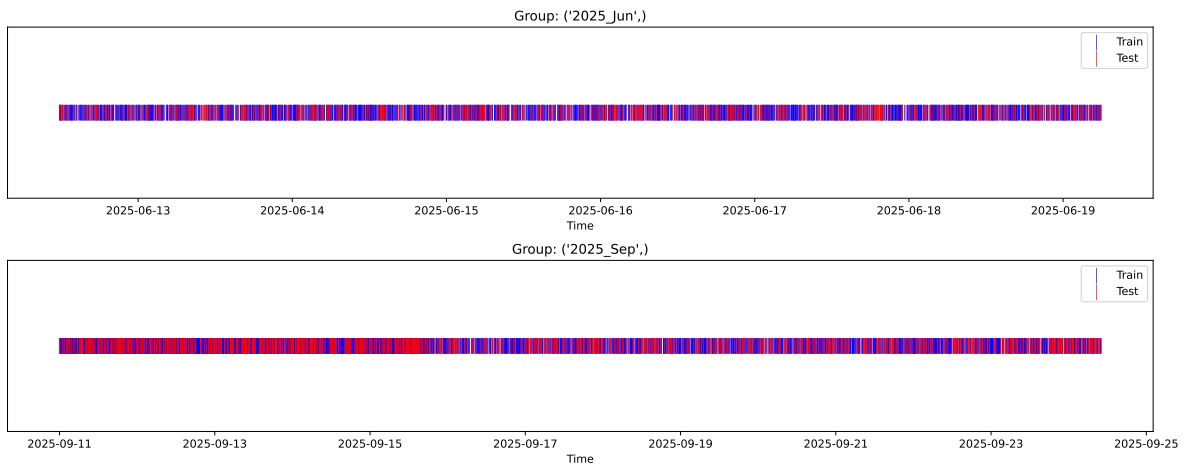
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



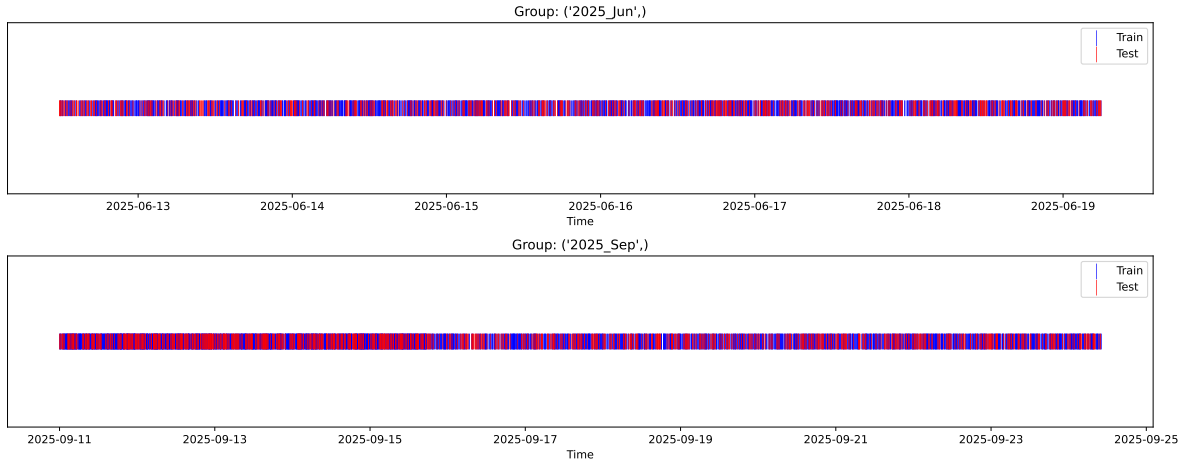
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



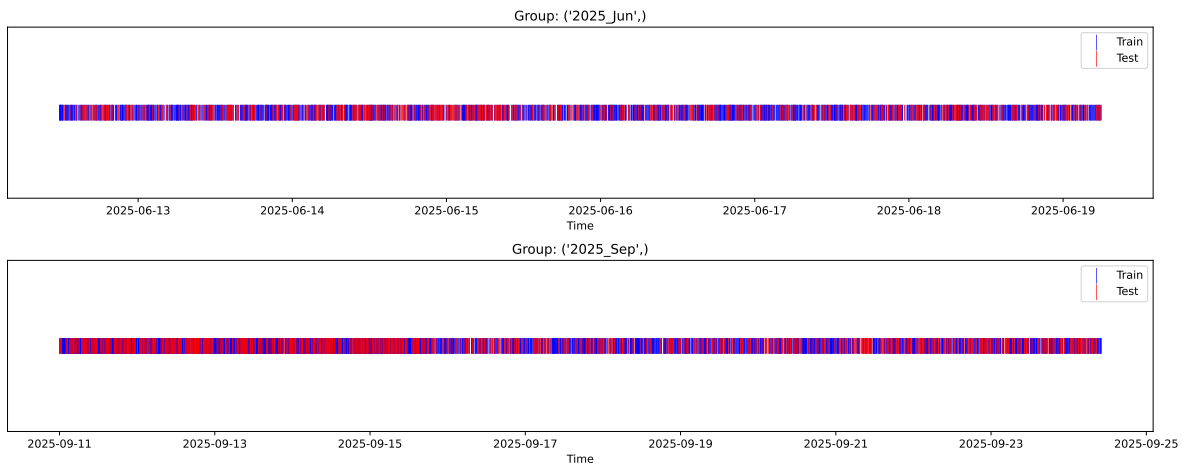
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



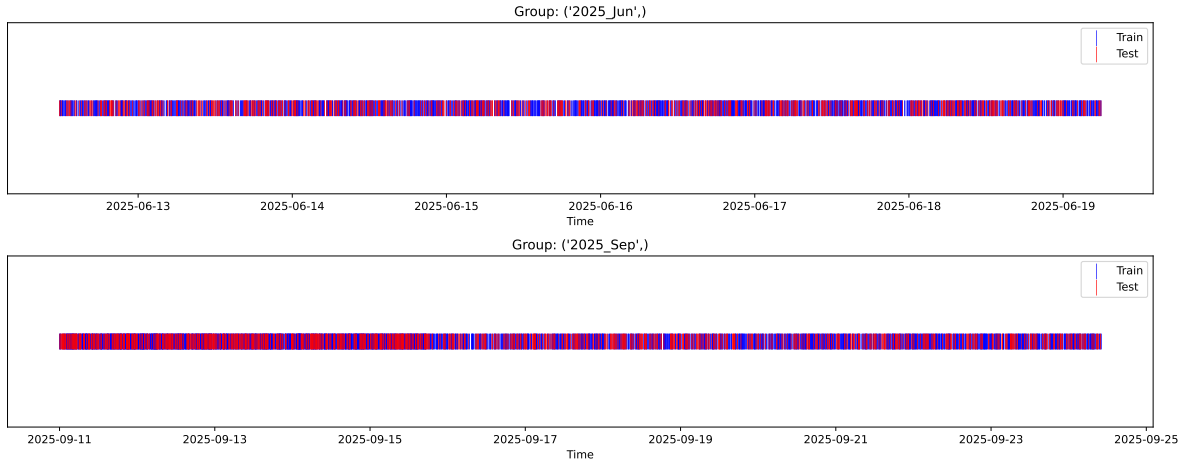
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 1

Len(group) : 5452, Len(training) : 3817, Len(test) : 1635

Distribution y_train vs y_test: y_train: mean=2.389, std=0.817, min=0.973, max=13.845
y_test: mean=2.385, std=0.771, min=1.088, max=7.277

Fold 0 — y_test: mean=2.37, std=0.77, y_train: mean=2.40, std=0.84

Fold 1 — y_test: mean=2.38, std=0.80, y_train: mean=2.39, std=0.82

Fold 2 — y_test: mean=2.39, std=0.83, y_train: mean=2.39, std=0.81

Fold 3 — y_test: mean=2.39, std=0.79, y_train: mean=2.39, std=0.83

Fold 4 — y_test: mean=2.40, std=0.86, y_train: mean=2.39, std=0.80

train scores CV: [0.43340769 0.4327579 0.42597048 0.43756683 0.4734002]

test scores CV: [0.44580332 0.44620218 0.46005545 0.43492237 0.36264806]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4406	0.4299	0.0346	0.4776	0.4782	0.0483	1.1123	1.0249

Coefficients:

$\text{coef_r_CO2} = -0.0821$

$\text{coef_r_NH3} = -0.1711$

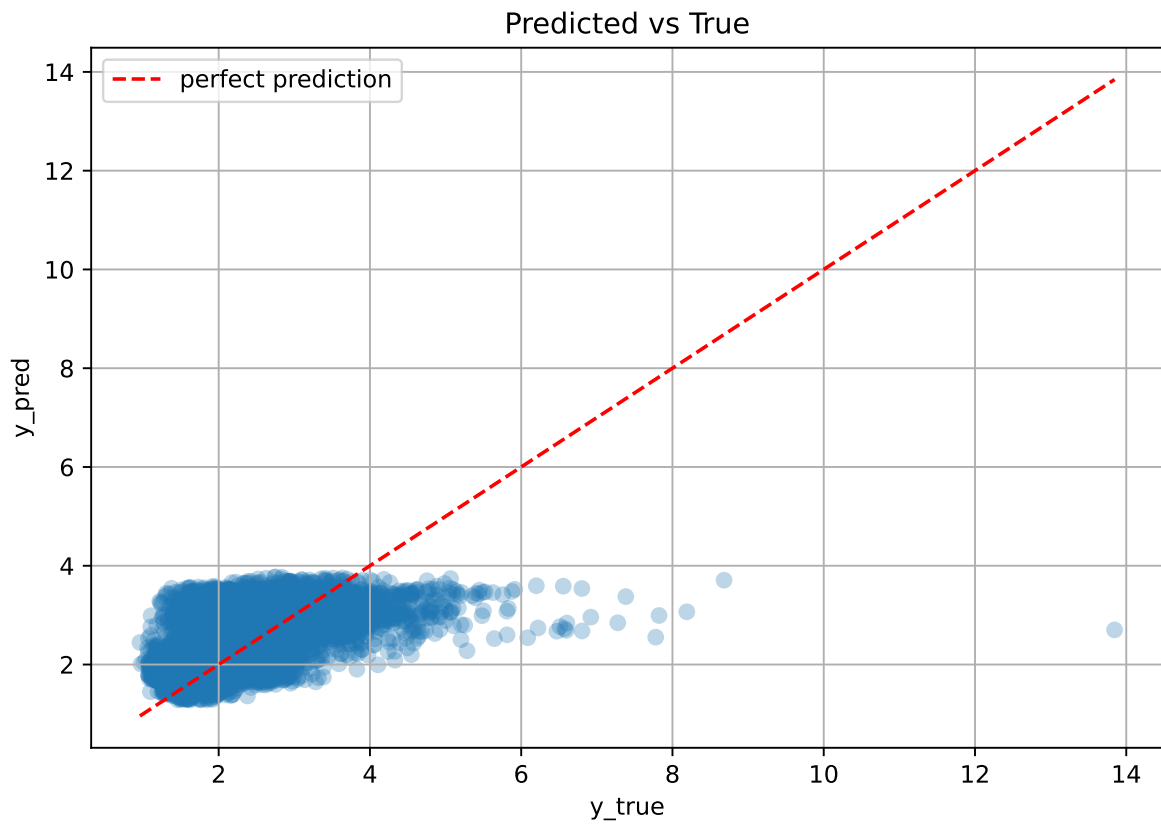
$\text{coef_r_THI} = 0.4407$

$\text{coef_r_temperature} = -0.4486$

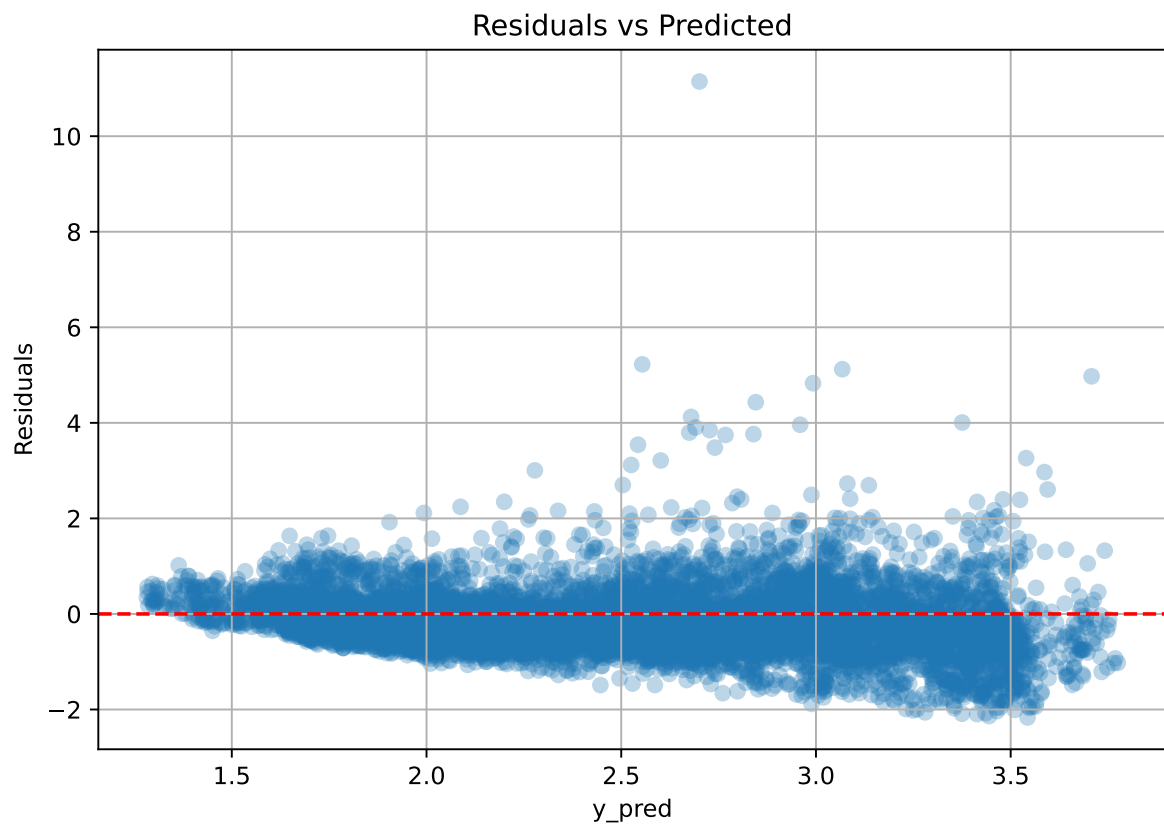
$\text{coef_r_umidity} = -0.5957$

$\text{intercept} = 2.3894$

2.3.2 Scatter



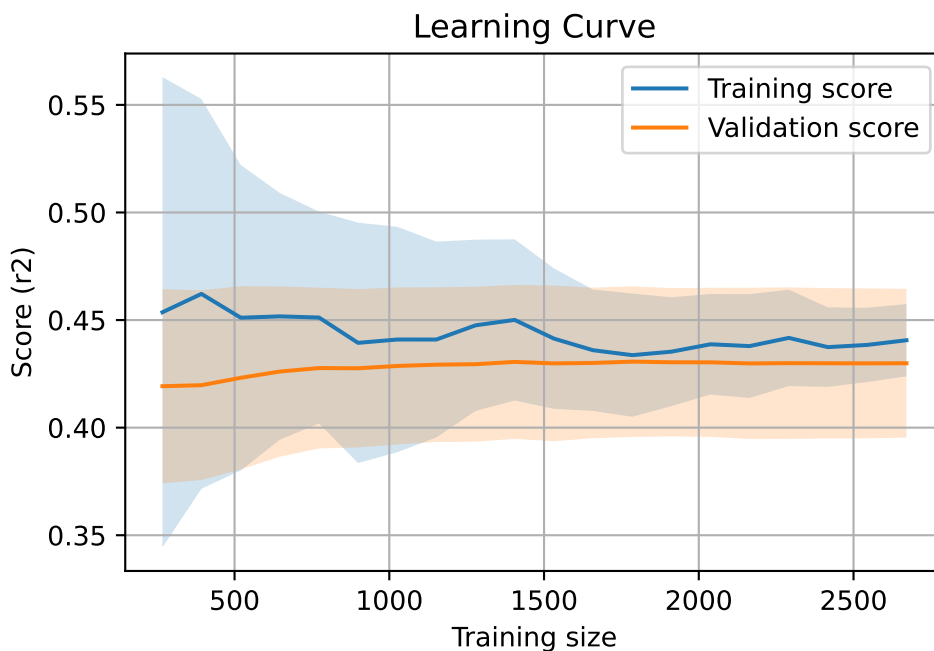
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 1

`Len(group)` : 5452, `Len(training)` : 3817,

`Len(training_real)` : 2672, `Len(test_of_training)` : 1145, `Len(test)` : 1635



2.3.5 Gridsearch — Group: 2

Len(group) : 2716, **Len(training)** : 1902, **Len(test)** : 814

Distribution y_train vs y_test: y_train: mean=2.260, std=0.774, min=0.959, max=7.822
y_test: mean=2.299, std=0.764, min=1.068, max=6.514

Fold 0 — y_test: mean=2.24, std=0.74, y_train: mean=2.27, std=0.79

Fold 1 — y_test: mean=2.27, std=0.74, y_train: mean=2.26, std=0.79

Fold 2 — y_test: mean=2.23, std=0.72, y_train: mean=2.27, std=0.80

Fold 3 — y_test: mean=2.26, std=0.77, y_train: mean=2.26, std=0.78

Fold 4 — y_test: mean=2.28, std=0.80, y_train: mean=2.25, std=0.76

train scores CV: [0.3588487 0.36320323 0.33531576 0.36111957 0.37208129]

test scores CV: [0.37354902 0.36389667 0.43899582 0.36954466 0.34431042]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3581	0.3781	0.0321	0.3225	0.3234	-0.0546	0.8555	0.9472

Coefficients:

$\text{coef_r_CO2} = -0.0293$

$\text{coef_r_NH3} = 0.0918$

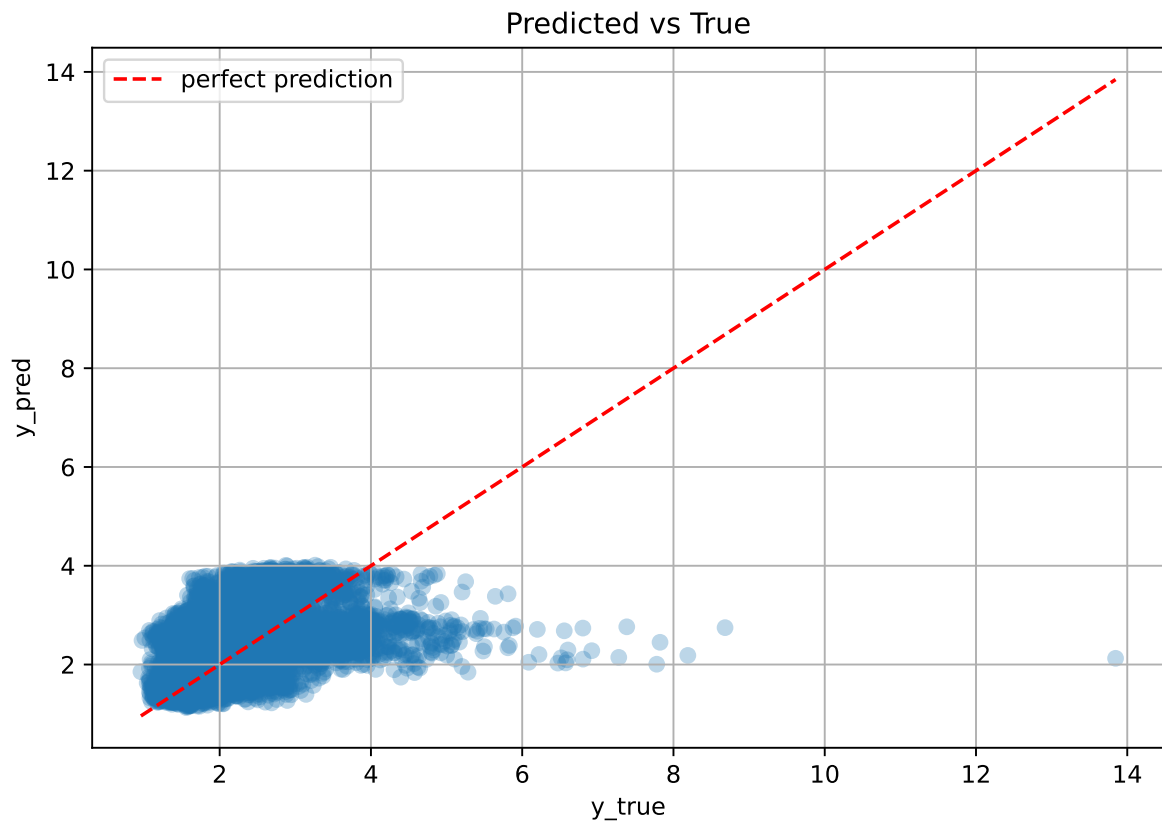
$\text{coef_r_THI} = 0.6273$

$\text{coef_r_temperature} = -0.8810$

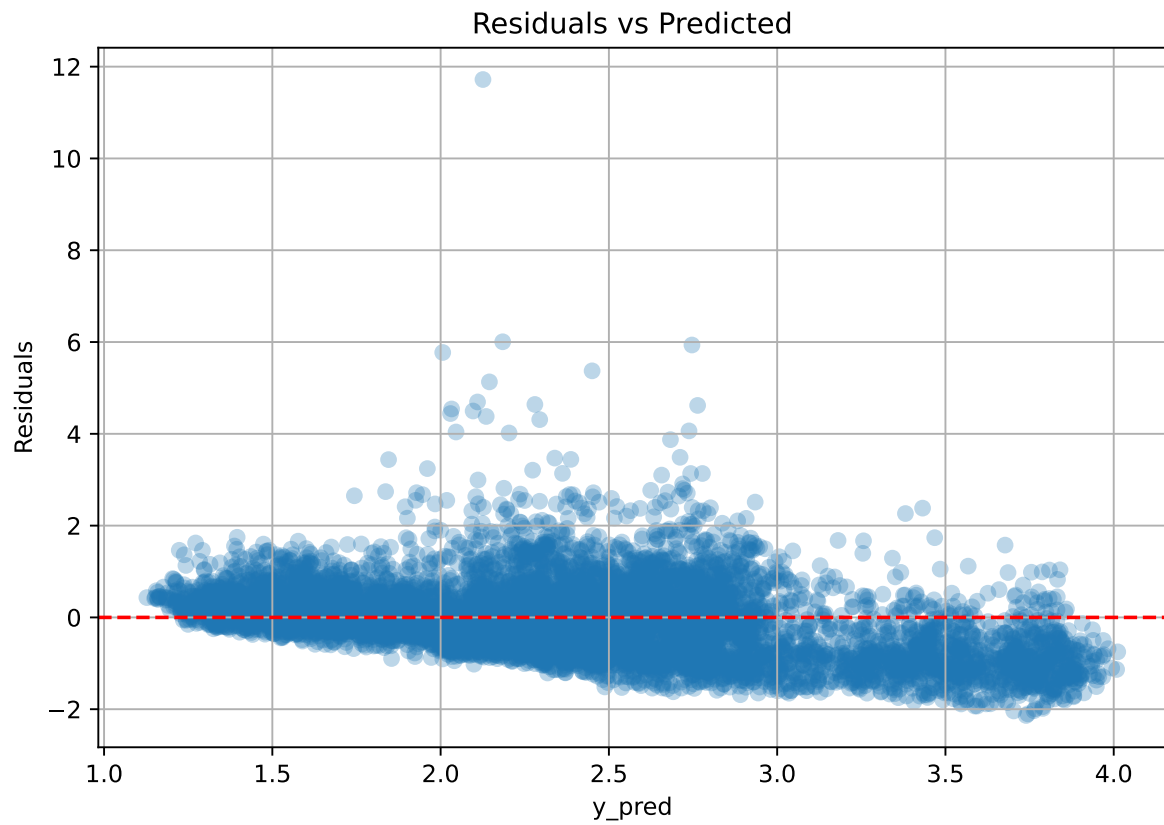
$\text{coef_r_umidity} = -0.7713$

$\text{intercept} = 2.2605$

2.3.6 Scatter



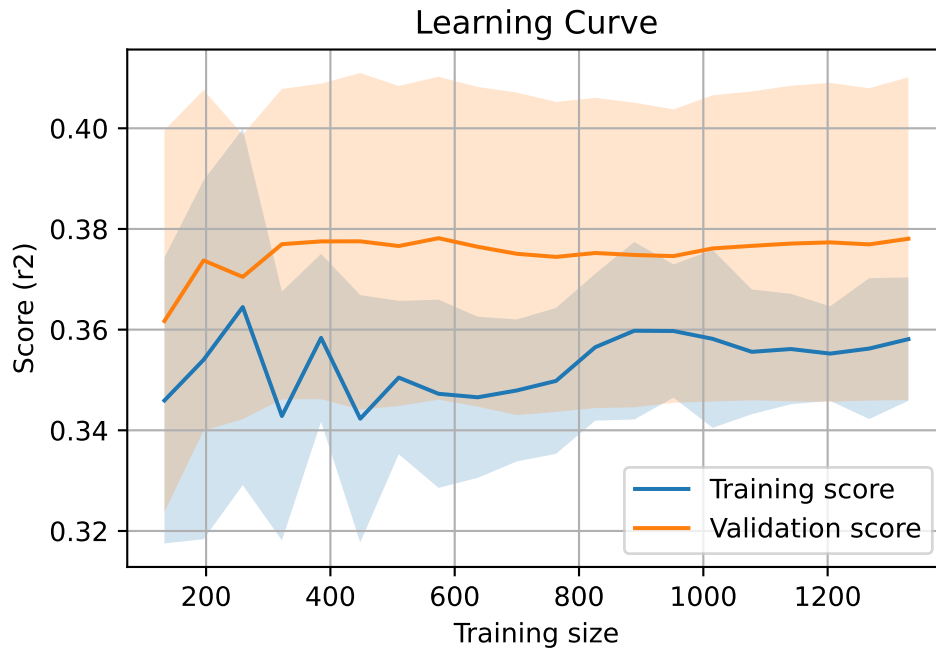
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2

`Len(group)` : 2716, `Len(training)` : 1902,

`Len(training_real)` : 1332, `Len(test_of_training)` : 570, `Len(test)` : 814



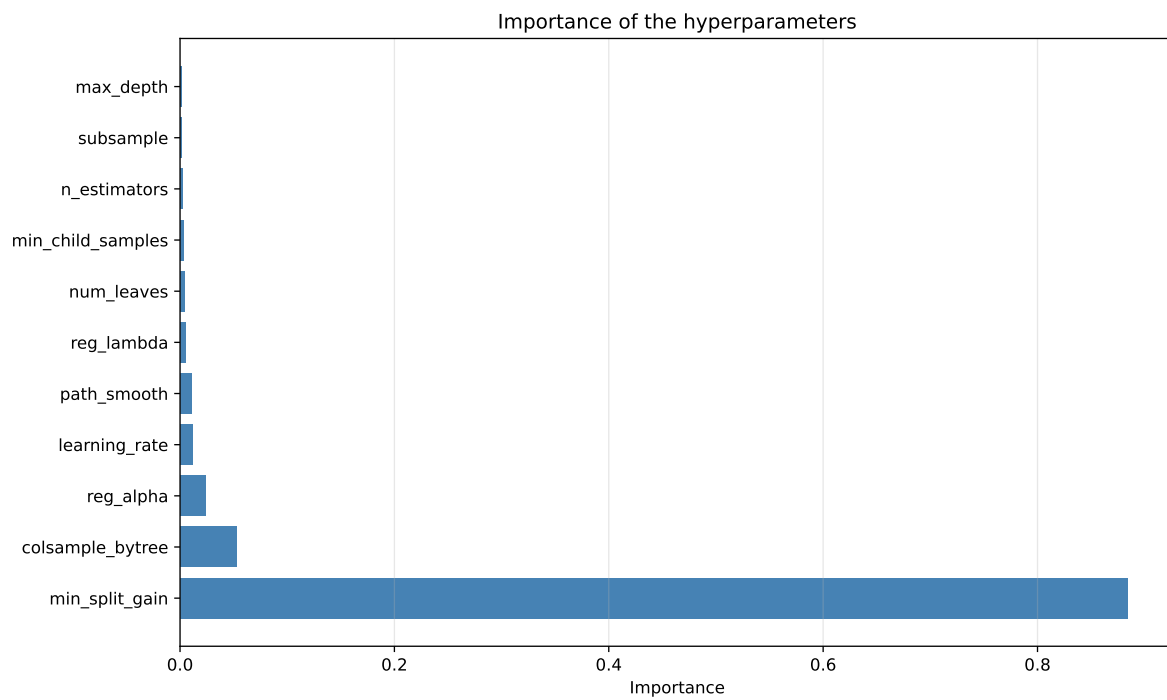
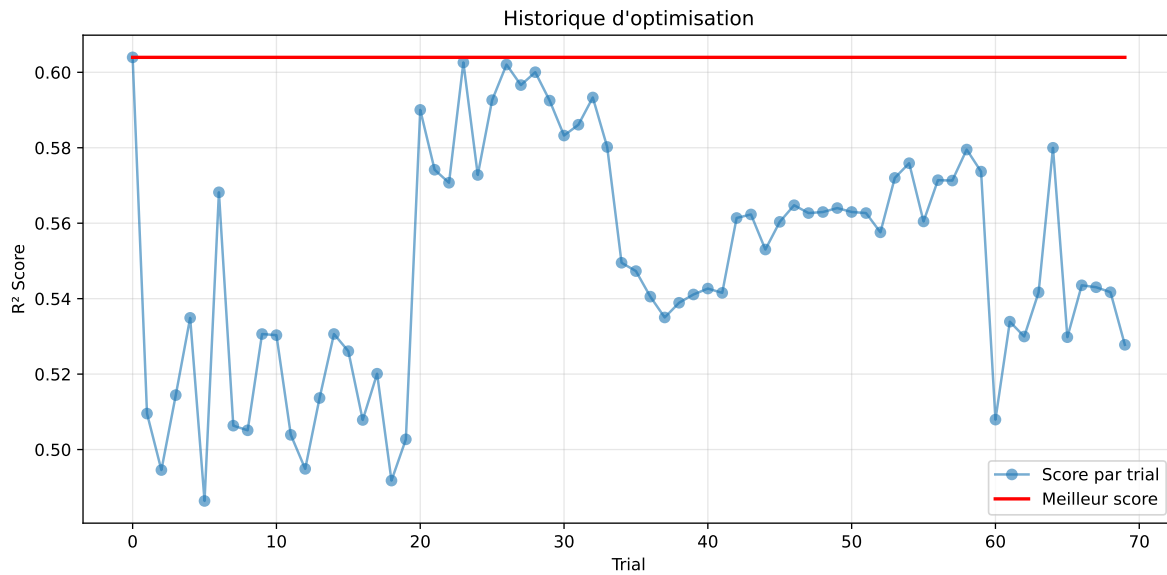
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 1

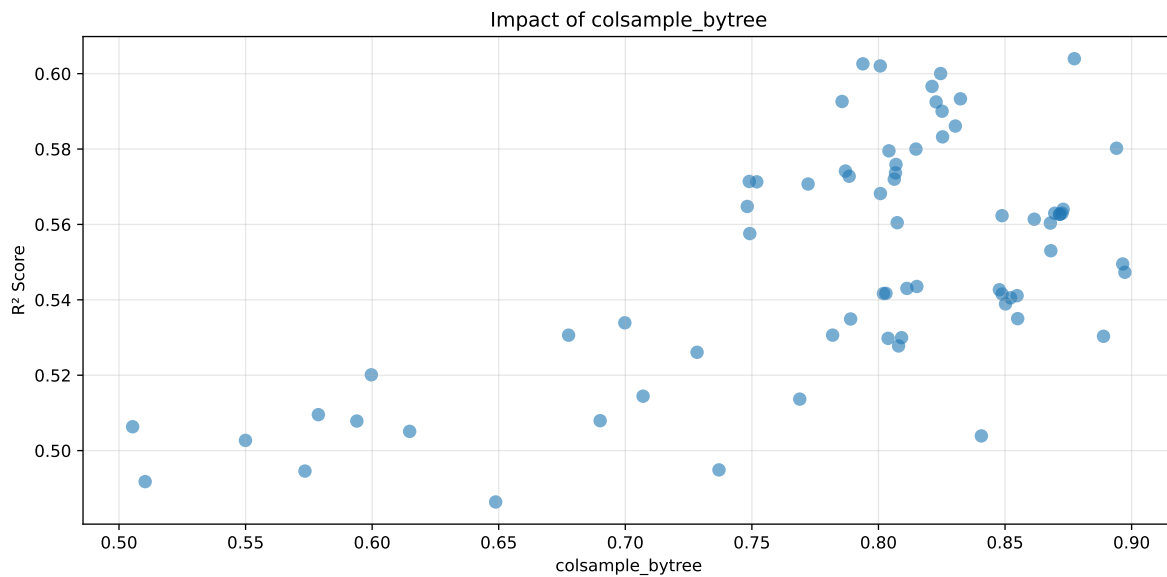
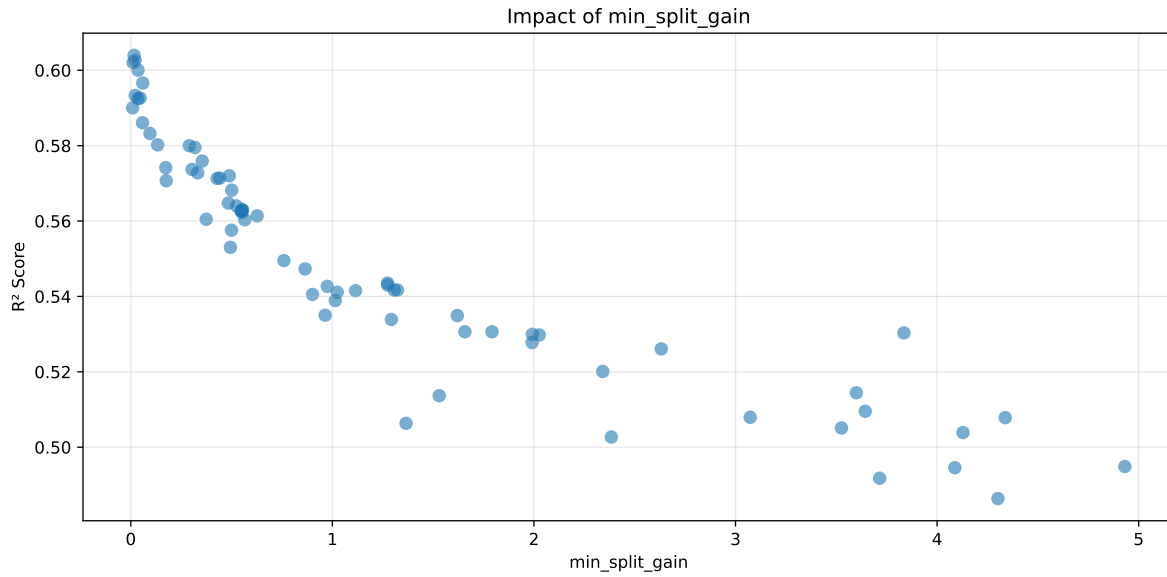
Len(group) : 5452, Len(training) : 3817, Len(test) : 1635

Distribution y_{train} vs y_{test} : y_{train} : mean=2.389, std=0.817, min=0.973, max=13.845
 y_{test} : mean=2.385, std=0.771, min=1.088, max=7.277

===== TOP Importance FEATURES =====
 feature importance r_NH3 160 r_umidity 133 r_CO2 95 r_temperature 46 r_THI 17

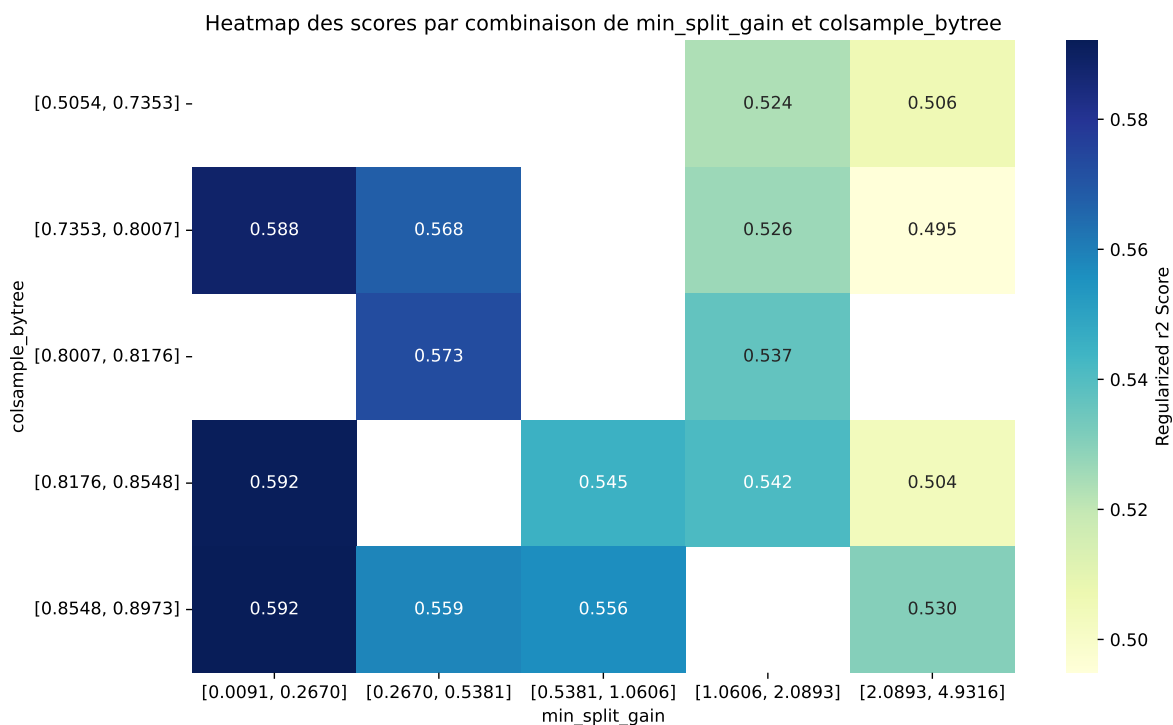


Paramètres avec >5% d'importance : ['min_split_gain', 'colsample_bytree']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5391	0.5144	0.5631	0.0375	0.5144	0.587	0.0726	1.1411	1.048
2	0.5319	0.5079	0.5565	0.0371	0.5079	0.5768	0.0688	1.1355	1.0472
3	0.5326	0.5078	0.5544	0.0362	0.5078	0.568	0.0602	1.1185	1.0487
4	0.5294	0.5051	0.5543	0.036	0.5051	0.5786	0.0735	1.1455	1.0481
5	0.5249	0.5039	0.5505	0.0359	0.5039	0.574	0.0701	1.1391	1.0416
6	0.5261	0.5027	0.5517	0.0361	0.5027	0.5762	0.0735	1.1461	1.0465
7	0.5145	0.4949	0.5406	0.0353	0.4949	0.5621	0.0672	1.1359	1.0397
8	0.5162	0.4946	0.5405	0.0345	0.4946	0.565	0.0705	1.1425	1.0437
9	0.5112	0.4918	0.5347	0.0362	0.4918	0.5525	0.0607	1.1234	1.0396
10	0.5054	0.4864	0.5309	0.0352	0.4864	0.5519	0.0655	1.1347	1.0392
---	---	---	---	---	---	---	---	---	---
70	0.7131	0.604	0.6499	0.0467	0.2765	0.6615	0.0576	1.0953	1.1807

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.045553956881323236, 'reg__num_leaves': 15, 'reg__subsample': 0.760820541120528, 'reg__colsample_bytree': 0.7070241783489023, 'reg__min_child_samples': 42, 'reg__reg_alpha': 8.237296042368857, 'reg__reg_lambda': 5.494689148051524, 'reg__min_split_gain': 3.5996408269770885, 'reg__path_smooth': 3.9720648463738675}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.0642126297861669, 'reg__num_leaves': 44, 'reg__subsample': 0.8675142703352609, 'reg__colsample_bytree': 0.6900647194591368, 'reg__min_child_samples': 34, 'reg__reg_alpha': 4.615254846043131, 'reg__reg_lambda': 44.32264913627063, 'reg__min_split_gain': 3.0728962619154236, 'reg__path_smooth': 0.9748594679111522}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.013521902964155242, 'reg__num_leaves': 18, 'reg__subsample': 0.607972126533715, 'reg__colsample_bytree': 0.5939608009485629, 'reg__min_child_samples': 15, 'reg__reg_alpha': 0.2068554384175076, 'reg__reg_lambda': 21.490035135853102, 'reg__min_split_gain': 4.337962657002311, 'reg__path_smooth': 1.0231813572821986}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.05525100990942551, 'reg__num_leaves': 41, 'reg__subsample': 0.627559599679662, 'reg__colsample_bytree': 0.6148056005902329, 'reg__min_child_samples': 49, 'reg__reg_alpha': 2.480667695381853, 'reg__reg_lambda': 48.2387811297051, 'reg__min_split_gain': 3.5256764825091738, 'reg__path_smooth': 4.441538657910312}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.01697359684341139, 'reg__num_leaves': 39, 'reg__subsample': 0.5221617984302172, 'reg__colsample_bytree': 0.8406537251213715, 'reg__min_child_samples': 47, 'reg__reg_alpha': 9.780044783099326, 'reg__reg_lambda': 11.137879917822113, 'reg__min_split_gain': 4.1282787489840445, 'reg__path_smooth': 3.6300696545649553}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.040728536456157705, 'reg__num_leaves': 20, 'reg__subsample': 0.6862143129046664, 'reg__colsample_bytree': 0.5499608491040999, 'reg__min_child_samples': 31, 'reg__reg_alpha': 9.332374349433994, 'reg__reg_lambda': 44.29871403759023, 'reg__min_split_gain': 2.384079800043345, 'reg__path_smooth': 4.2388100854098365}

Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 8, 'reg__learning_rate': 0.08297783972094375, 'reg__num_leaves': 50, 'reg__subsample': 0.8607235233098808, 'reg__colsample_bytree': 0.7370040809822034, 'reg__min_child_samples': 32, 'reg__reg_alpha': 8.55826492729749, 'reg__reg_lambda': 32.00795050961301, 'reg__min_split_gain': 4.93157841719068, 'reg__path_smooth': 3.1327269102050455}

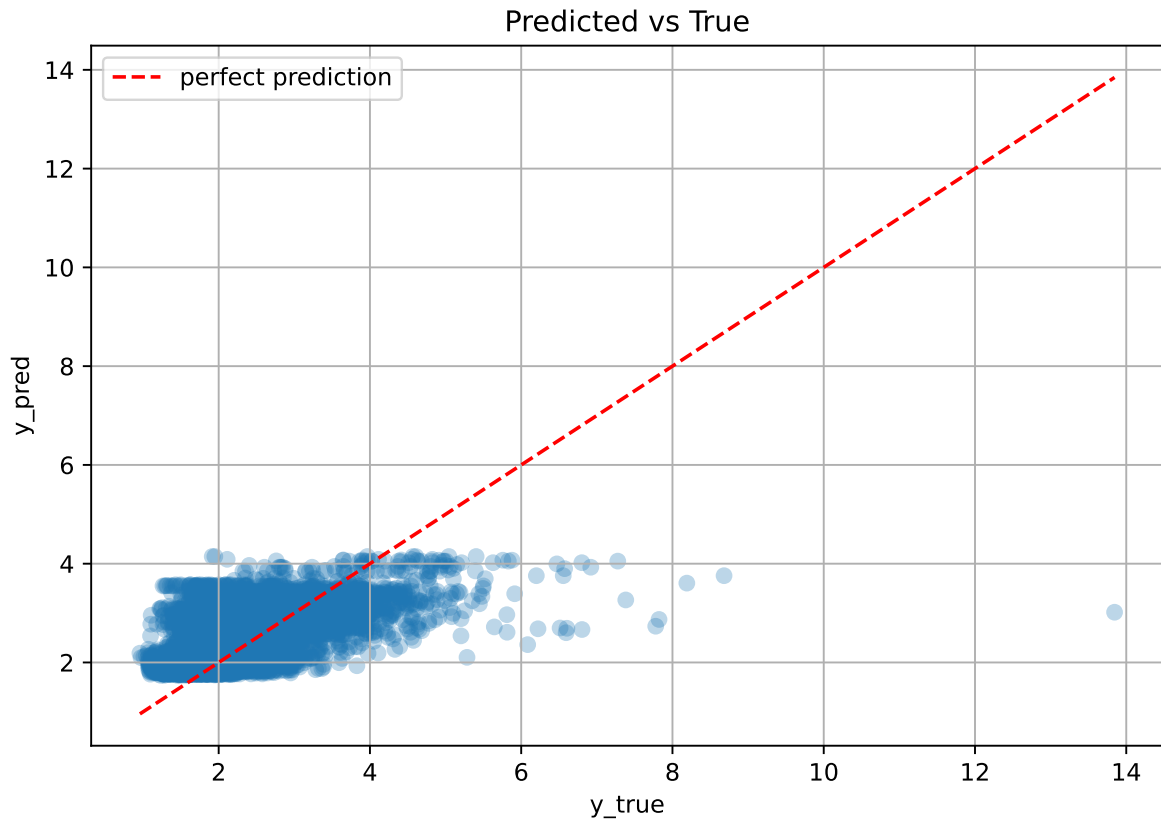
Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.05776503429902865, 'reg__num_leaves': 52, 'reg__subsample': 0.6681494828797923, 'reg__colsample_bytree': 0.5734559965604878, 'reg__min_child_samples': 20, 'reg__reg_alpha': 8.517465666400366, 'reg__reg_lambda': 31.255049954904468, 'reg__min_split_gain': 4.08822726945109, 'reg__path_smooth': 3.851480238010865}

Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.0938396454682987, 'reg__num_leaves': 26, 'reg__subsample': 0.6837894712899061,

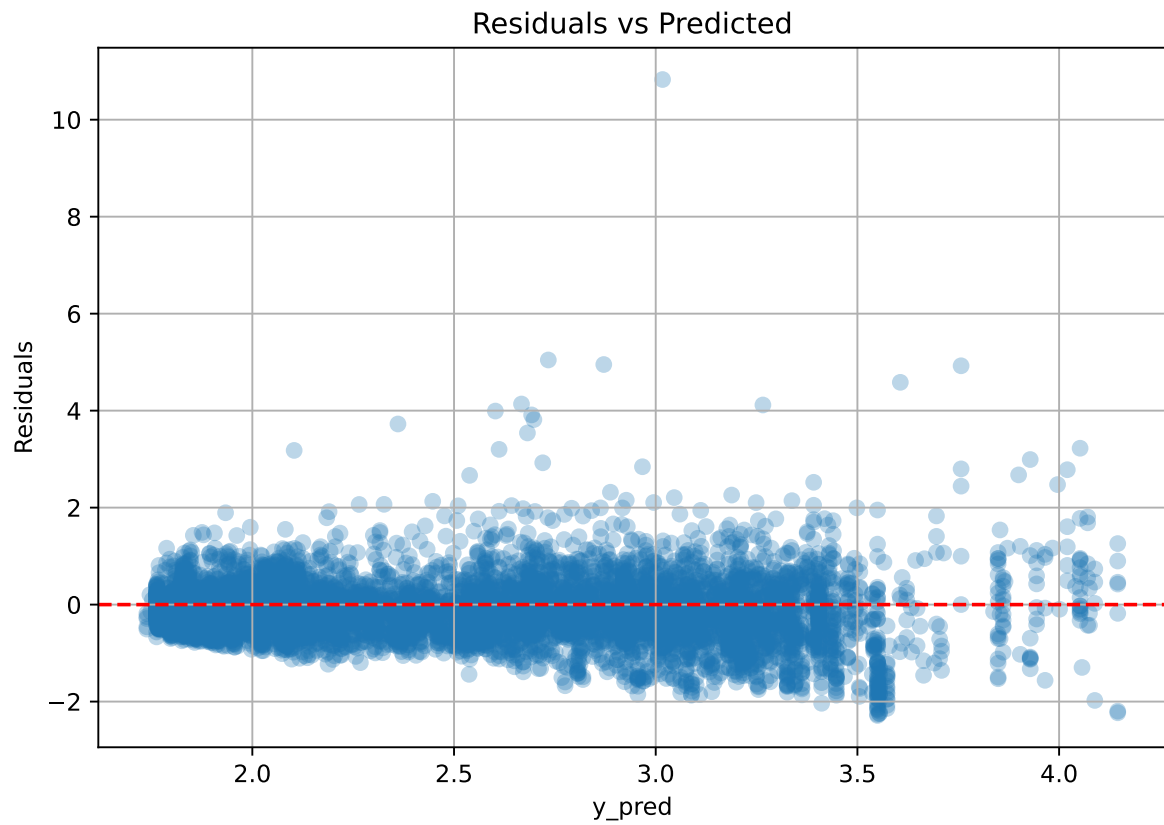
'reg__colsample_bytree': 0.5103307164877043, 'reg__min_child_samples': 26, 'reg__reg_alpha': 4.709121955397178, 'reg__reg_lambda': 34.4932719072213, 'reg__min_split_gain': 3.715040407210224, 'reg__path_smooth': 2.0945961414919494}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.09252529088127708, 'reg__num_leaves': 51, 'reg__subsample': 0.6368201223472186, 'reg__colsample_bytree': 0.6487803572898326, 'reg__min_child_samples': 38, 'reg__reg_alpha': 8.467278574038199, 'reg__reg_lambda': 48.68511231758062, 'reg__min_split_gain': 4.3017866037525785, 'reg__path_smooth': 4.520888294583504}

2.4.2 Scatter



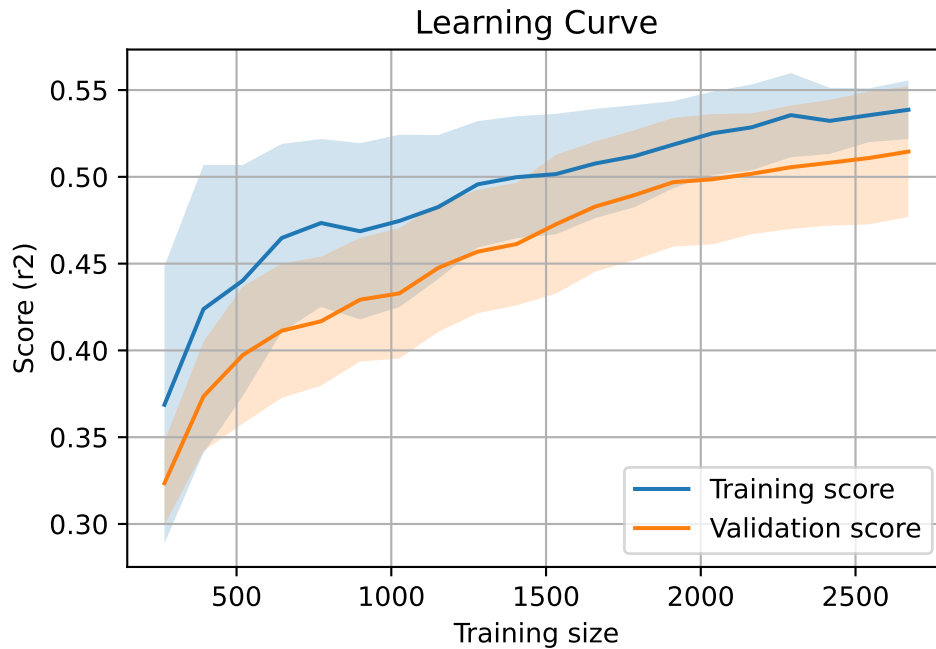
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 1

`Len(group)` : 5452, `Len(training)` : 3817,

`Len(training_real)` : 2672, `Len(test_of_training)` : 1145, `Len(test)` : 1635

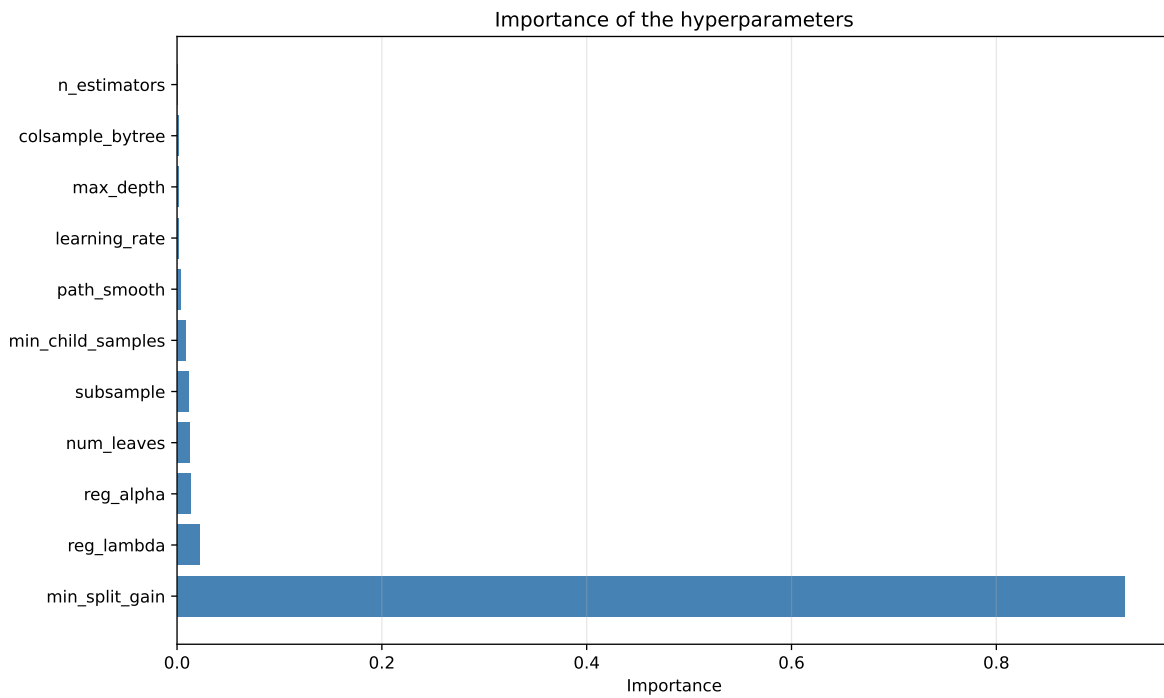
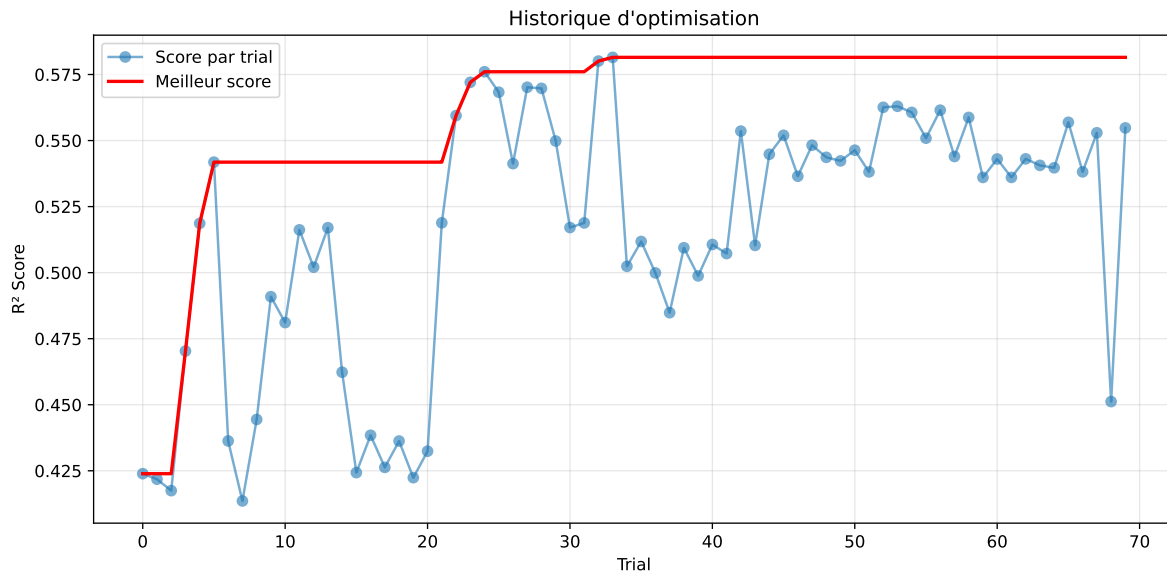


2.4.5 Gridsearch — Group: 2

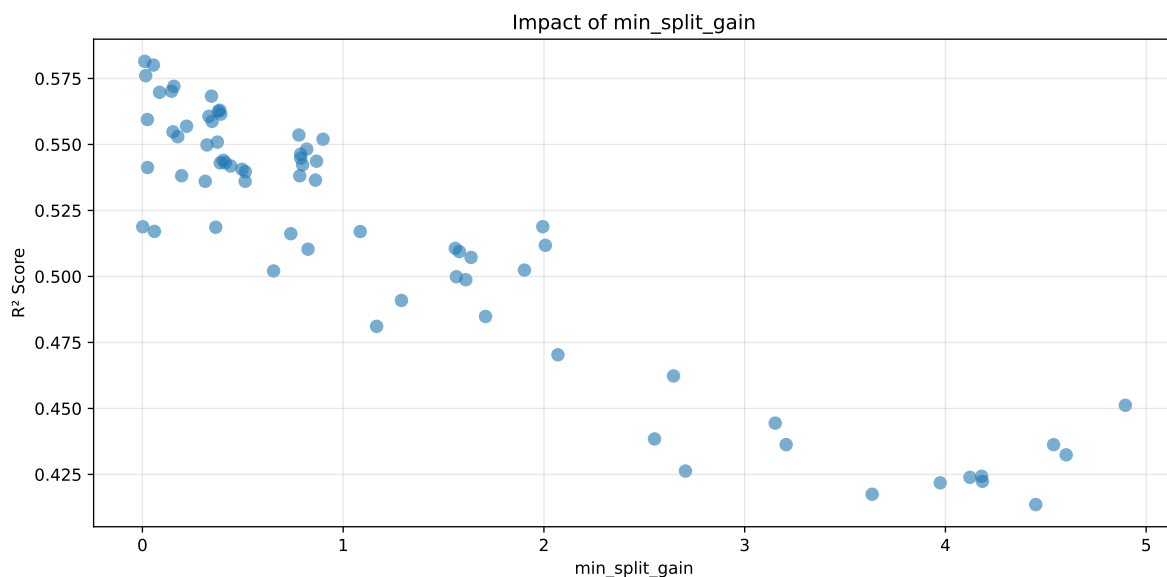
Len(group) : 2716, Len(training) : 1902, Len(test) : 814

Distribution y_train vs y_test: y_train: mean=2.260, std=0.774, min=0.959, max=7.822
y_test: mean=2.299, std=0.764, min=1.068, max=6.514

===== TOP Importance FEATURES =====
feature importance r_umidity 310 r_CO2 250 r_temperature 191 r_THI 189 r_NH3 158



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5087	0.4848	0.4355	0.0387	0.4848	0.4642	-0.0206	0.9575	1.0492
2	0.5045	0.4811	0.4317	0.0368	0.4811	0.4611	-0.02	0.9584	1.0486
3	0.4913	0.4703	0.4227	0.0388	0.4703	0.4478	-0.0226	0.952	1.0447
4	0.4759	0.4623	0.4097	0.0379	0.4623	0.4328	-0.0295	0.9362	1.0293
5	0.465	0.4512	0.4037	0.036	0.4512	0.4304	-0.0208	0.9538	1.0306
6	0.4531	0.4444	0.3905	0.0364	0.4444	0.4144	-0.03	0.9324	1.0196
7	0.4475	0.4384	0.3891	0.0364	0.4384	0.4157	-0.0227	0.9482	1.0206
8	0.4451	0.4363	0.3869	0.0354	0.4363	0.4099	-0.0264	0.9394	1.0203
9	0.4461	0.4363	0.3905	0.0357	0.4363	0.4123	-0.0239	0.9452	1.0226
10	0.4409	0.4324	0.3872	0.0345	0.4324	0.4104	-0.022	0.9492	1.0196
---	---	---	---	---	---	---	---	---	---
70	0.8135	0.5815	0.5664	0.0318	-0.1146	0.5938	0.0124	1.0212	1.3991

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.020586295291299562, 'reg__num_leaves': 53, 'reg__subsample': 0.6001593094136645, 'reg__colsample_bytree': 0.7986689820869165, 'reg__min_child_samples': 16, 'reg__reg_alpha': 1.6574153461273413, 'reg__reg_lambda': 47.82200284733577, 'reg__min_split_gain': 1.709067031996451, 'reg__path_smooth': 2.335840809178459}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.09813175628195285, 'reg__num_leaves': 18, 'reg__subsample': 0.53223056527957,

'reg__colsample_bytree': 0.8853270519522018, 'reg__min_child_samples': 49, 'reg__reg_alpha': 4.69156426422782, 'reg__reg_lambda': 39.93937373934531, 'reg__min_split_gain': 1.166538288039165, 'reg__path_smooth': 2.6086156646614707}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.012068479952709548, 'reg__num_leaves': 42, 'reg__subsample': 0.6949843312792702, 'reg__colsample_bytree': 0.6243610836036592, 'reg__min_child_samples': 28, 'reg__reg_alpha': 3.351563796653323, 'reg__reg_lambda': 14.40624645170172, 'reg__min_split_gain': 2.0697505008049717, 'reg__path_smooth': 4.829635570381862}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.01841001058035537, 'reg__num_leaves': 47, 'reg__subsample': 0.6319089968077147, 'reg__colsample_bytree': 0.7577651769994262, 'reg__min_child_samples': 31, 'reg__reg_alpha': 3.3330946468849785, 'reg__reg_lambda': 34.49699791726819, 'reg__min_split_gain': 2.645649979469751, 'reg__path_smooth': 0.9701385325696088}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.023272507064090897, 'reg__num_leaves': 18, 'reg__subsample': 0.6577229797209729, 'reg__colsample_bytree': 0.7467484826991344, 'reg__min_child_samples': 12, 'reg__reg_alpha': 0.69866296782789, 'reg__reg_lambda': 10.192380987794085, 'reg__min_split_gain': 4.8962806774846985, 'reg__path_smooth': 4.60537659280396}

Rank 6: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.02529126626652188, 'reg__num_leaves': 45, 'reg__subsample': 0.5413000760052342, 'reg__colsample_bytree': 0.8868052231813521, 'reg__min_child_samples': 26, 'reg__reg_alpha': 8.066974544724705, 'reg__reg_lambda': 11.712939845698255, 'reg__min_split_gain': 3.1520449504736496, 'reg__path_smooth': 1.9783692797007508}

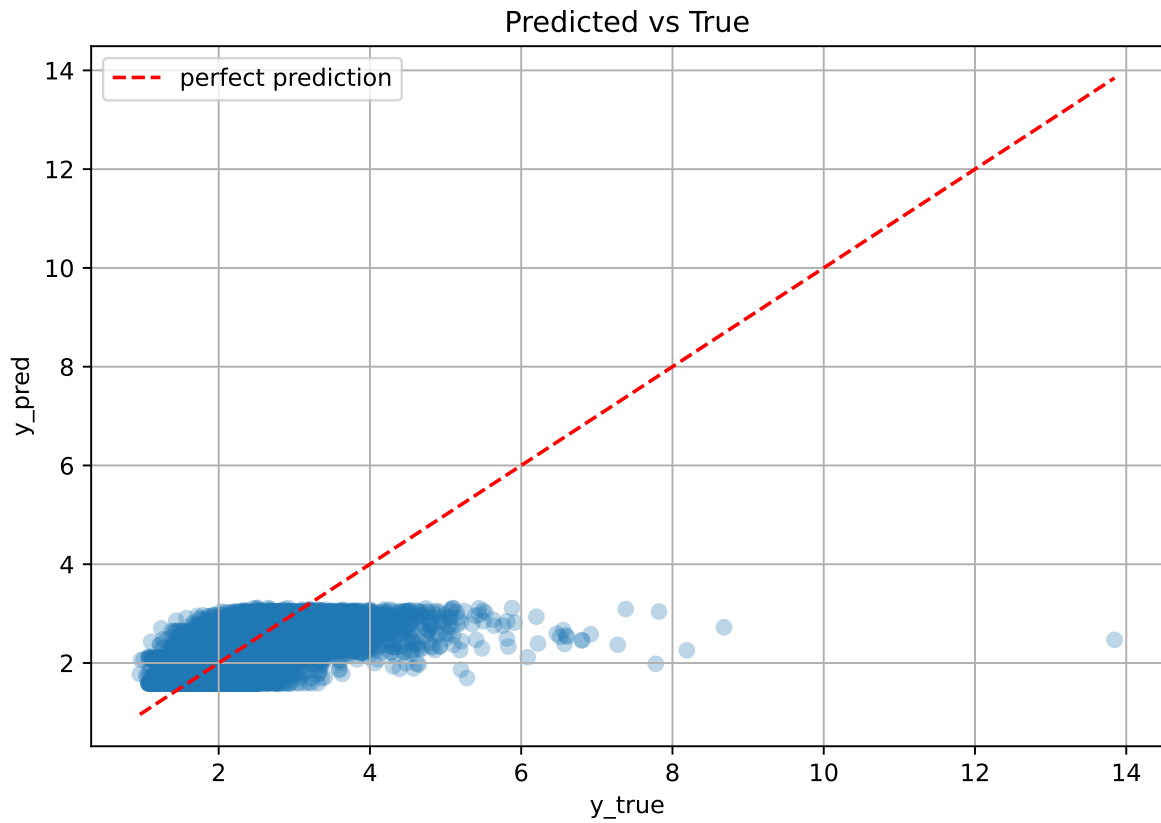
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.02315239258442772, 'reg__num_leaves': 26, 'reg__subsample': 0.7895497540515392, 'reg__colsample_bytree': 0.5814311879611065, 'reg__min_child_samples': 30, 'reg__reg_alpha': 8.711368325550998, 'reg__reg_lambda': 1.6395079570048265, 'reg__min_split_gain': 2.550902020770749, 'reg__path_smooth': 1.8781703319604466}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.025690075965979615, 'reg__num_leaves': 29, 'reg__subsample': 0.6924245942887055, 'reg__colsample_bytree': 0.5143401791049425, 'reg__min_child_samples': 22, 'reg__reg_alpha': 3.852080360554706, 'reg__reg_lambda': 31.512523278987363, 'reg__min_split_gain': 3.2065435137268428, 'reg__path_smooth': 0.9476294812319241}

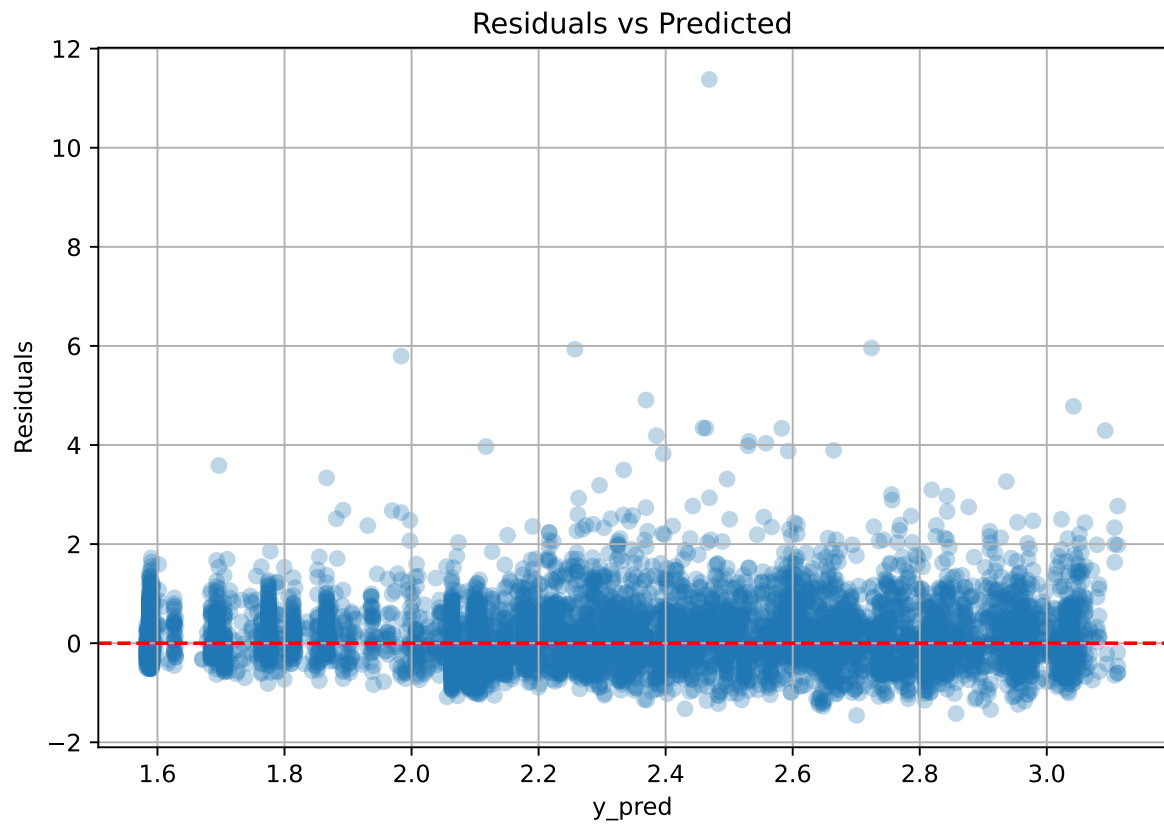
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.03698743318429457, 'reg__num_leaves': 60, 'reg__subsample': 0.8812335511926765, 'reg__colsample_bytree': 0.6099853187642361, 'reg__min_child_samples': 13, 'reg__reg_alpha': 1.414614089622579, 'reg__reg_lambda': 9.958995655801338, 'reg__min_split_gain': 4.538690806556759, 'reg__path_smooth': 0.9810856561084436}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.03526209143272994, 'reg__num_leaves': 37, 'reg__subsample': 0.8167081307668036, 'reg__colsample_bytree': 0.5302876654217741, 'reg__min_child_samples': 46, 'reg__reg_alpha': 1.0056116308867036, 'reg__reg_lambda': 16.74344675913138, 'reg__min_split_gain': 4.601551630739035, 'reg__path_smooth': 3.6361783583828116}

2.4.6 Scatter



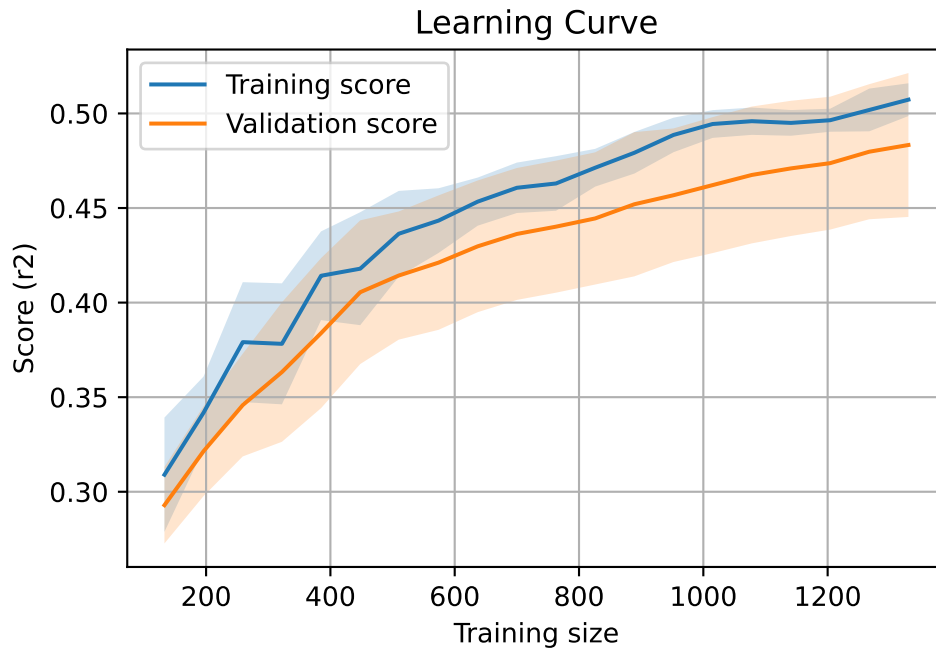
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2

`Len(group)` : 2716, `Len(training)` : 1902,

`Len(training_real)` : 1332, `Len(test_of_training)` : 570, `Len(test)` : 814



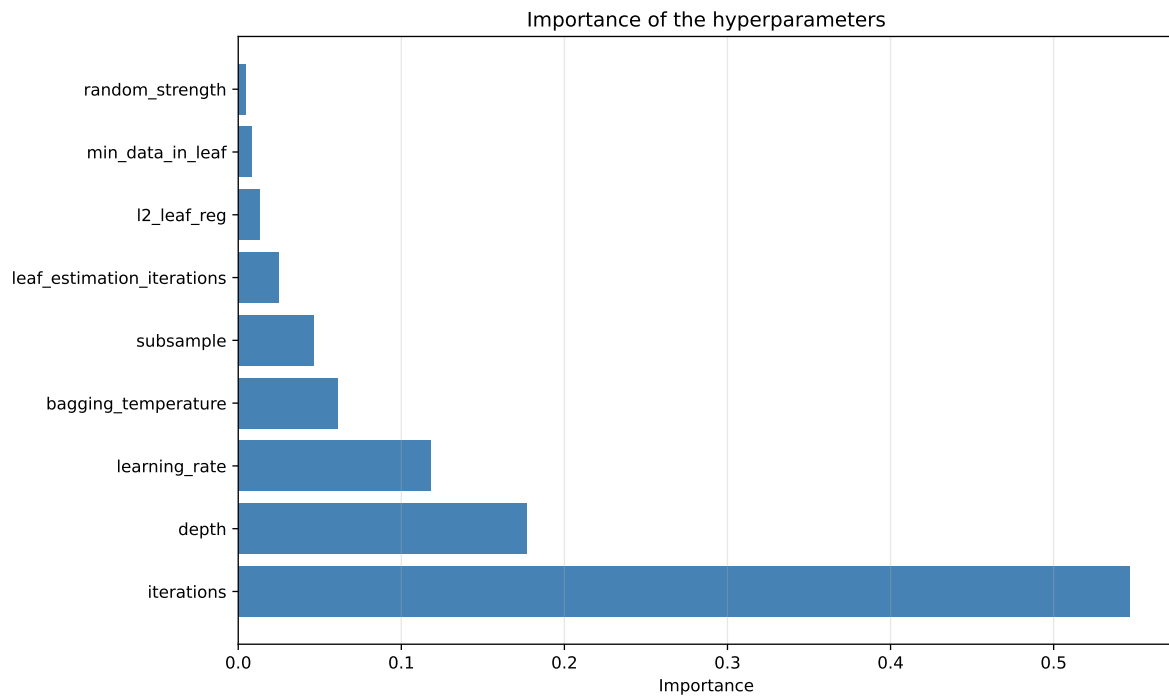
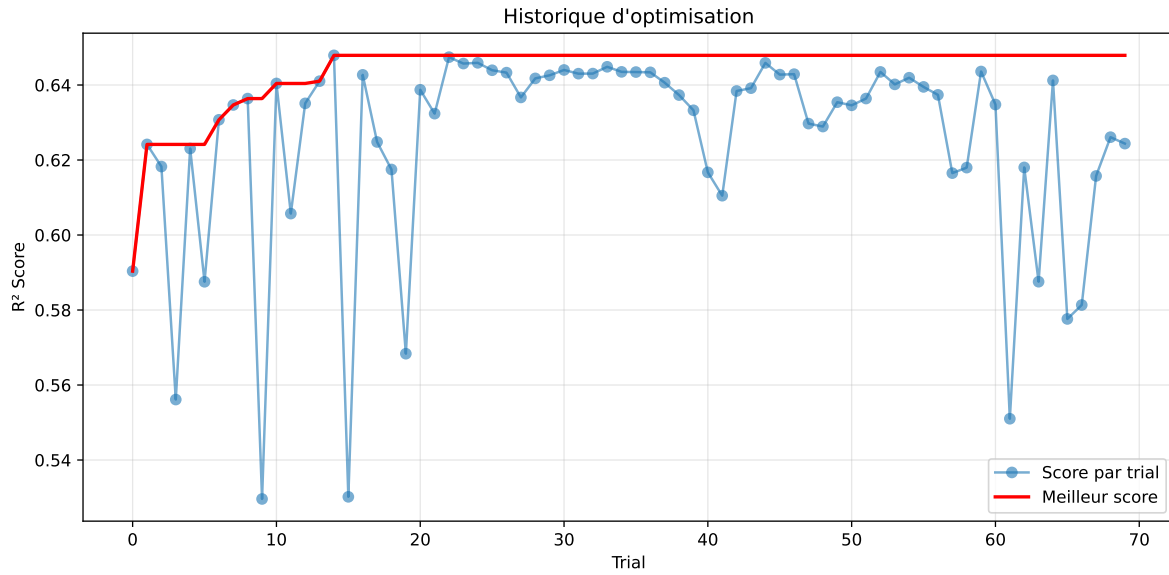
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 1

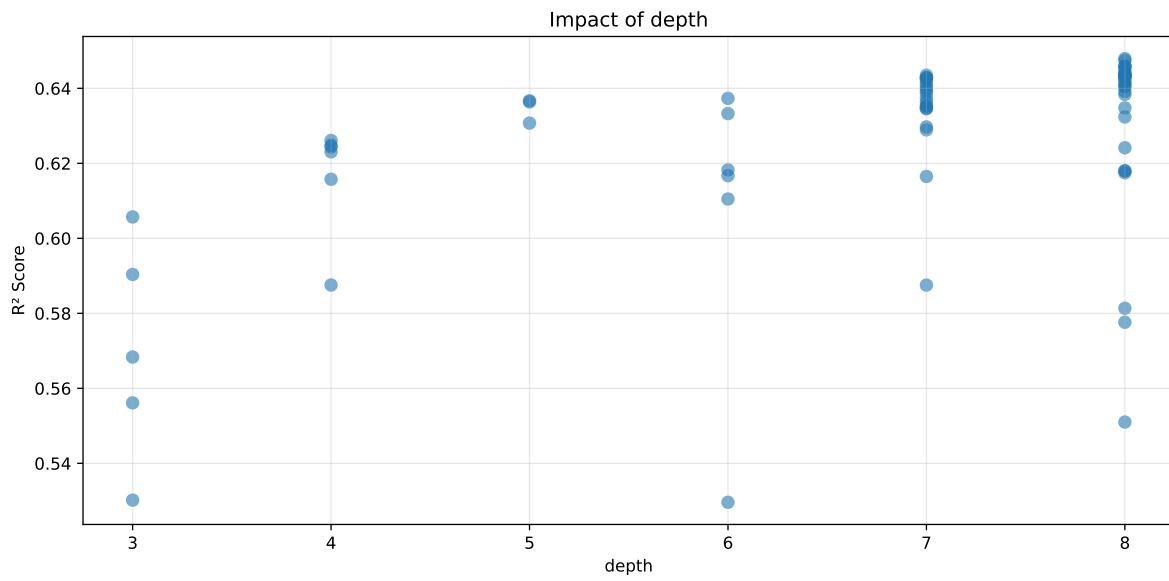
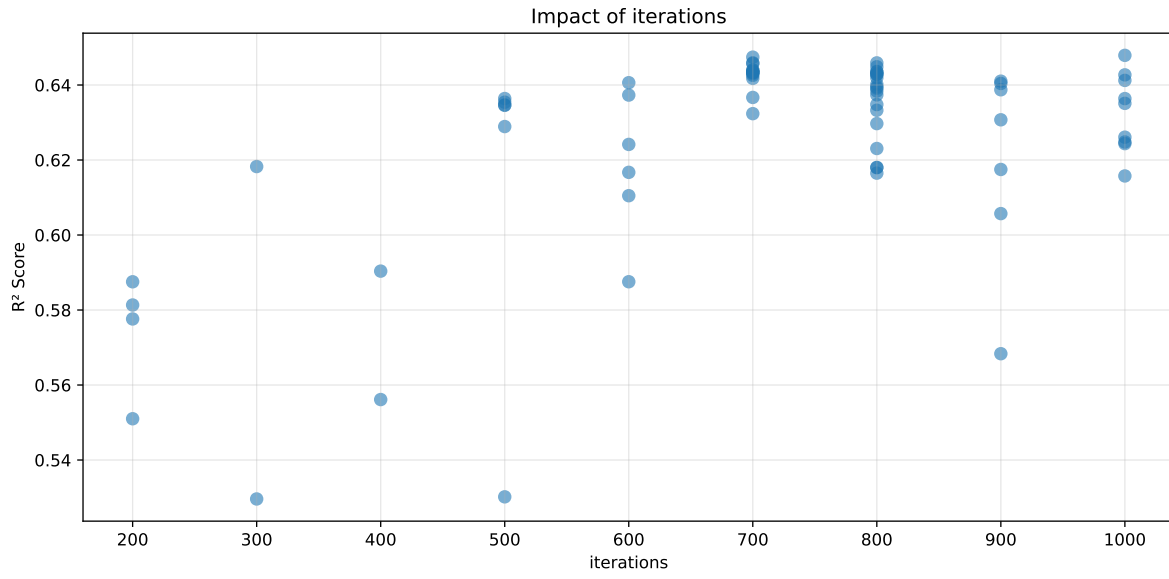
Len(group) : 5452, **Len(training) :** 3817, **Len(test) :** 1635

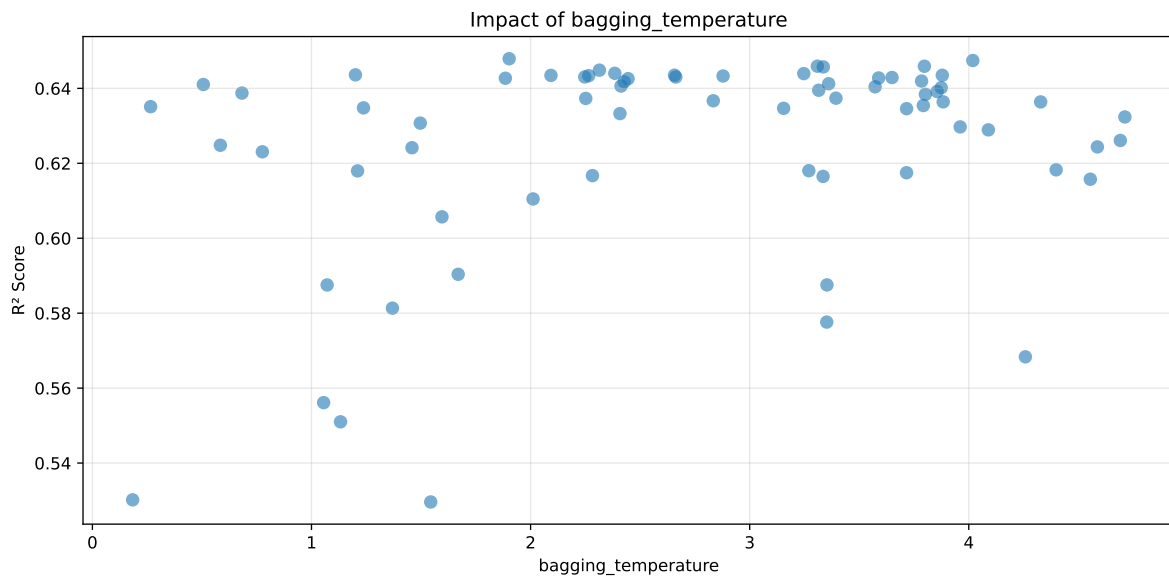
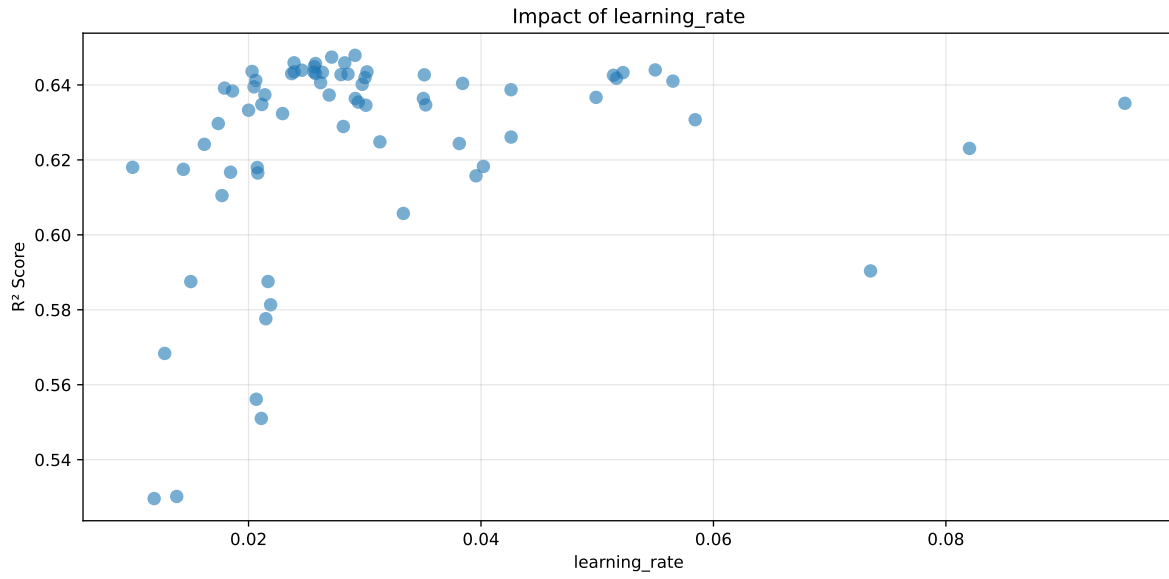
Distribution y_train vs y_test: y_train: mean=2.389, std=0.817, min=0.973, max=13.845
y_test: mean=2.385, std=0.771, min=1.088, max=7.277

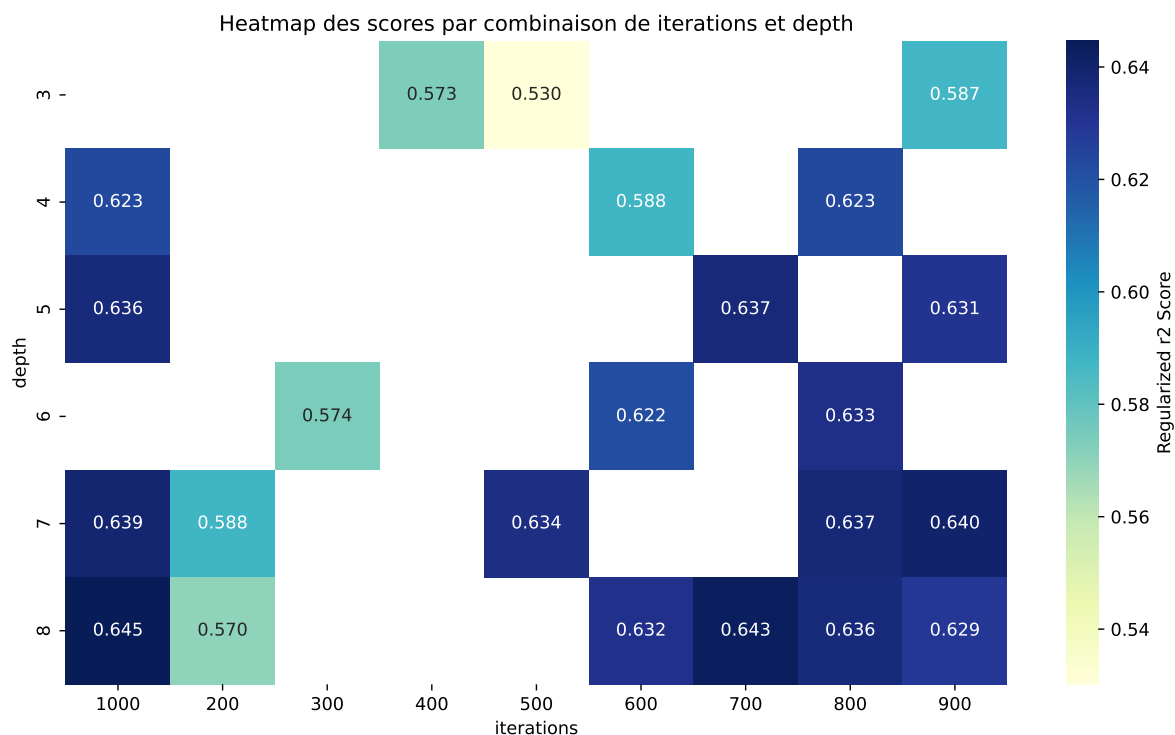
===== TOP Importance FEATURES =====
feature importance r_umidity 41.609755 r_NH3 29.438466 r_CO2 12.566107 r_temperature
10.171550 r_THI 6.214122



Paramètres avec >5% d'importance : ['iterations', 'depth', 'learning_rate', 'bagging_temperature']







rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5839	0.5561	0.6019	0.0411	0.5561	0.6078	0.0517	1.0929	1.0499
2	0.5491	0.5302	0.578	0.0381	0.5302	0.5868	0.0566	1.1067	1.0356
3	0.547	0.5296	0.5771	0.0369	0.5296	0.5884	0.0588	1.111	1.0328
4	0.581	0.551	0.6037	0.0382	0.461	0.6204	0.0694	1.126	1.0544
5	0.6048	0.5683	0.6146	0.0426	0.4589	0.6212	0.0529	1.0931	1.0642
6	0.6199	0.5776	0.6257	0.0406	0.4509	0.6357	0.0581	1.1005	1.0731
7	0.6268	0.5813	0.625	0.0424	0.445	0.6332	0.0519	1.0892	1.0782
8	0.6357	0.5875	0.6271	0.0452	0.4432	0.6377	0.0502	1.0854	1.0819
9	0.647	0.5904	0.6319	0.0478	0.4204	0.6445	0.0542	1.0918	1.0959
10	0.674	0.6105	0.646	0.0479	0.4198	0.6517	0.0412	1.0675	1.1041
---	---	---	---	---	---	---	---	---	---
70	0.9326	0.6426	0.6707	0.0551	-0.2274	0.6918	0.0492	1.0766	1.4513

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.020663407155360893, 'reg__l2_leaf_reg': 15.220723710228956, 'reg__bagging_temperature': 1.0553611027022543, 'reg__random_strength': 1.7061733873199025, 'reg__subsample': 0.5966545910632073, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 9}

Rank 2: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.013821243807776645, 'reg__l2_leaf_reg': 18.286126032400386, 'reg__bagging_temperature': 0.18360908119202657, 'reg__random_strength': 1.8517011389253046, 'reg__subsample': 0.5308710782456004, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 1}

Rank 3: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.01187900916123026, 'reg__l2_leaf_reg': 45.62065470955194, 'reg__bagging_temperature': 1.5442223582445547, 'reg__random_strength': 2.7442632769895225, 'reg__subsample': 0.5364803838344195, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 2}

Rank 4: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.021099269659127902, 'reg__l2_leaf_reg': 13.528709253076249, 'reg__bagging_temperature': 1.132459556677613, 'reg__random_strength': 4.018813004530381, 'reg__subsample': 0.81682703249107, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.012784364869179395, 'reg__l2_leaf_reg': 30.921859255437877, 'reg__bagging_temperature': 4.258401469465897, 'reg__random_strength': 2.5127572059028287, 'reg__subsample': 0.5806637214620127, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 9}

Rank 6: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.021486048403061852, 'reg__l2_leaf_reg': 12.9045485668371, 'reg__bagging_temperature': 3.3517289041516465, 'reg__random_strength': 3.013839177837304, 'reg__subsample': 0.6178466414162468, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 2}

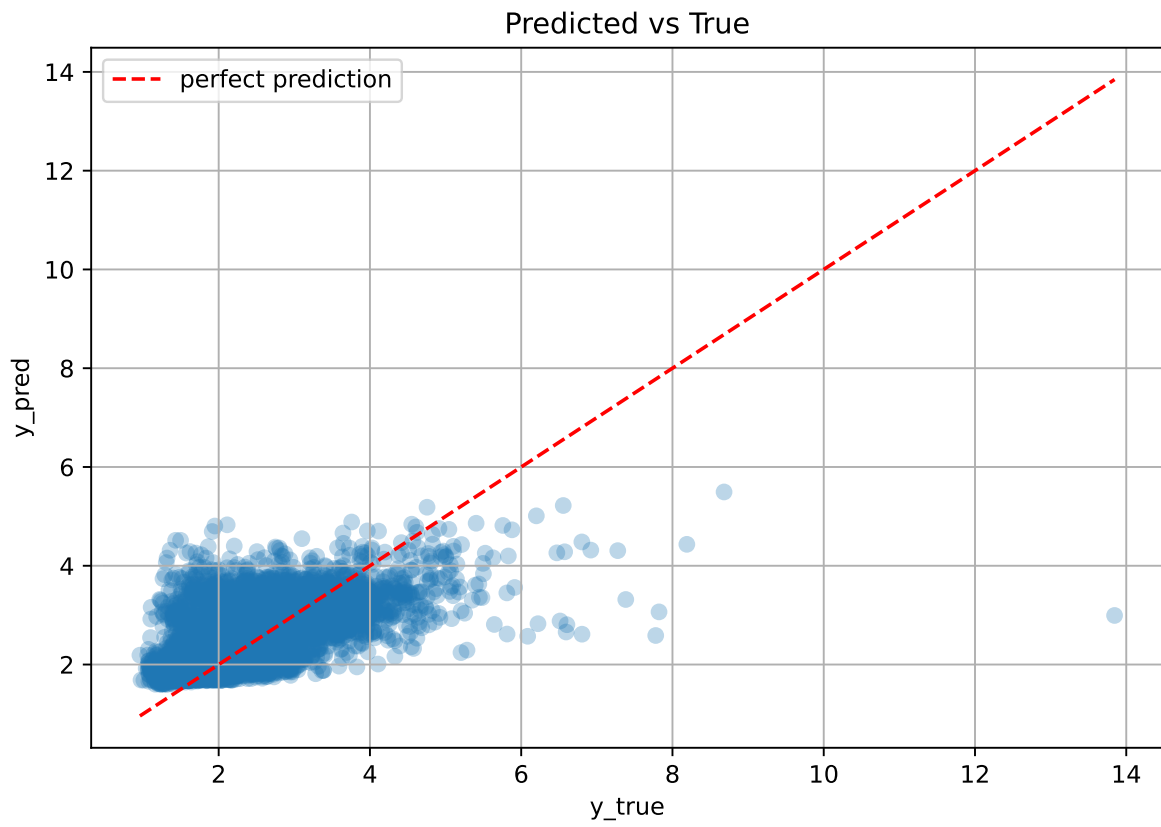
Rank 7: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.021897794651837207, 'reg__l2_leaf_reg': 4.994749182666861, 'reg__bagging_temperature': 1.3694250895296554, 'reg__random_strength': 4.5443799103165485, 'reg__subsample': 0.6397055281187832, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.02167790688473931, 'reg__l2_leaf_reg': 13.362756210344473, 'reg__bagging_temperature': 3.3526919855095274, 'reg__random_strength': 4.028012672533617, 'reg__subsample': 0.6329308940060324, 'reg__min_data_in_leaf': 29, 'reg__leaf_estimation_iterations': 2}

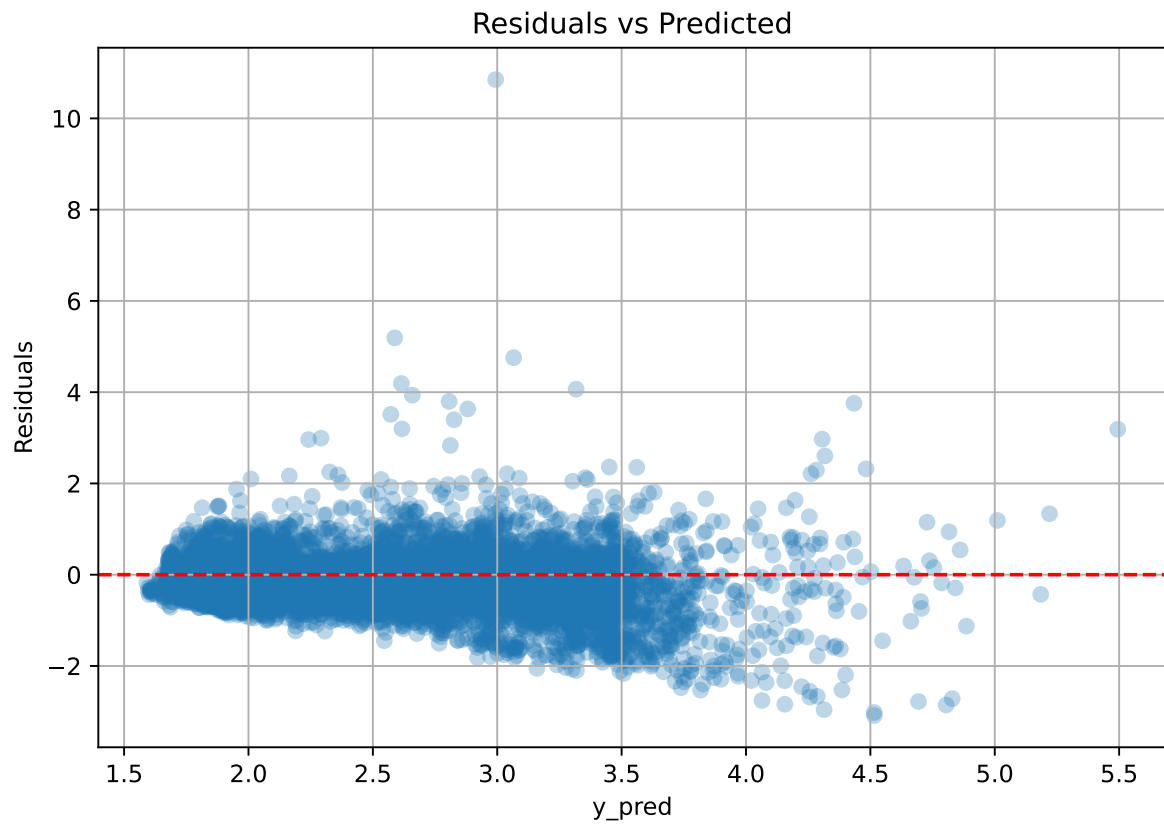
Rank 9: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.07351595829786904, 'reg__l2_leaf_reg': 48.89658398332616, 'reg__bagging_temperature': 1.6694106489045013, 'reg__random_strength': 4.153014219802057, 'reg__subsample': 0.8276821217431678, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 2}

Rank 10: {'reg__iterations': 600, 'reg__depth': 6, 'reg__learning_rate': 0.017713025528719523, 'reg__l2_leaf_reg': 9.361284182749584, 'reg__bagging_temperature': 2.0111410324391725, 'reg__random_strength': 4.510347069099751, 'reg__subsample': 0.7654291888031804, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 3}

2.5.2 Scatter



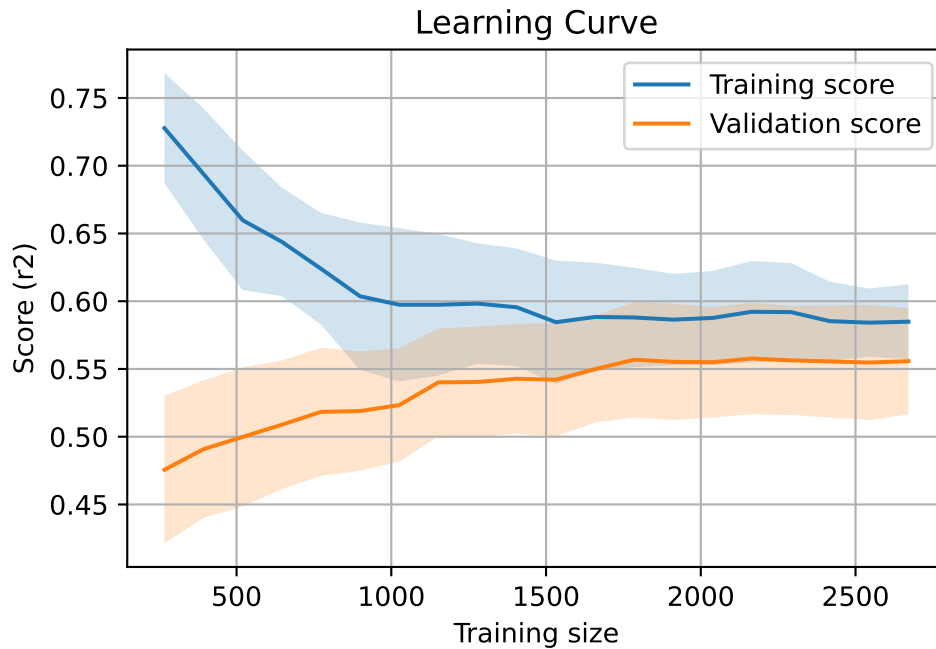
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 1

`Len(group)` : 5452, `Len(training)` : 3817,

`Len(training_real)` : 2672, `Len(test_of_training)` : 1145, `Len(test)` : 1635

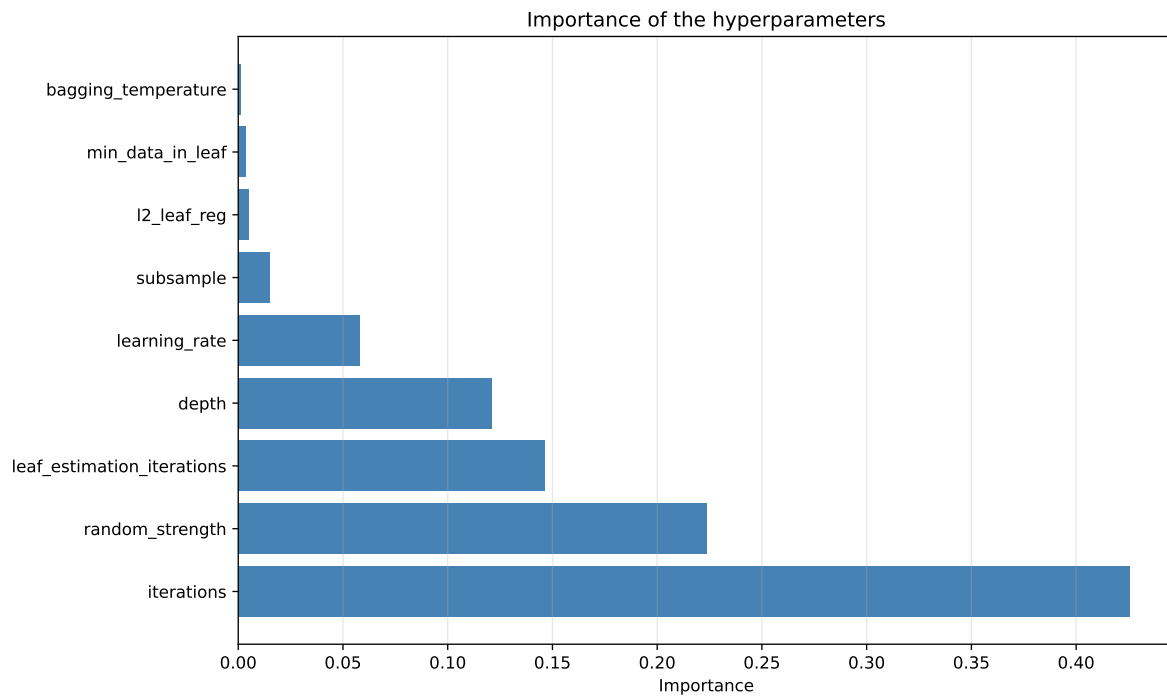
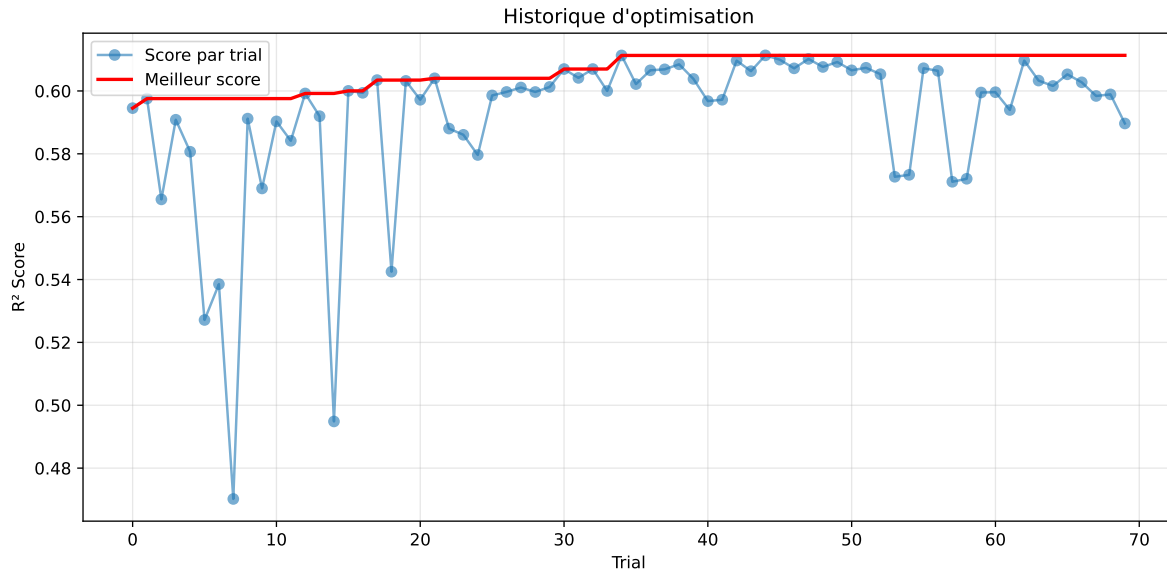


2.5.5 Gridsearch — Group: 2

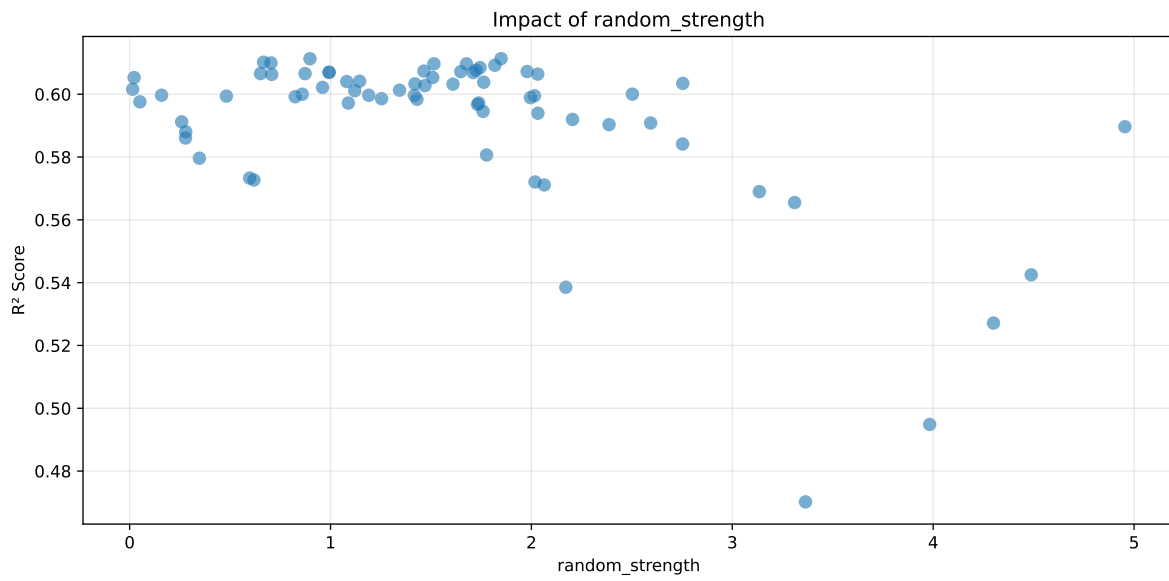
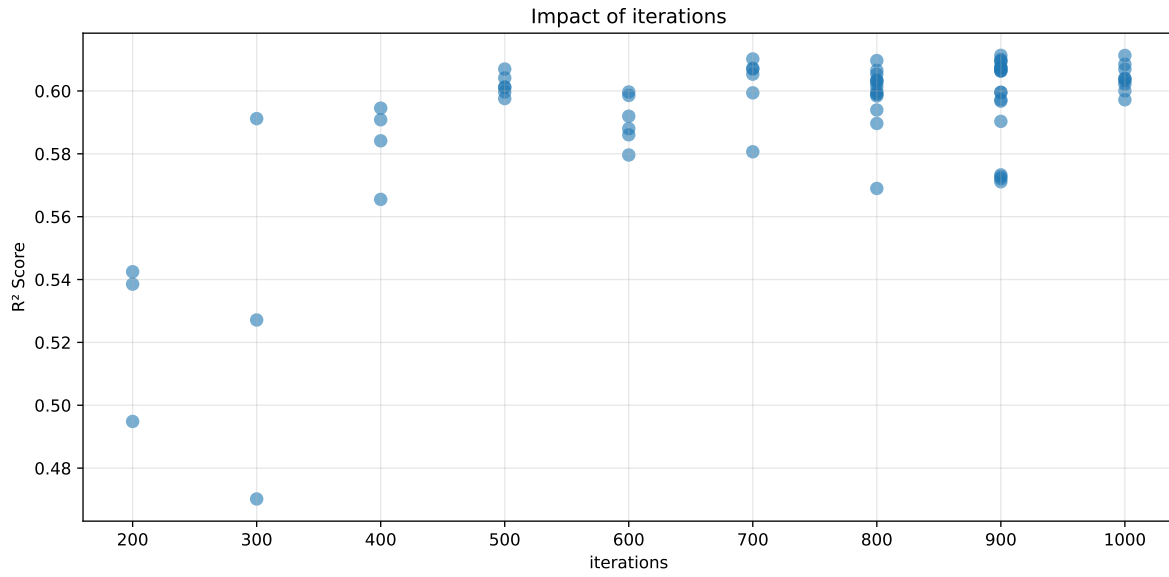
Len(group) : 2716, Len(training) : 1902, Len(test) : 814

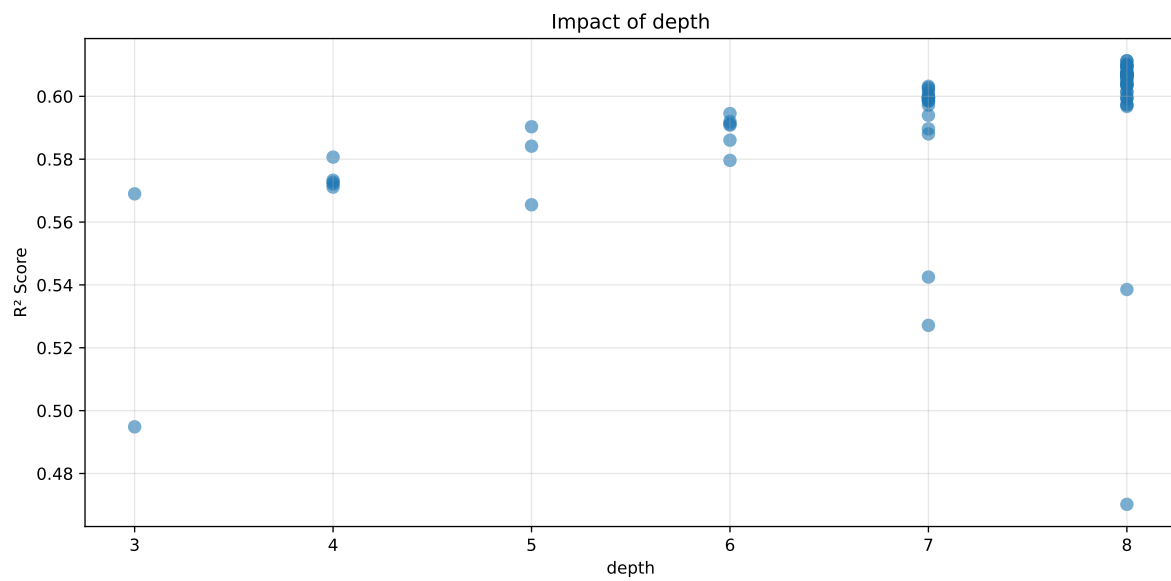
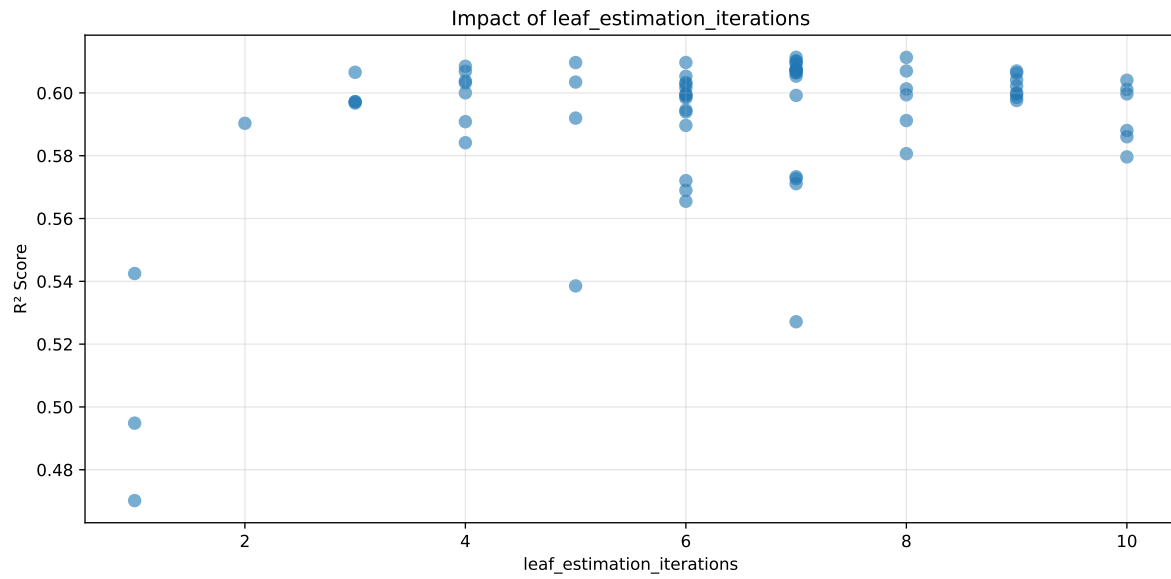
Distribution y_train vs y_test: y_train: mean=2.260, std=0.774, min=0.959, max=7.822
y_test: mean=2.299, std=0.764, min=1.068, max=6.514

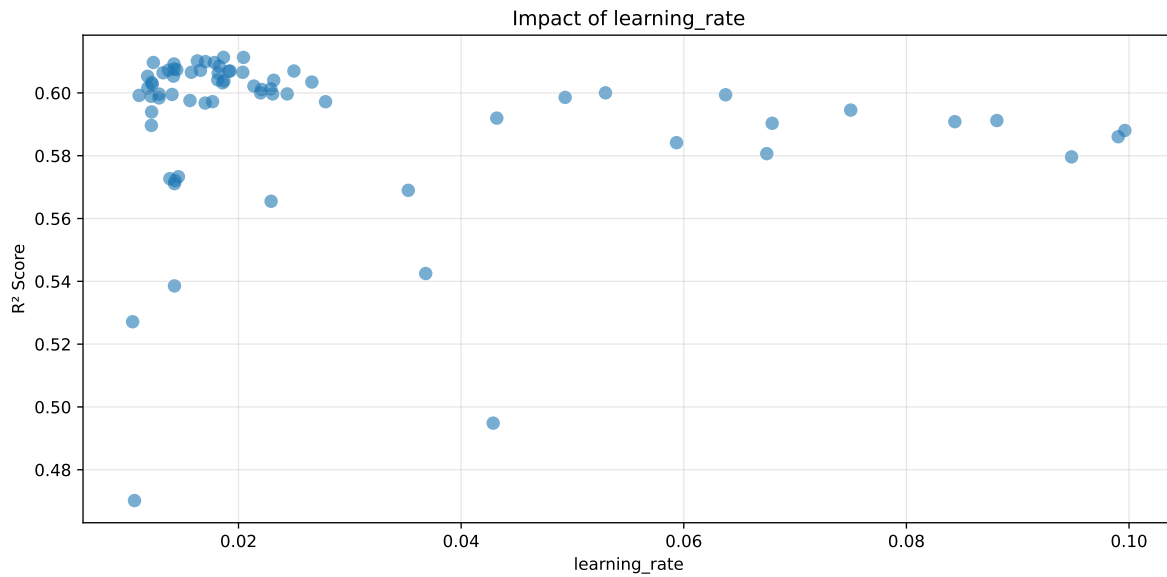
===== TOP Importance FEATURES =====
feature importance r_umidity 61.477636 r_THI 12.753615 r_CO2 10.951181 r_NH3 7.783379
r_temperature 7.034189



Paramètres avec >5% d'importance : ['iterations', 'random_strength', 'leaf_estimation_iterations', 'depth', 'learning_rate']

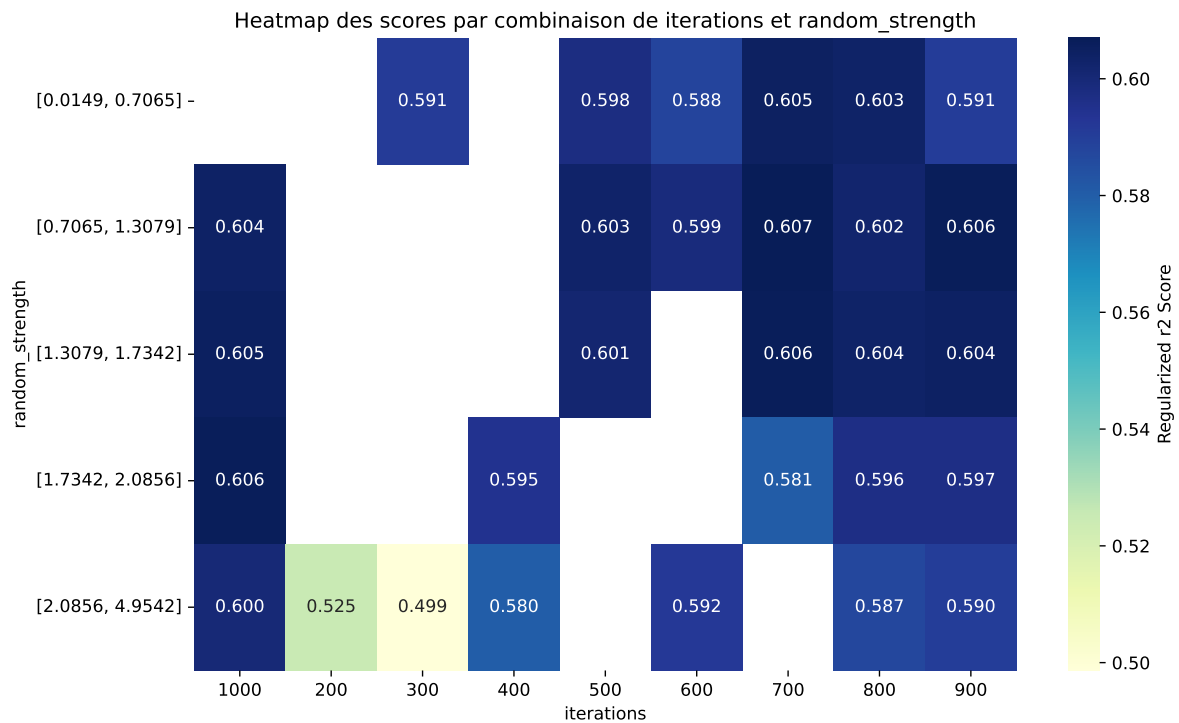






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.504	0.4948	0.4646	0.0338	0.4948	0.4729	-0.022	0.9556	1.0184
2	0.4797	0.4702	0.4369	0.0318	0.4702	0.4558	-0.0144	0.9693	1.0202
3	0.579	0.5425	0.5245	0.0344	0.4331	0.5416	-0.0009	0.9984	1.0672
4	0.5597	0.5271	0.5077	0.0347	0.4295	0.5136	-0.0135	0.9744	1.0617
5	0.5858	0.5385	0.5244	0.0346	0.3966	0.5308	-0.0077	0.9857	1.0878
6	0.6446	0.5655	0.5709	0.0333	0.3282	0.5758	0.0103	1.0182	1.1399
7	0.6741	0.5721	0.5778	0.0343	0.2658	0.5894	0.0174	1.0304	1.1785
8	0.6755	0.5711	0.5771	0.0342	0.2579	0.5878	0.0167	1.0293	1.1828
9	0.7144	0.5897	0.6006	0.0338	0.2154	0.6108	0.0211	1.0358	1.2116
10	0.7029	0.5727	0.5783	0.0336	0.1822	0.5922	0.0195	1.034	1.2273
---	---	---	---	---	---	---	---	---	---
70	0.9851	0.588	0.5687	0.0256	-0.603	0.584	-0.0041	0.9931	1.6752

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.04287838806454703, 'reg__l2_leaf_reg': 43.419799894578546, 'reg__bagging_temperature': 3.3823843486890306, 'reg__random_strength': 3.982704118480764, 'reg__subsample': 0.793481908451882, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.010656408744115893, 'reg__l2_leaf_reg': 45.33449877733599, 'reg__bagging_temperature': 4.786774966515343, 'reg__random_strength': 3.3643950222961463, 'reg__subsample': 0.8072652400339834, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 1}

Rank 3: {'reg__iterations': 200, 'reg__depth': 7, 'reg__learning_rate': 0.03681500504530205, 'reg__l2_leaf_reg': 31.212365759674988, 'reg__bagging_temperature': 4.2911300722417085, 'reg__random_strength': 4.4878344279205375, 'reg__subsample': 0.7098337291751224, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 1}

Rank 4: {'reg__iterations': 300, 'reg__depth': 7, 'reg__learning_rate': 0.010488915772122975, 'reg__l2_leaf_reg': 29.146382058276274, 'reg__bagging_temperature': 0.3393477587714344, 'reg__random_strength': 4.300335757612682, 'reg__subsample': 0.6848041782772833, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 7}

Rank 5: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.014246758879722975, 'reg__l2_leaf_reg': 25.198023900152055, 'reg__bagging_temperature': 2.816833946224073, 'reg__random_strength': 2.1710765854908773, 'reg__subsample': 0.5888891223028461, 'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 5}

Rank 6: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.022928126950253203, 'reg__l2_leaf_reg': 23.45729163116856, 'reg__bagging_temperature': 1.910913452218967, 'reg__random_strength': 3.3101926089977747, 'reg__subsample': 0.6078123392389316, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 6}

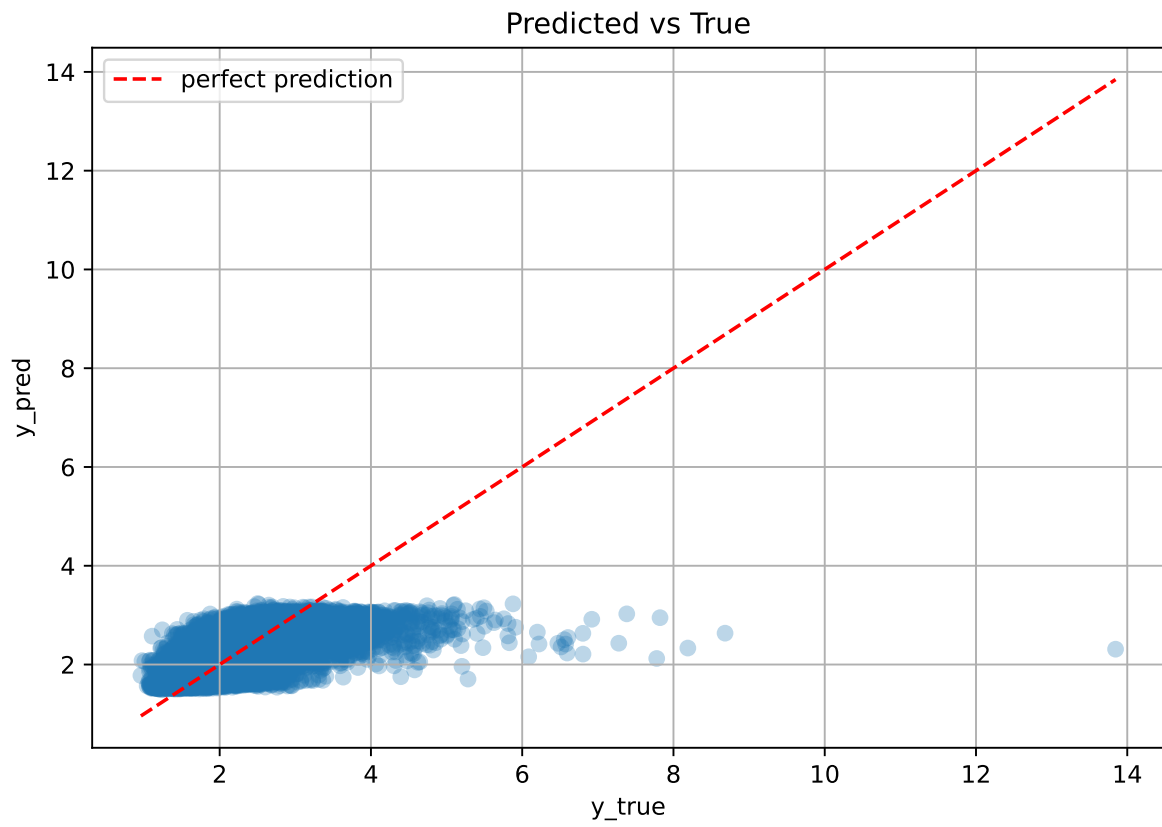
Rank 7: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.01430354440384331, 'reg__l2_leaf_reg': 21.306687933791995, 'reg__bagging_temperature': 4.9776759566122415, 'reg__random_strength': 2.0180565475487695, 'reg__subsample': 0.6236626874461331, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 6}

Rank 8: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.014247729665558763, 'reg__l2_leaf_reg': 21.487228784042706, 'reg__bagging_temperature': 1.8525139062409752, 'reg__random_strength': 2.0641966738870945, 'reg__subsample': 0.6012841282859321, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 7}

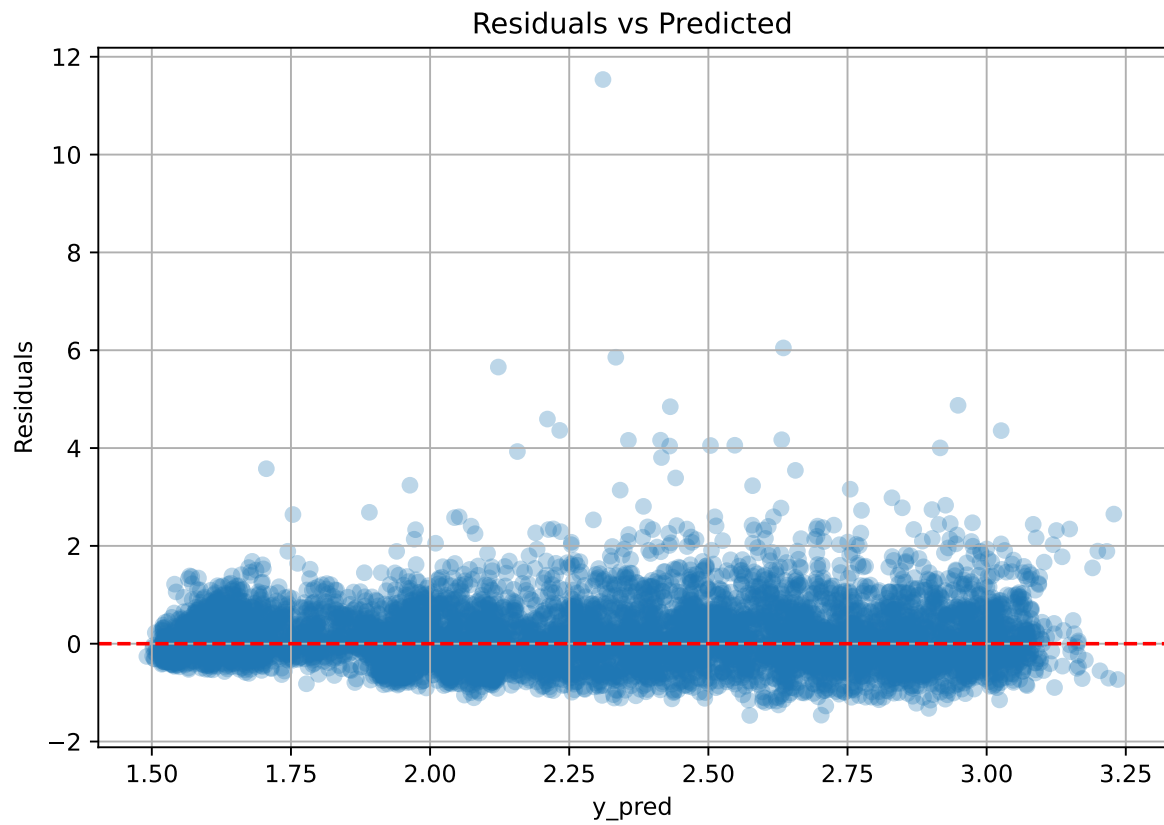
Rank 9: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.012171049874540311, 'reg__l2_leaf_reg': 17.11707147974743, 'reg__bagging_temperature': 1.2104272130727947, 'reg__random_strength': 4.9542140745934375, 'reg__subsample': 0.6688253820080352, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 6}

Rank 10: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.013830737745921487, 'reg__l2_leaf_reg': 14.770710568415925, 'reg__bagging_temperature': 1.505754879652021, 'reg__random_strength': 0.6187166489848481, 'reg__subsample': 0.6032379936413081, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 7}

2.5.6 Scatter



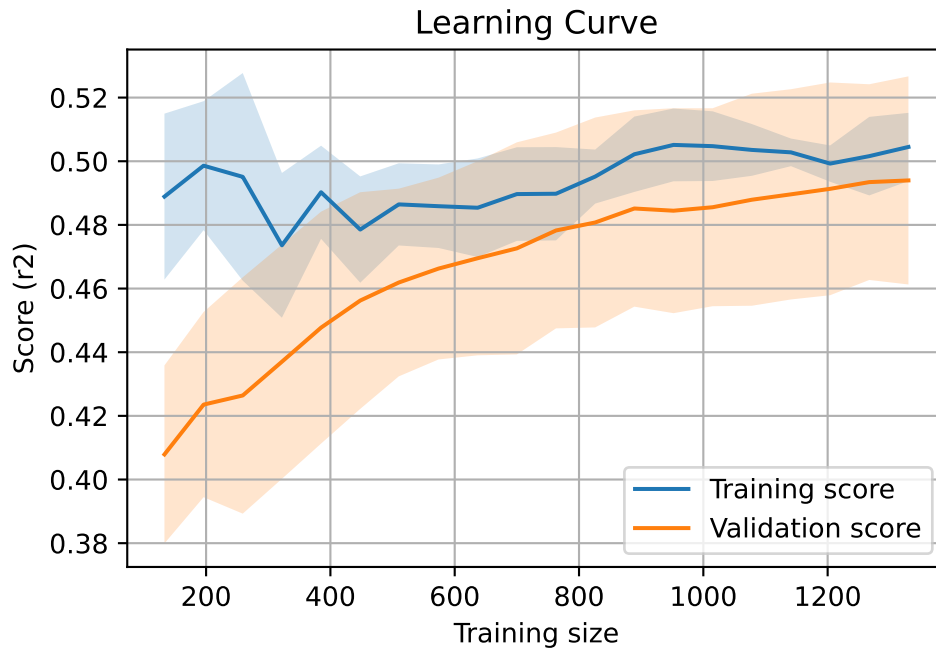
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2

`Len(group)` : 2716, `Len(training)` : 1902,

`Len(training_real)` : 1332, `Len(test_of_training)` : 570, `Len(test)` : 814



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	1	3817	1635	5452	0.441	0.43	0.035	0.478	0.478	0.048	1.112	1.025	nan
Linear	2	1902	814	2716	0.358	0.378	0.032	0.322	0.323	-0.055	0.856	0.947	nan
Linear	MEAN	5719	2449	8168	0.399	0.404	0.033	0.4	0.401	-0.003	0.984	0.986	nan
LGBM	1	3817	1635	5452	0.539	0.514	0.038	0.563	0.587	0.073	1.141	1.048	0.5144
LGBM	2	1902	814	2716	0.509	0.485	0.039	0.436	0.464	-0.021	0.958	1.049	0.4848
LGBM	MEAN	5719	2449	8168	0.524	0.5	0.038	0.499	0.526	0.026	1.049	1.049	0.5
CatBoost	1	3817	1635	5452	0.584	0.556	0.041	0.602	0.608	0.052	1.093	1.05	0.5561
CatBoost	2	1902	814	2716	0.504	0.495	0.034	0.465	0.473	-0.022	0.956	1.018	0.4948
CatBoost	MEAN	5719	2449	8168	0.544	0.525	0.037	0.533	0.54	0.015	1.024	1.034	0.525

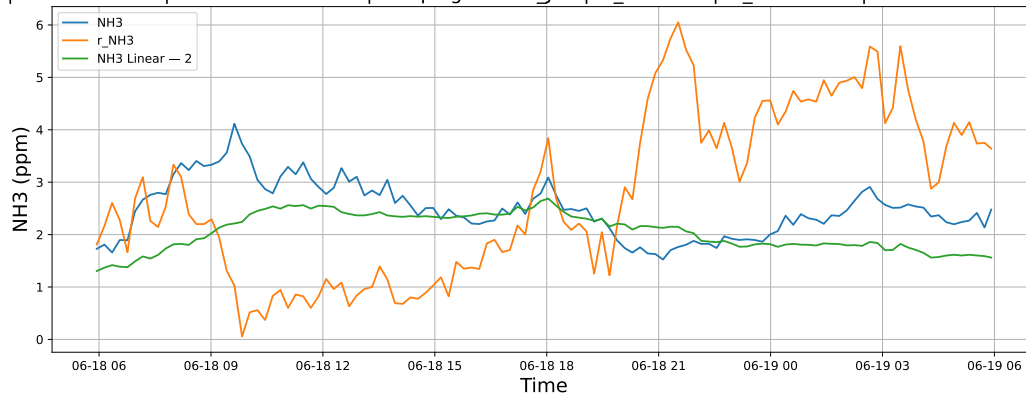
3 Plot

3.1 Standalone plots

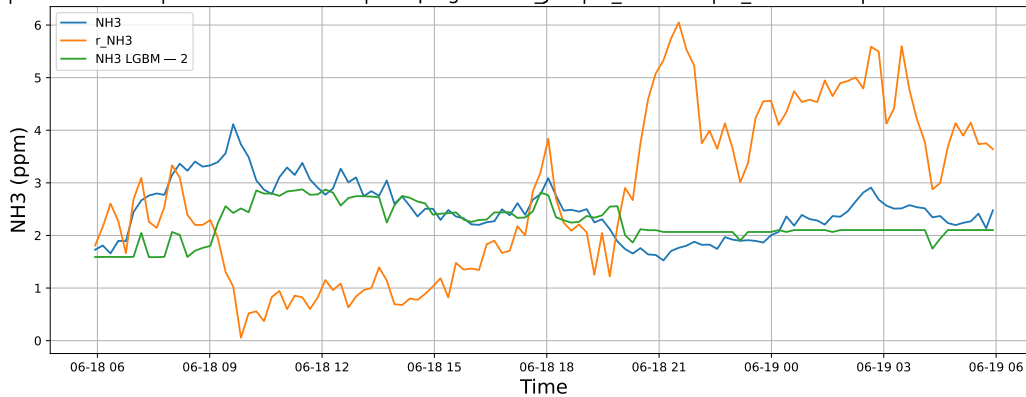
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

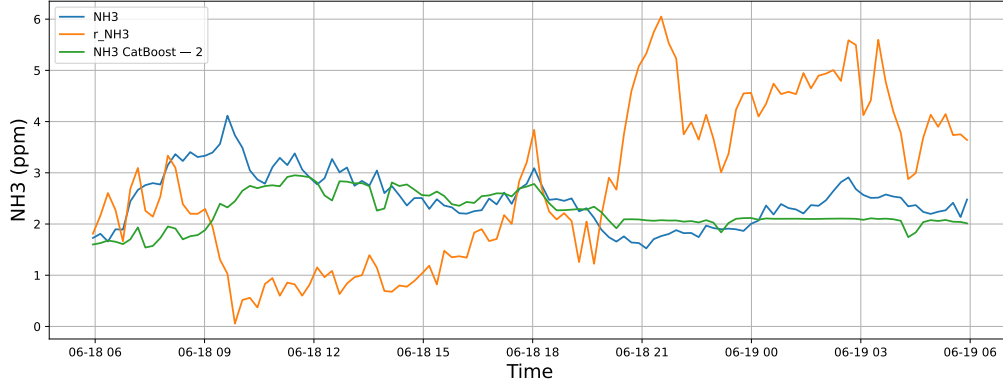
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

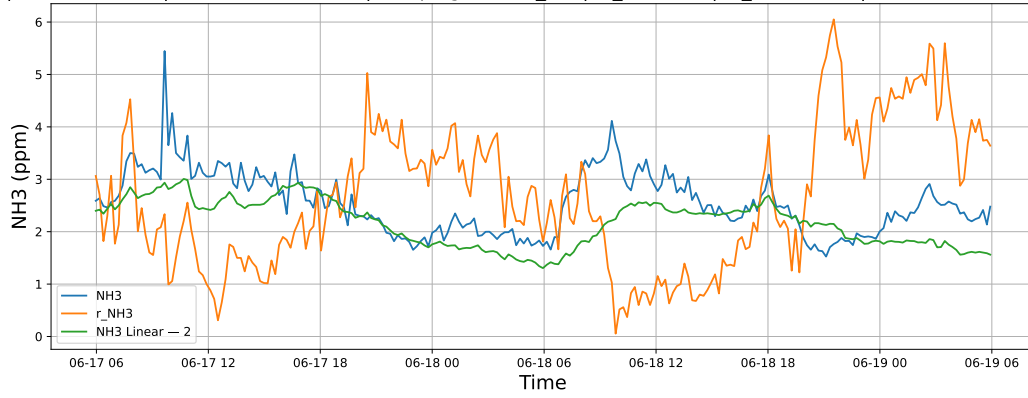


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

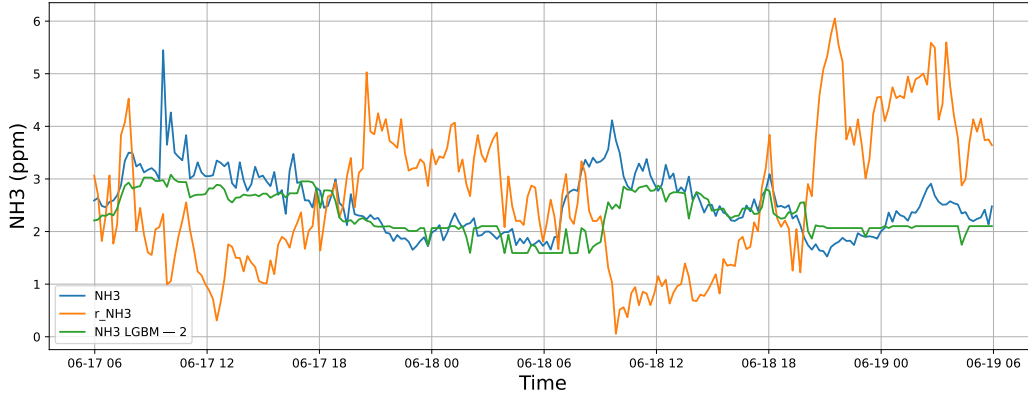


3.1.1.2 Time window : 48

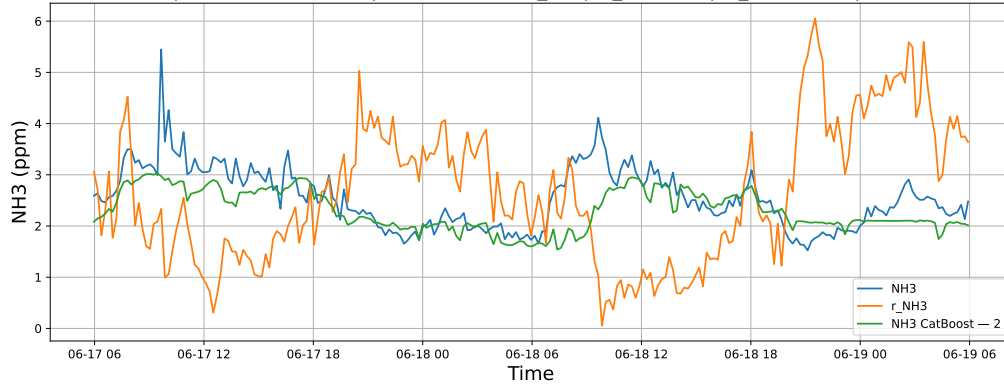
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

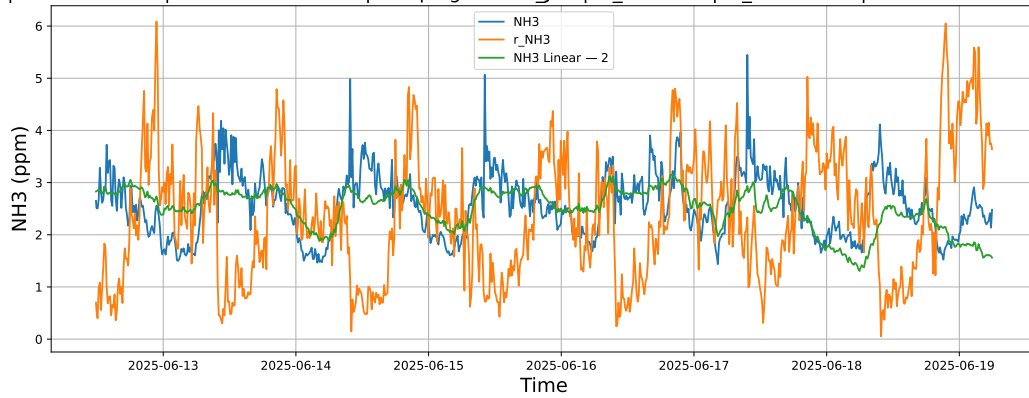


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

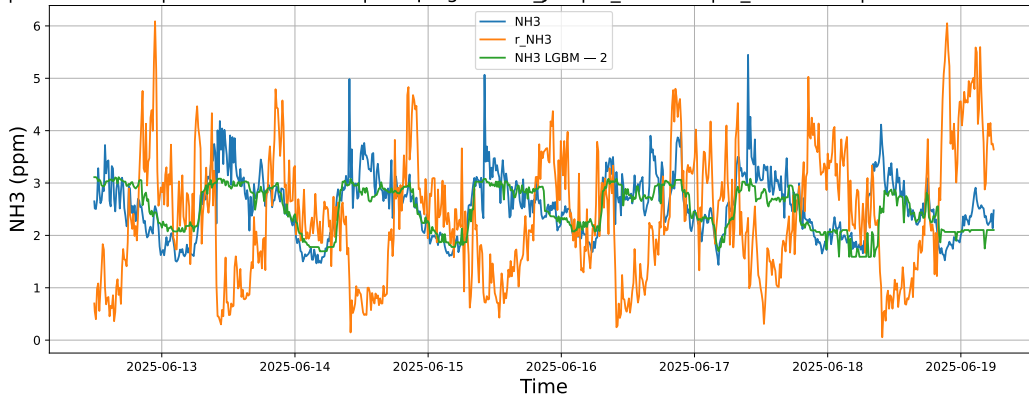


3.1.1.3 Time window : ALL

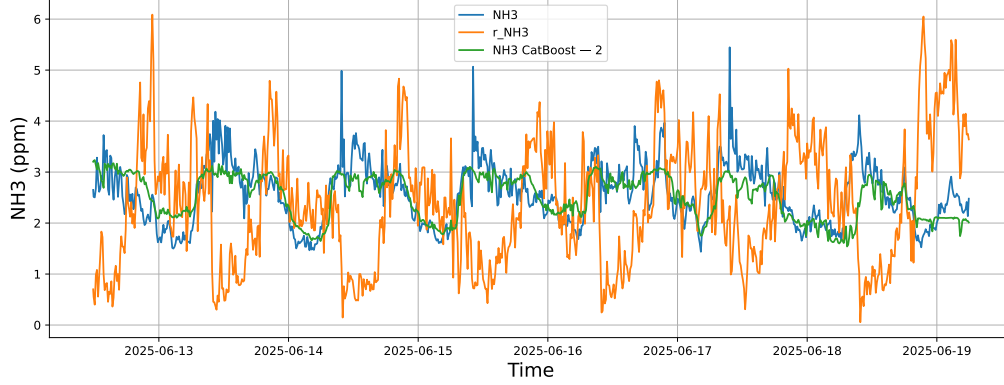
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



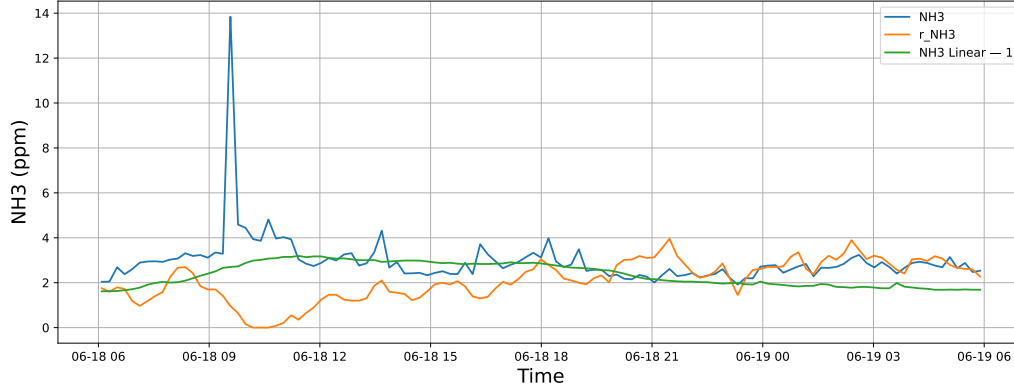
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



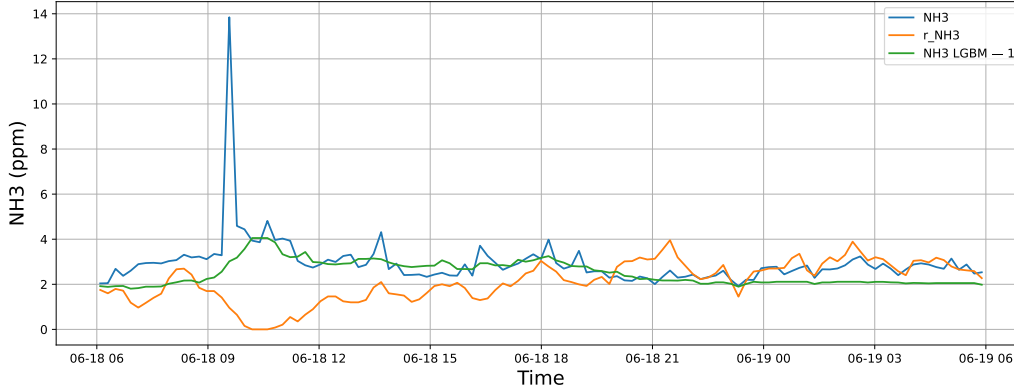
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

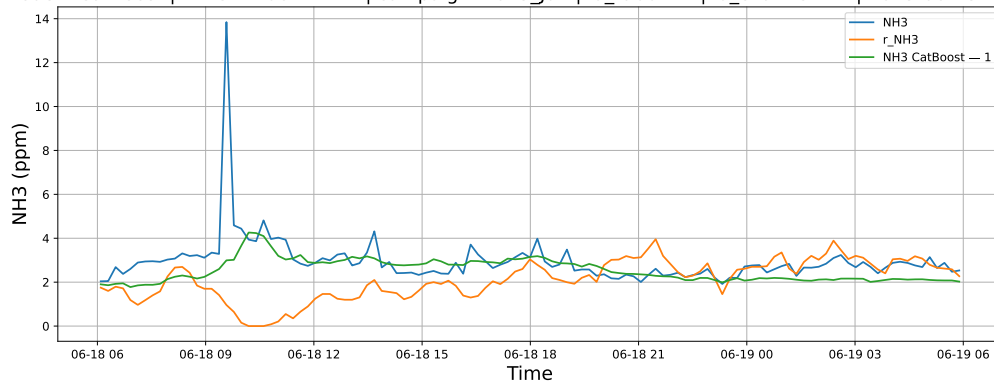
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

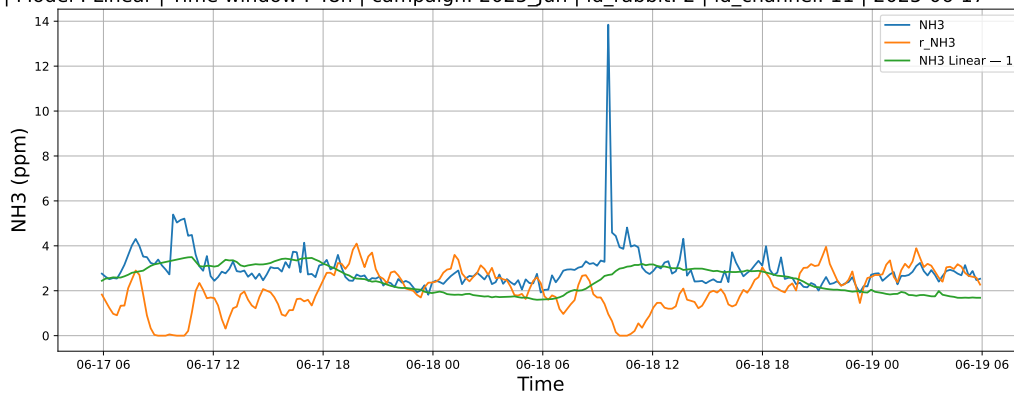


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

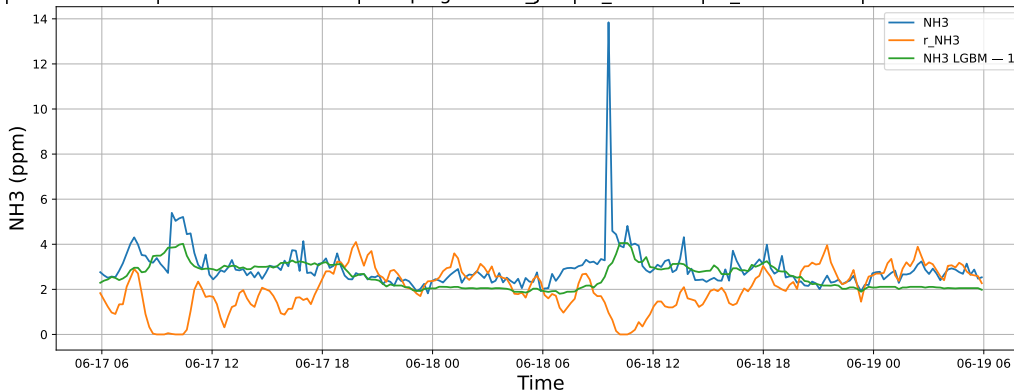


3.1.2.2 Time window : 48

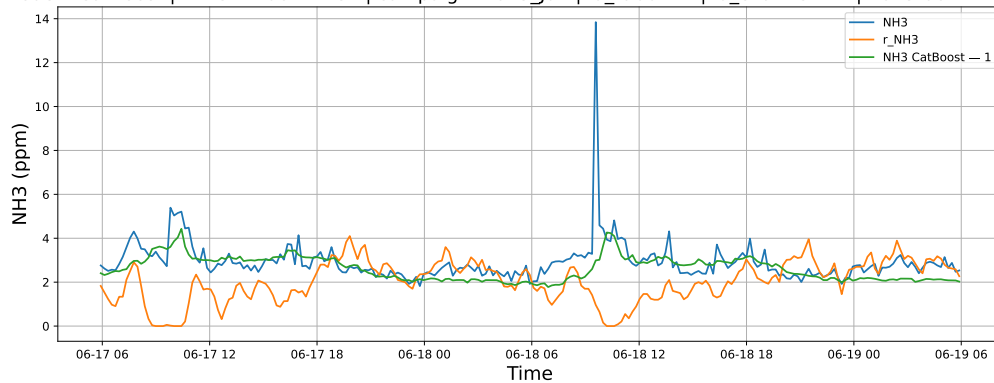
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

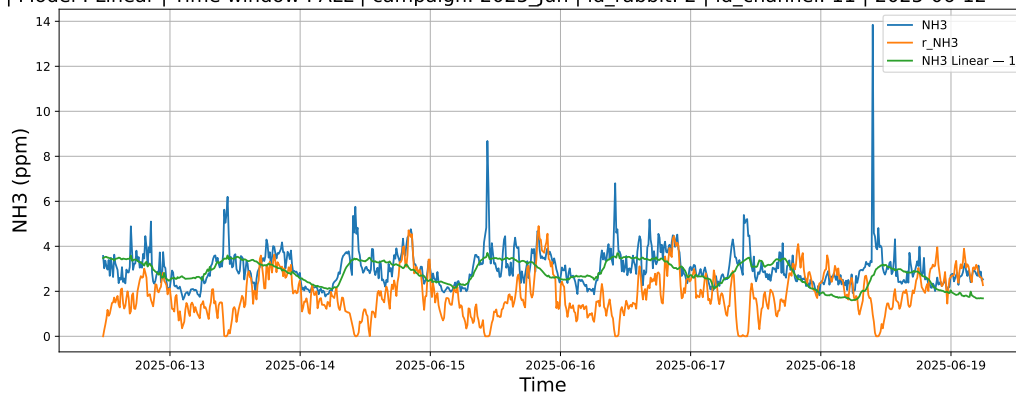


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

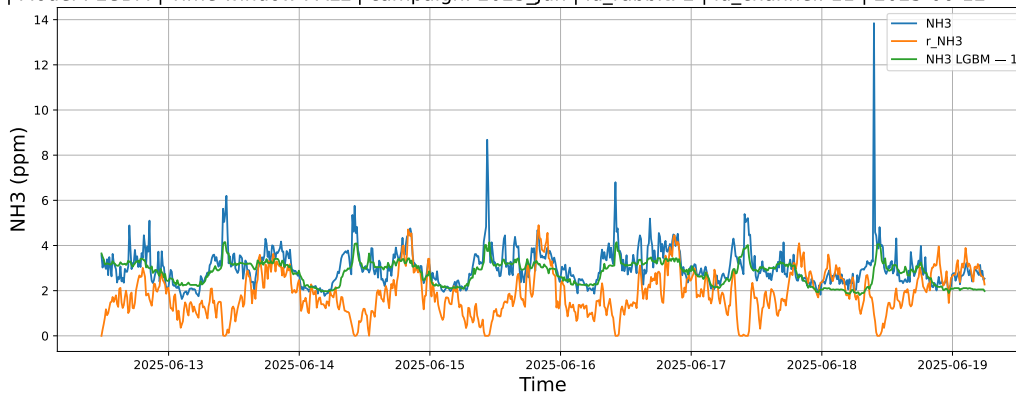


3.1.2.3 Time window : ALL

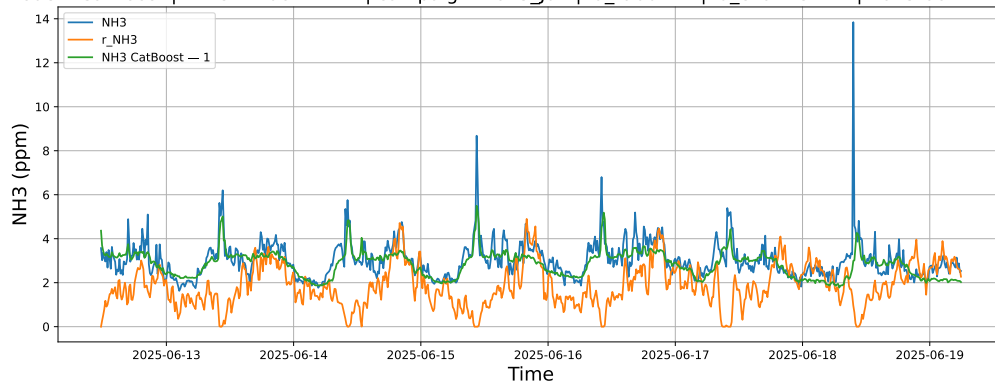
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



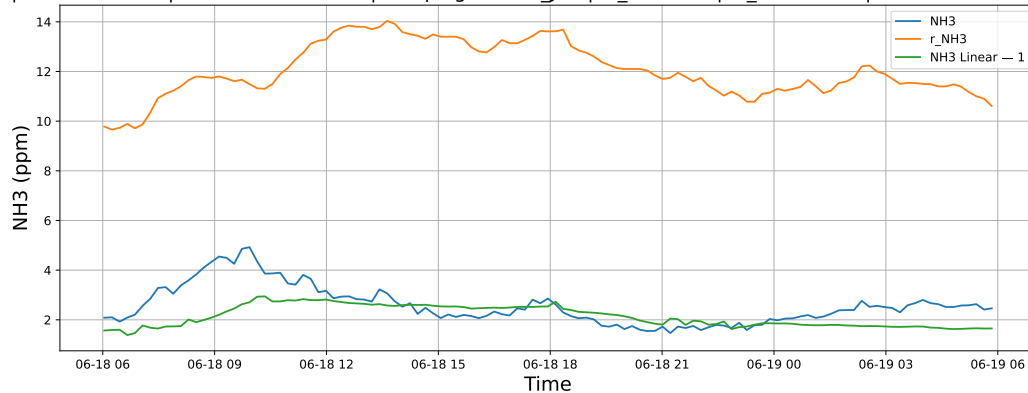
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



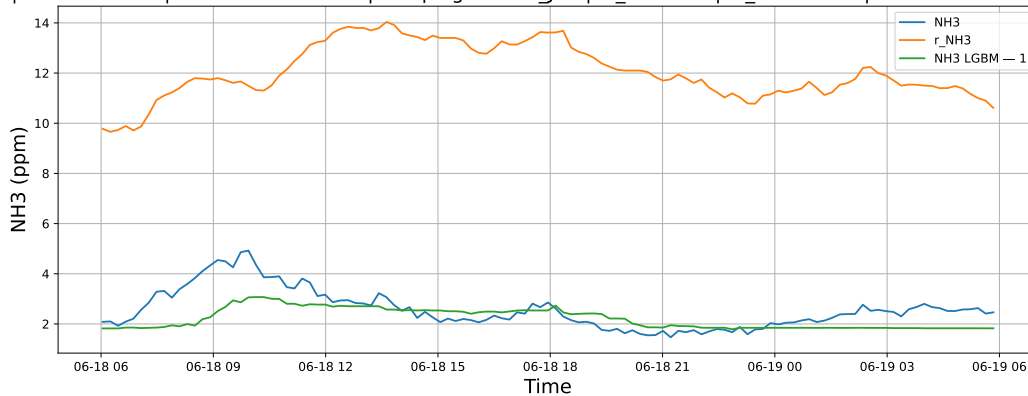
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

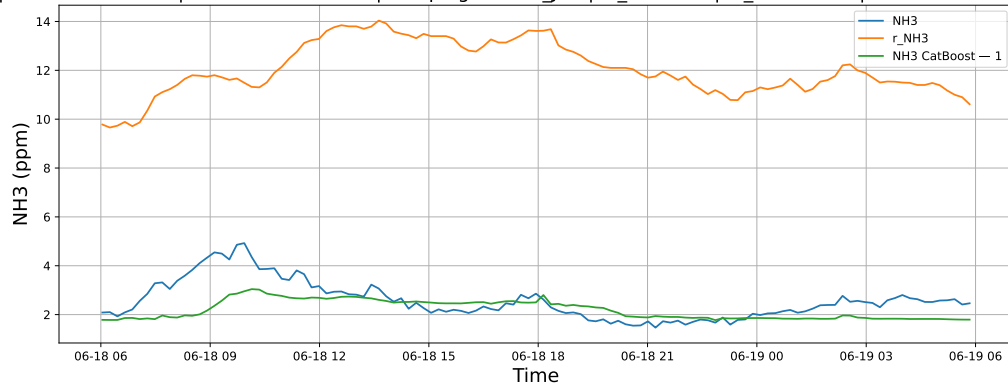
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

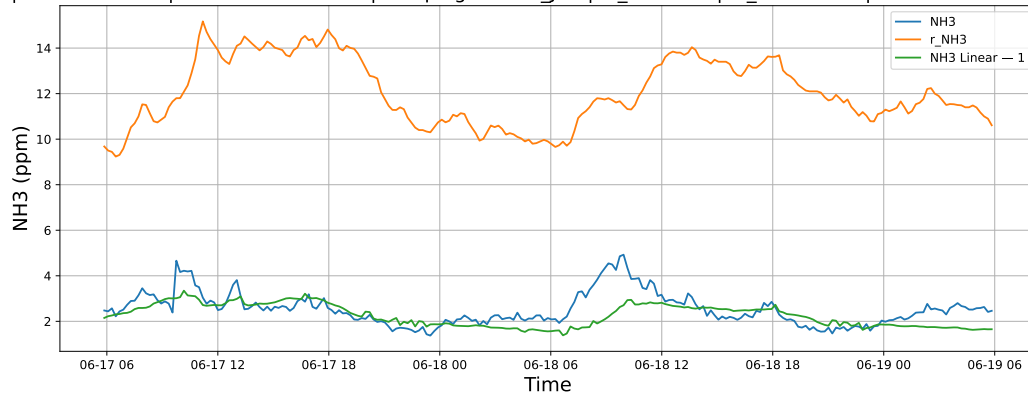


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

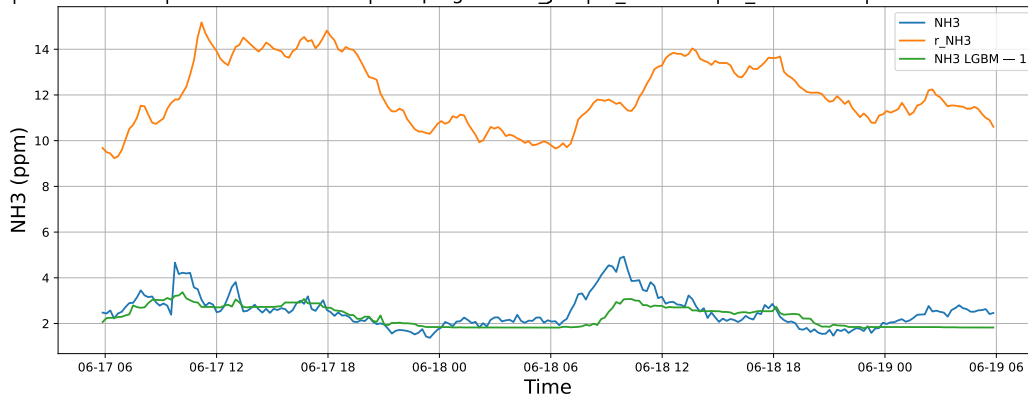


3.1.3.2 Time window : 48

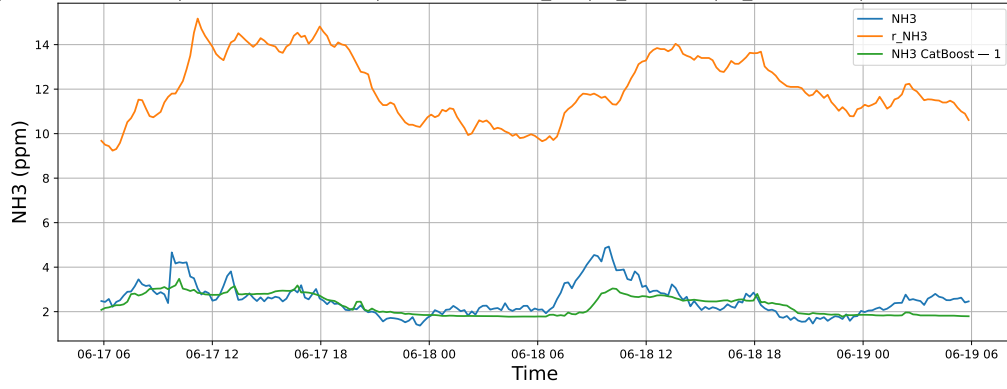
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

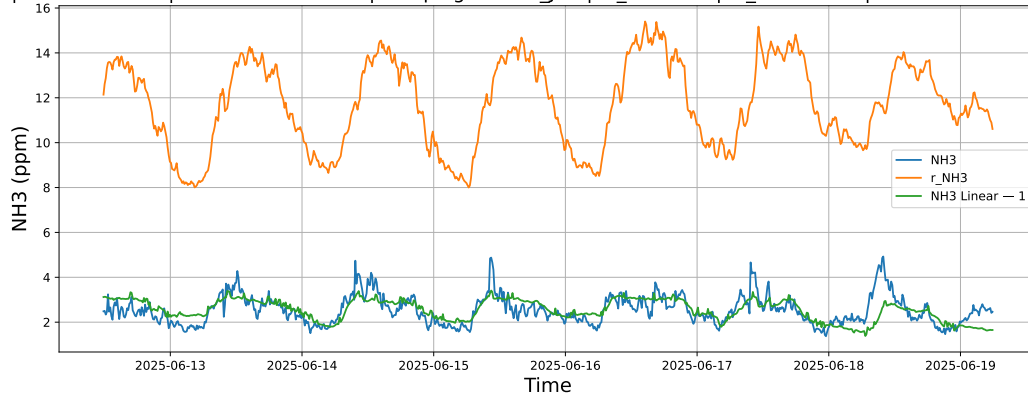


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

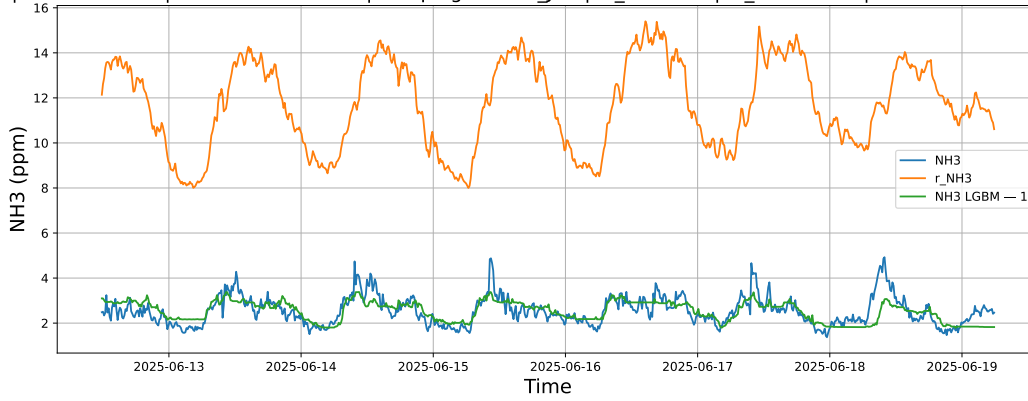


3.1.3.3 Time window : ALL

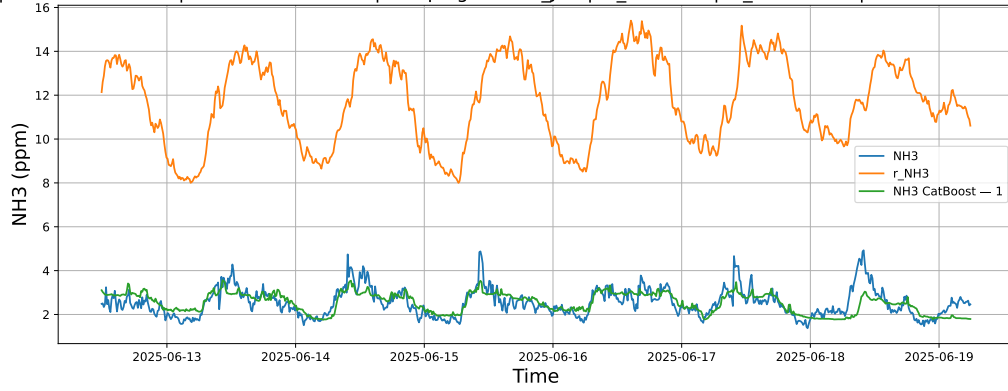
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



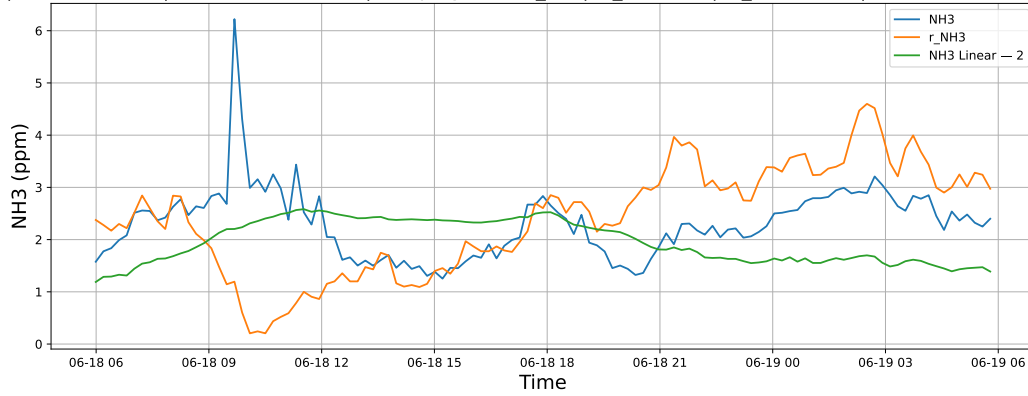
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



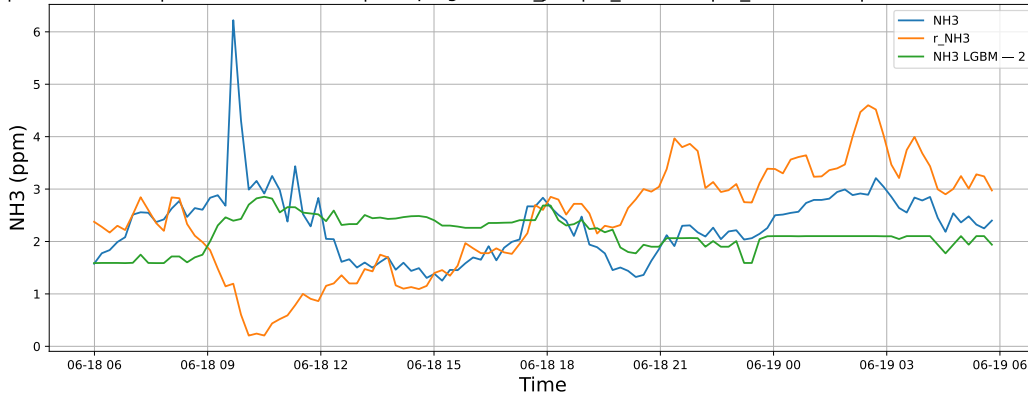
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

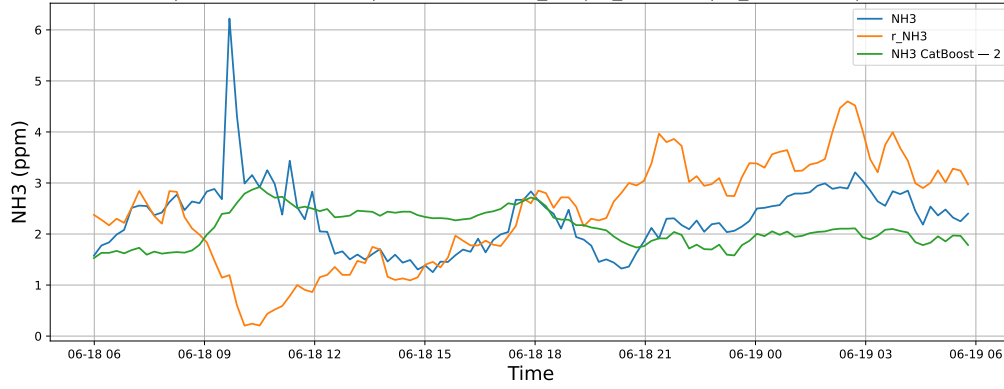
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

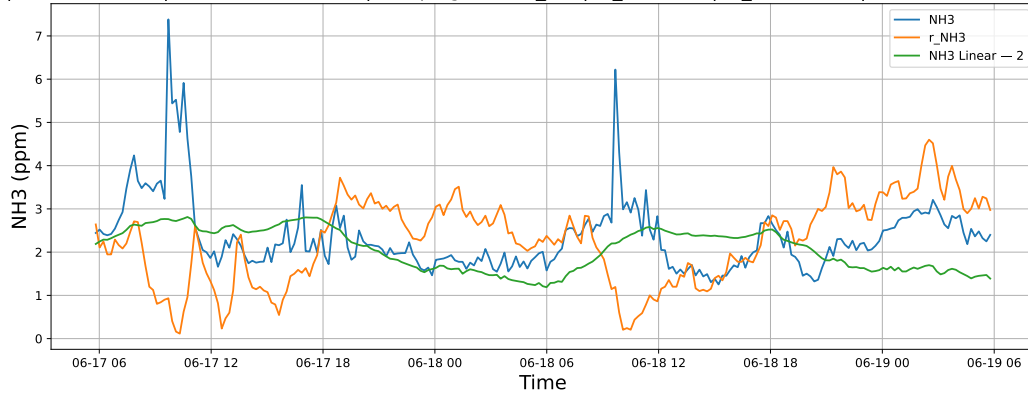


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

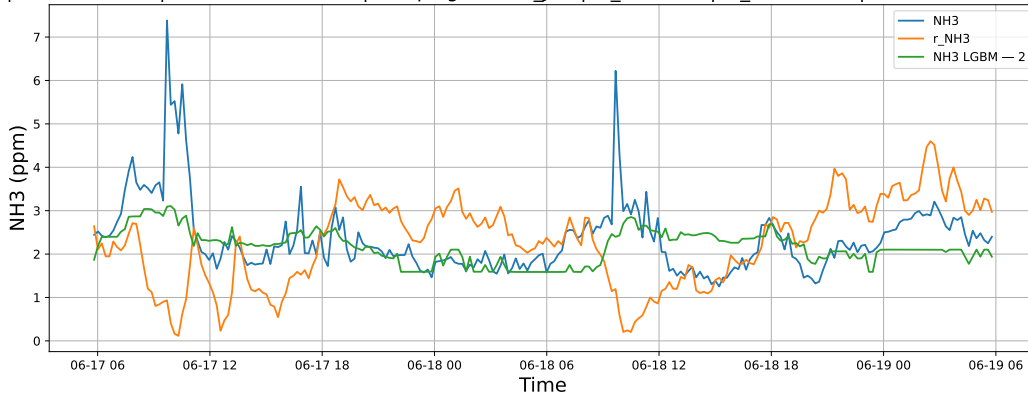


3.1.4.2 Time window : 48

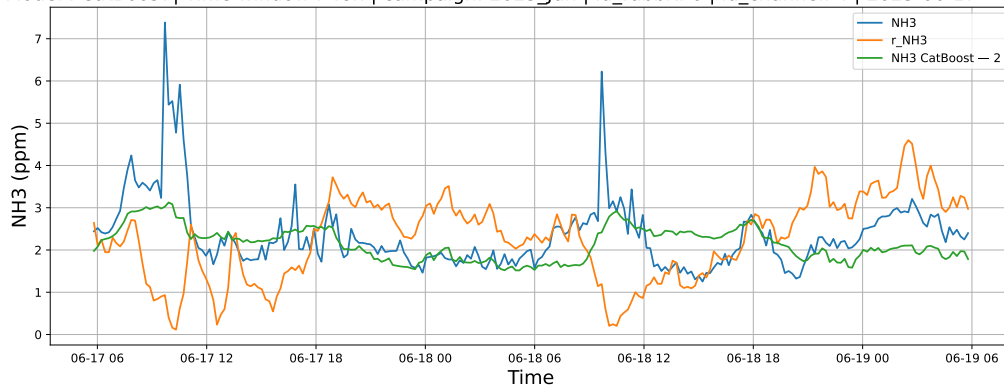
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

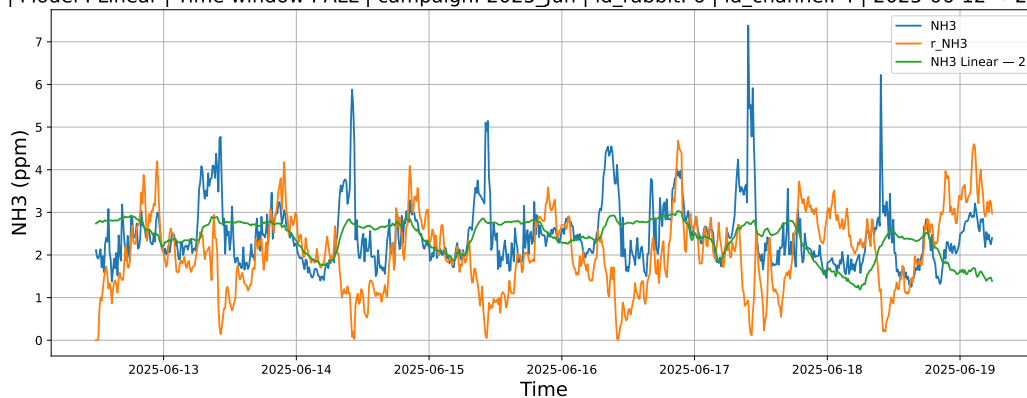


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

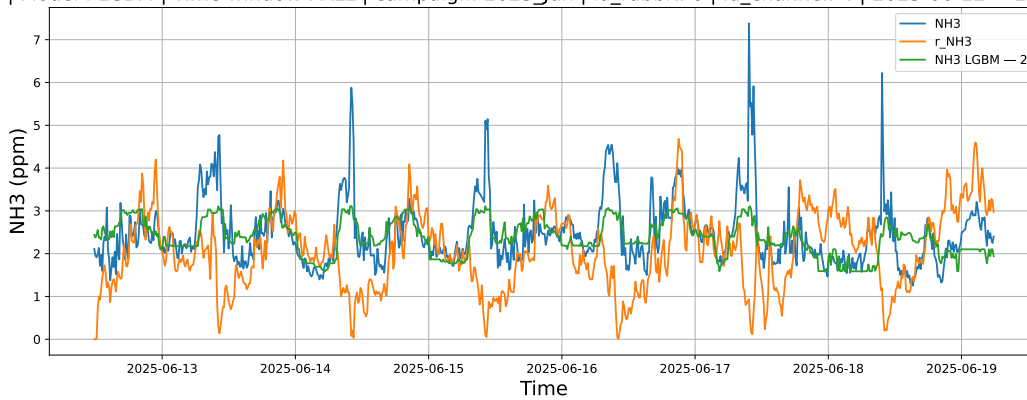


3.1.4.3 Time window : ALL

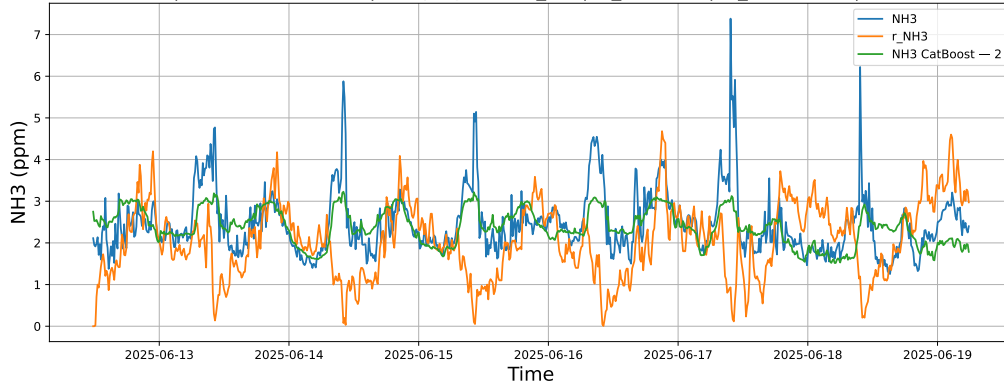
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



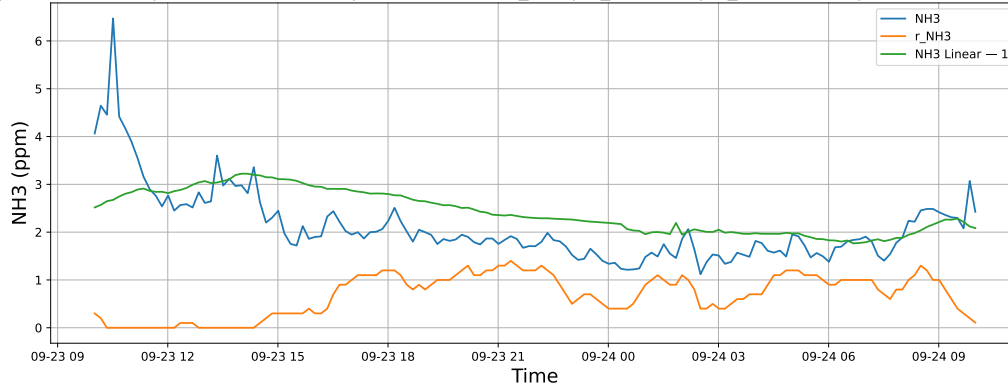
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



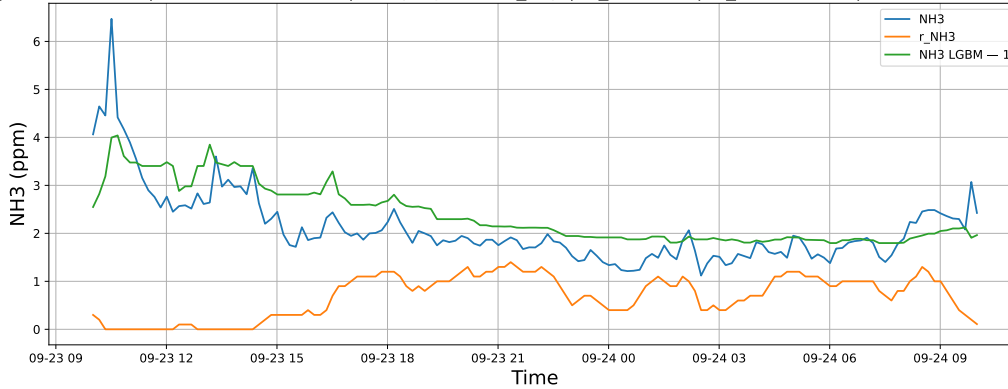
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

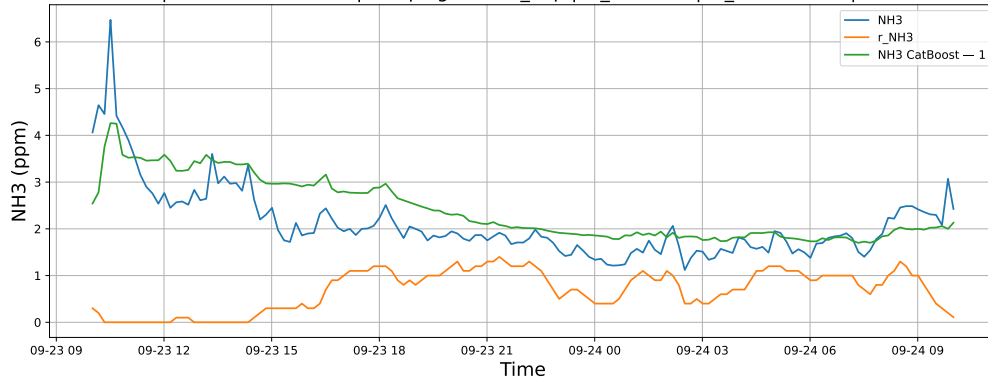
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

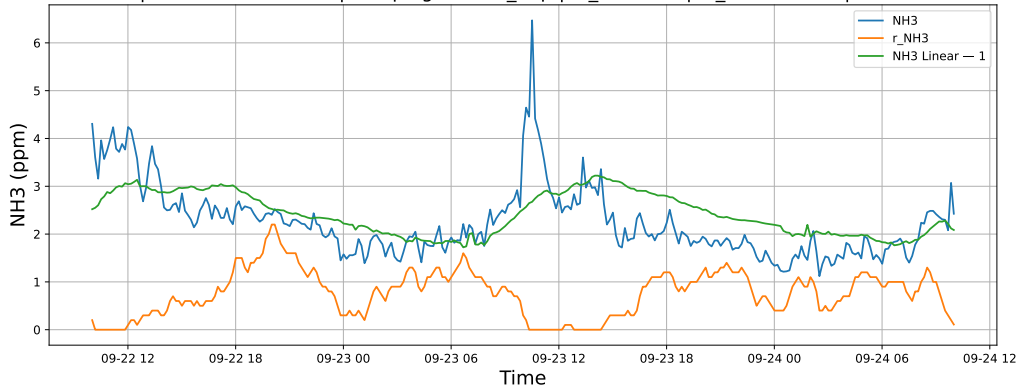


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

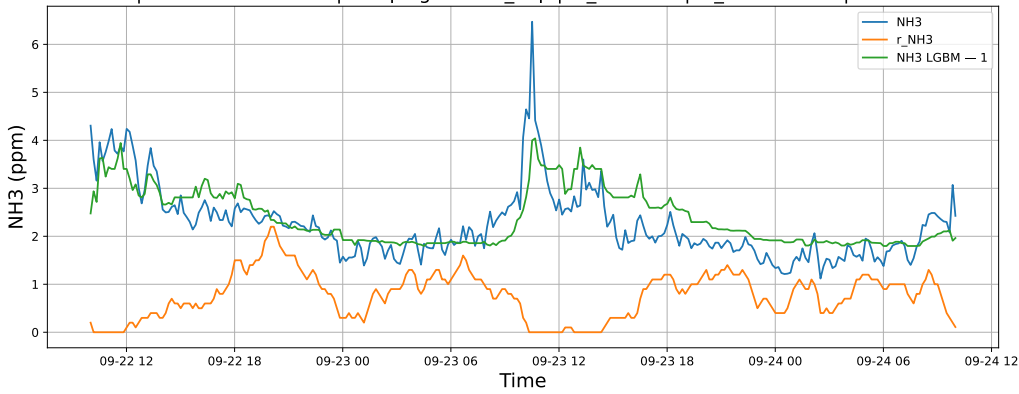


3.1.5.2 Time window : 48

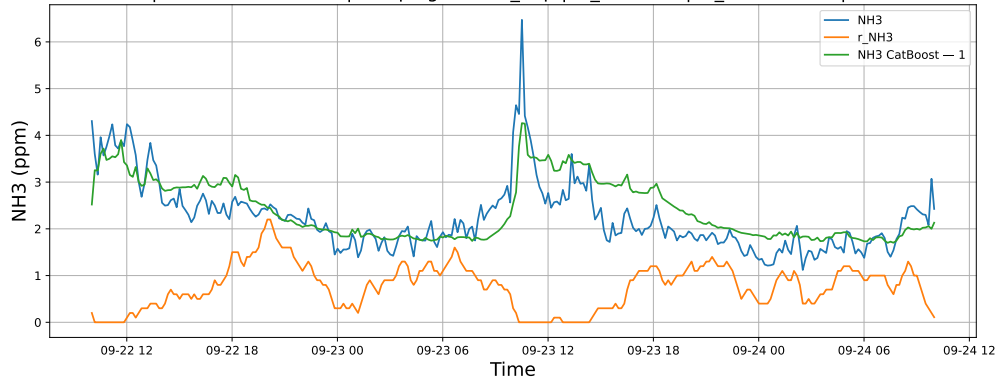
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

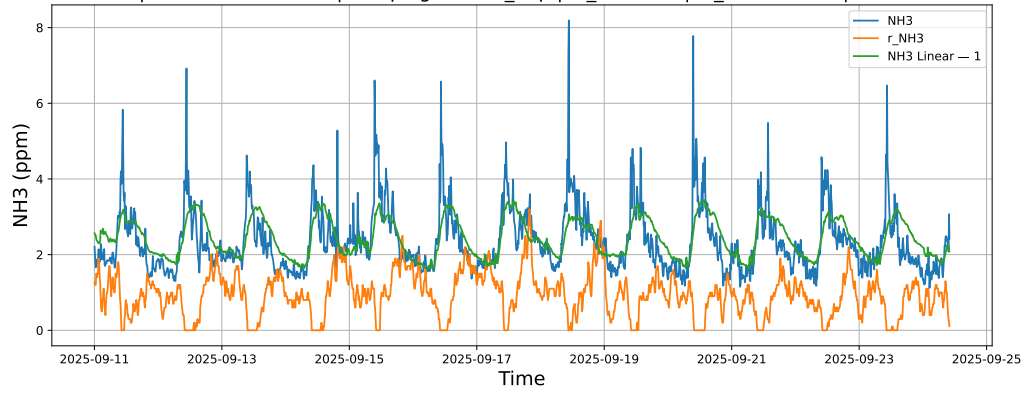


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

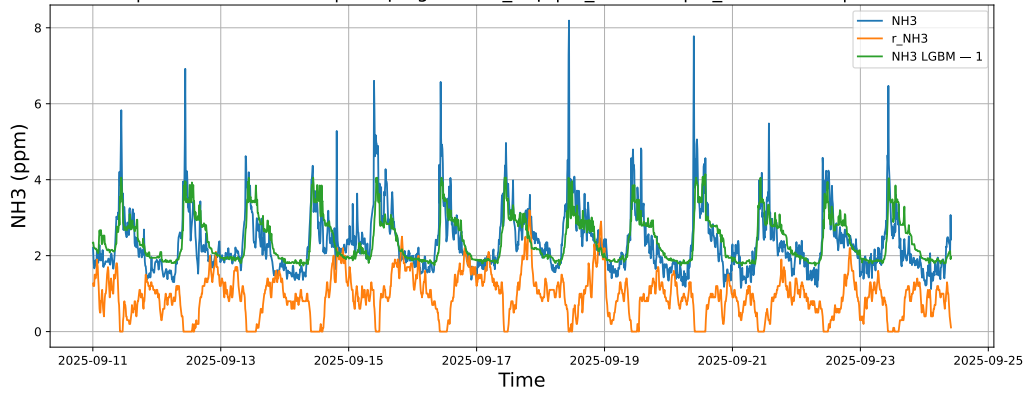


3.1.5.3 Time window : ALL

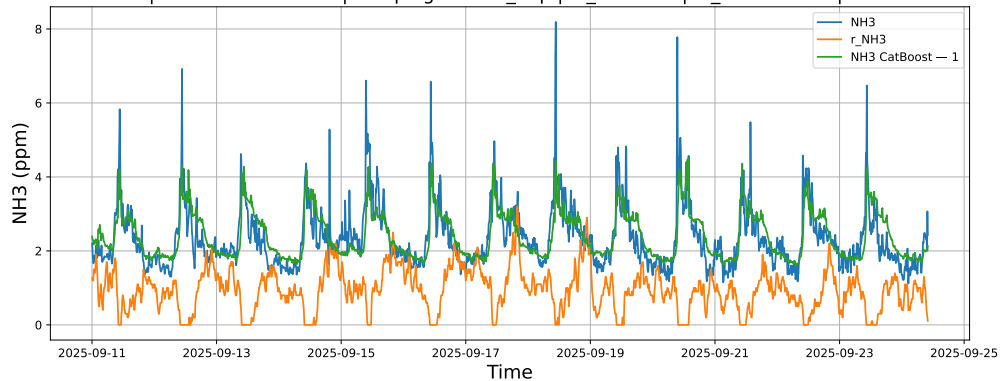
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



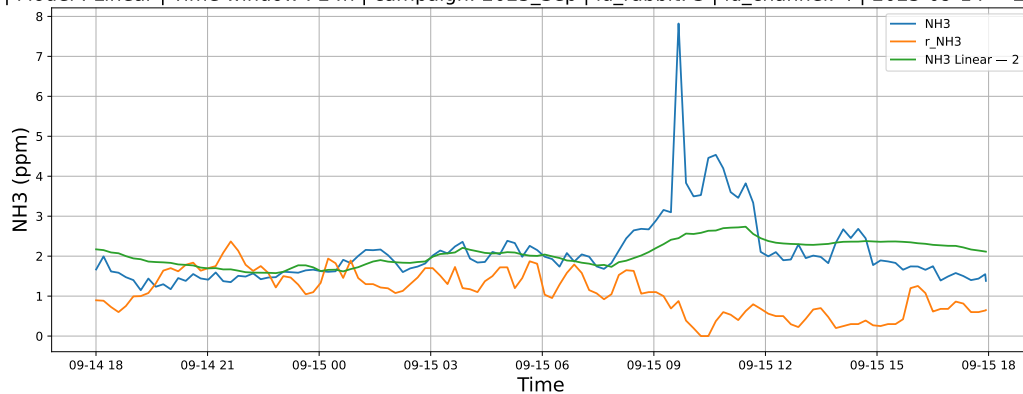
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



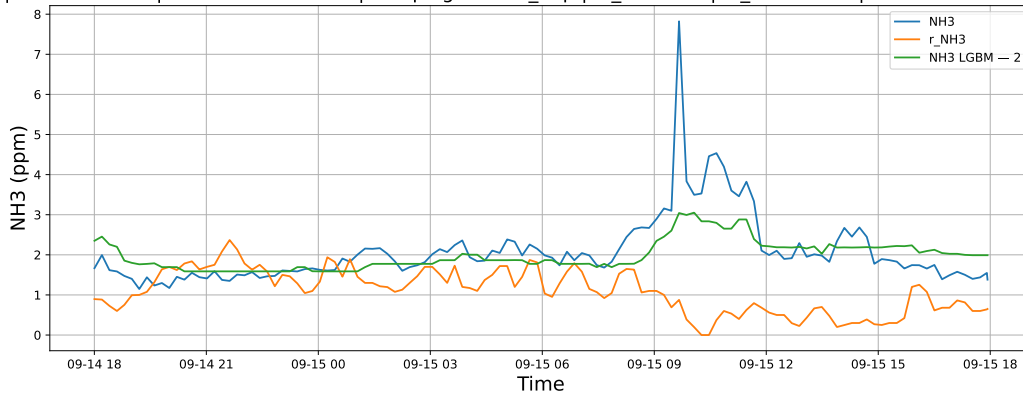
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

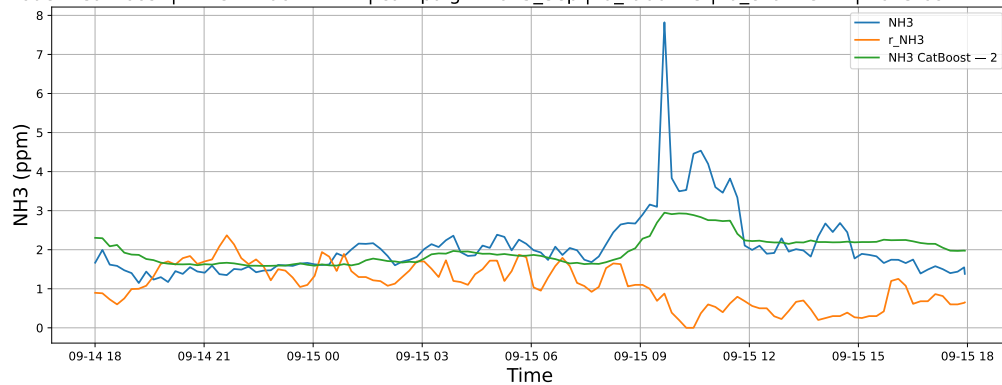
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

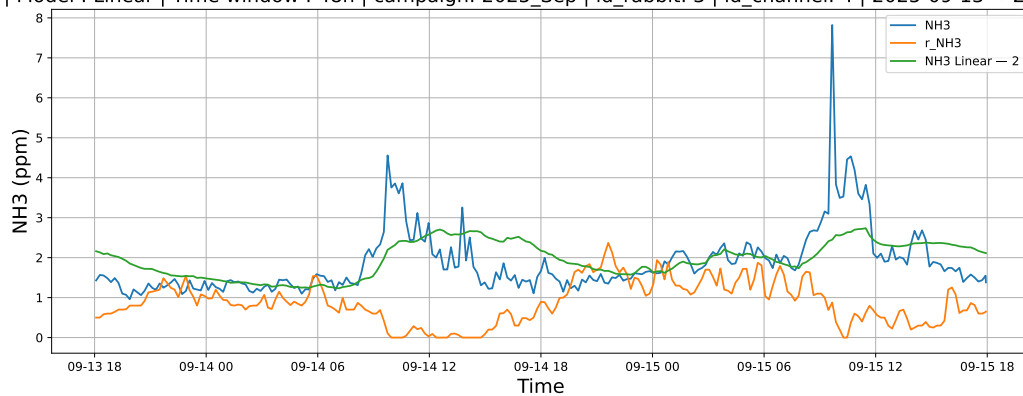


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

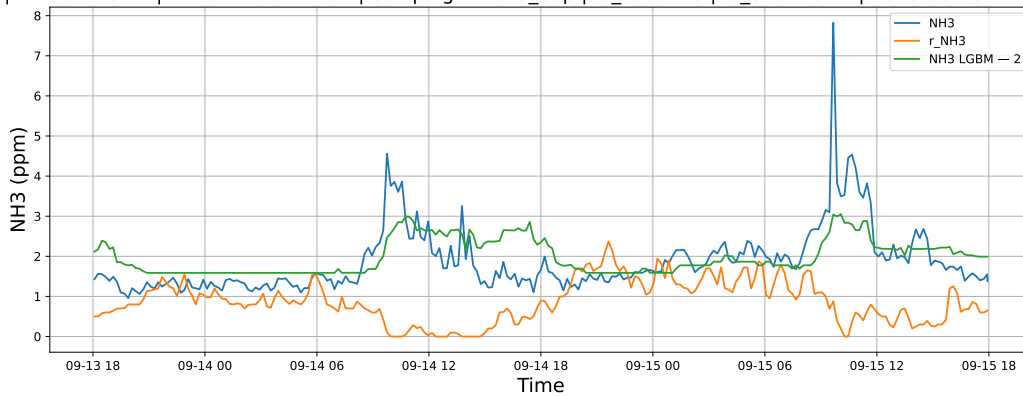


3.1.6.2 Time window : 48

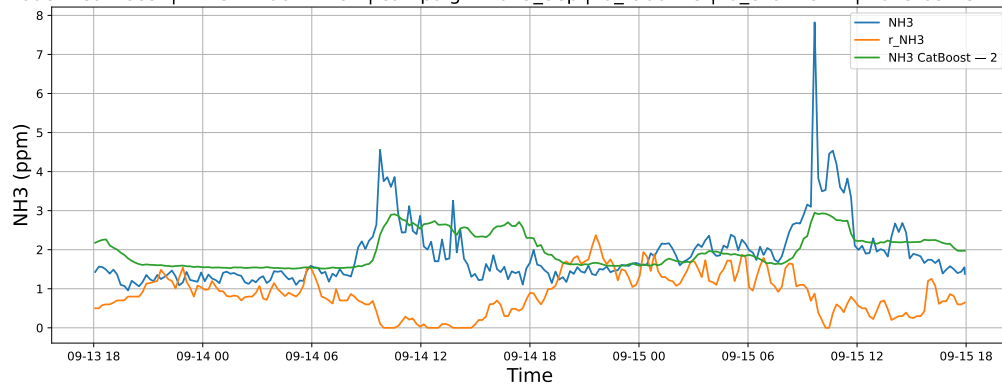
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

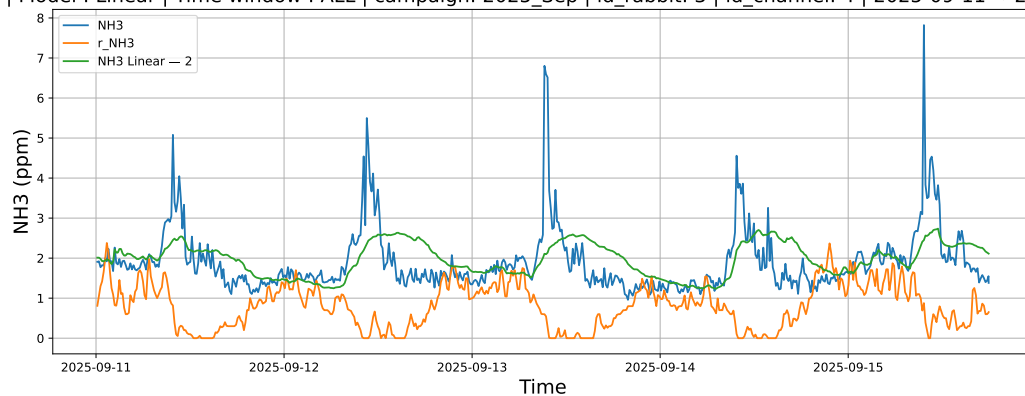


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

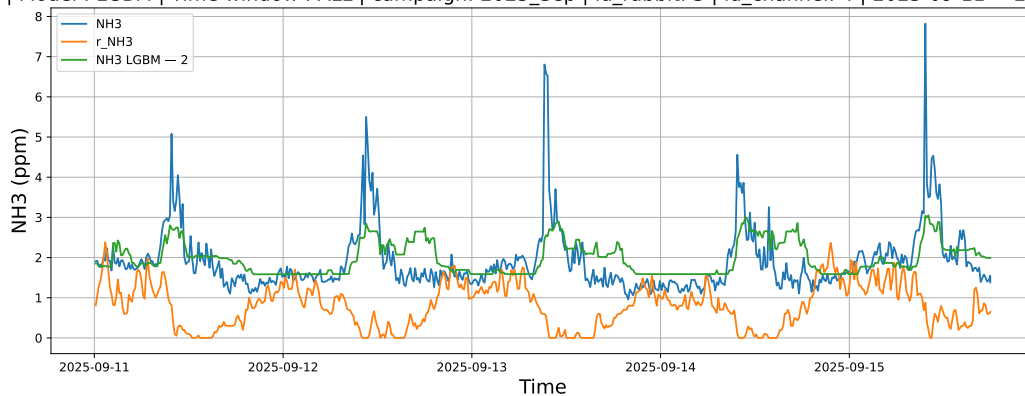


3.1.6.3 Time window : ALL

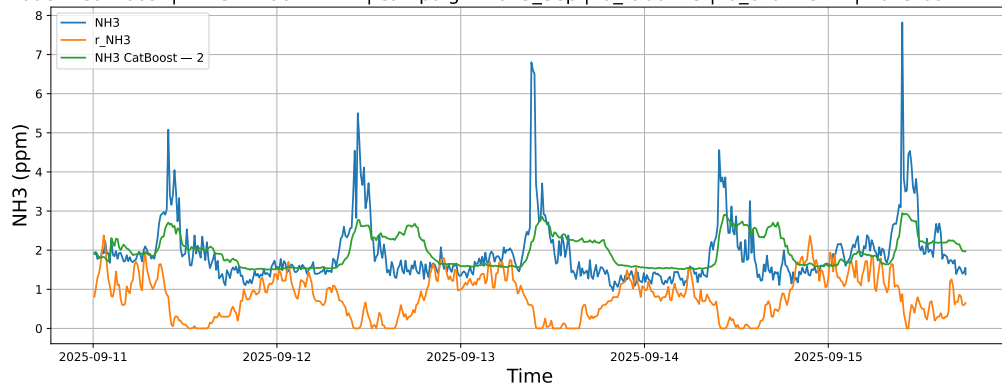
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



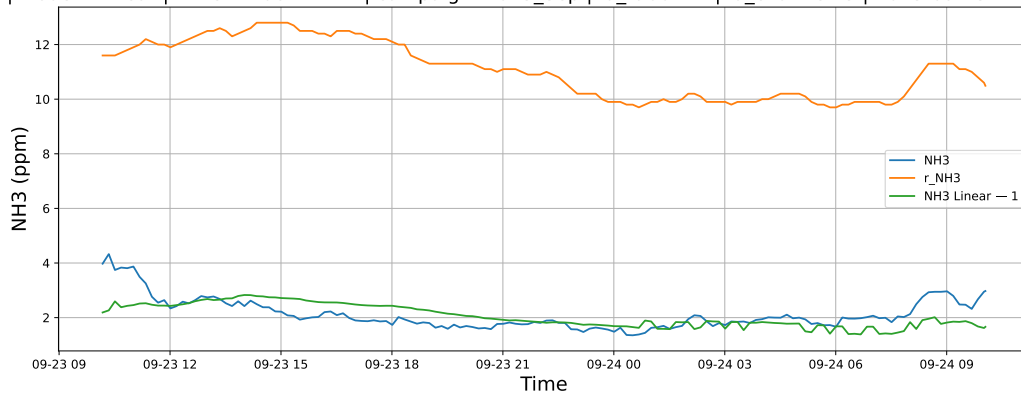
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



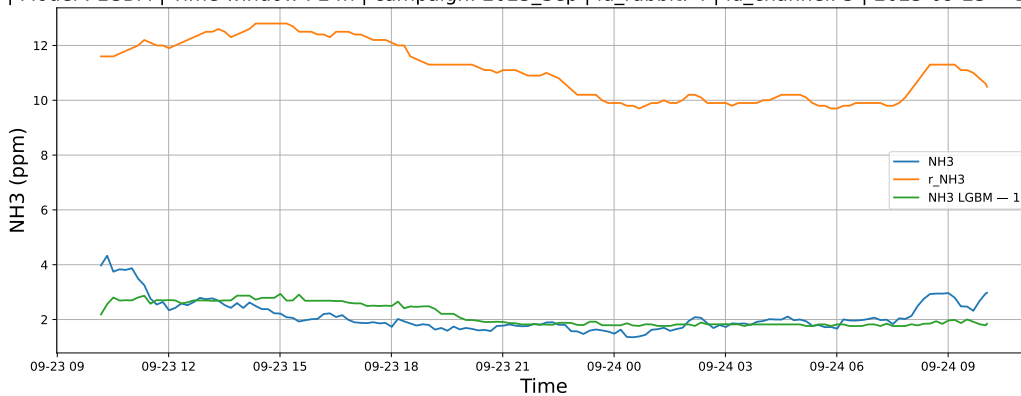
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

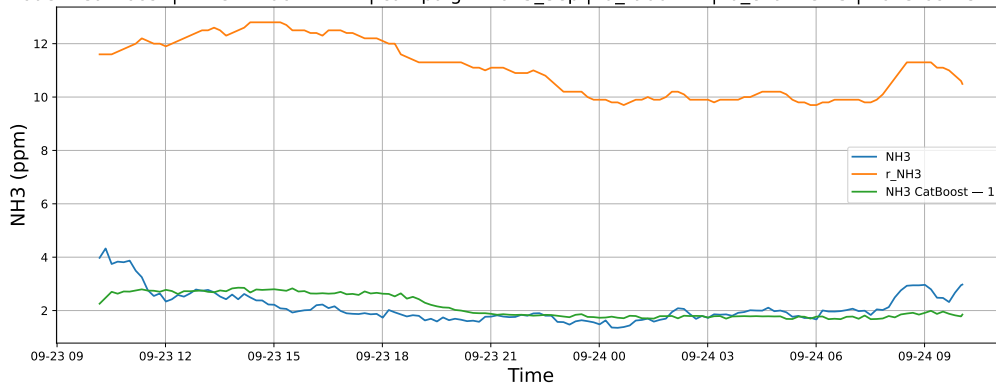
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

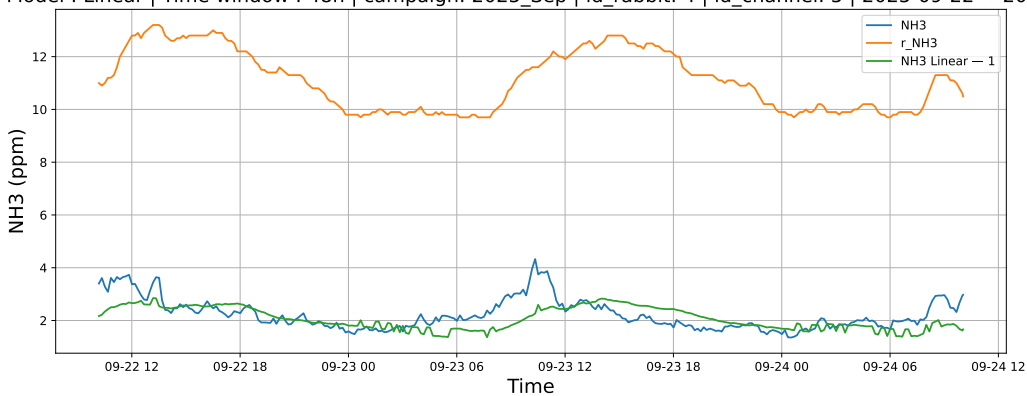


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

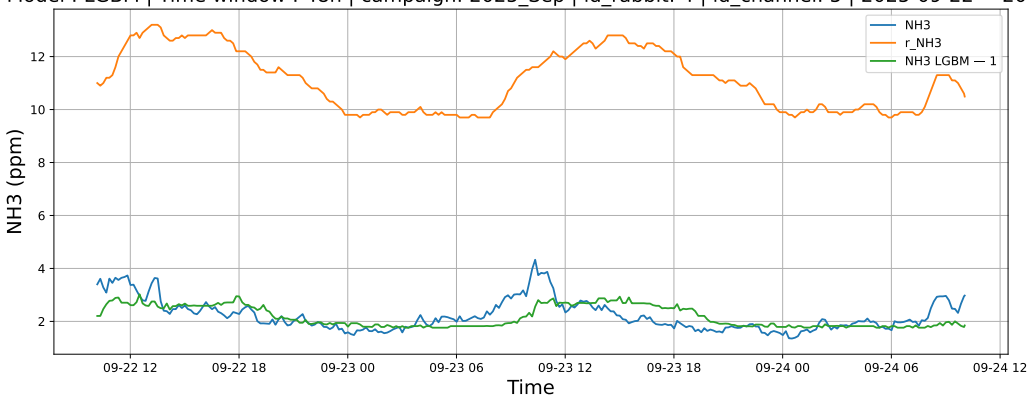


3.1.7.2 Time window : 48

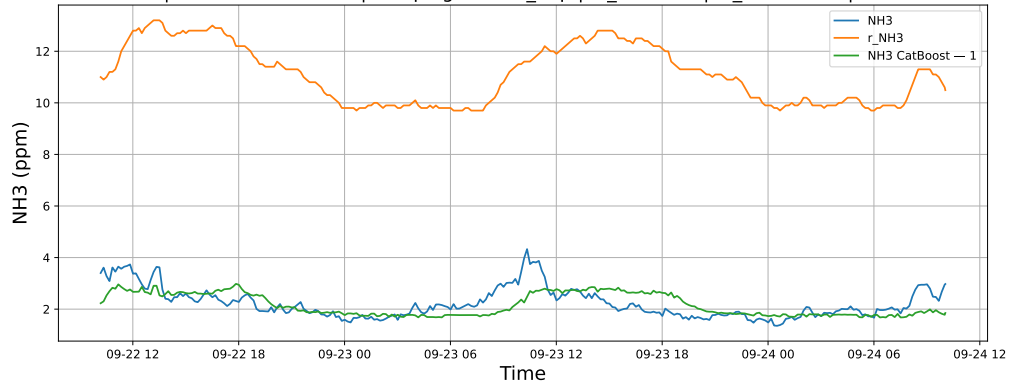
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

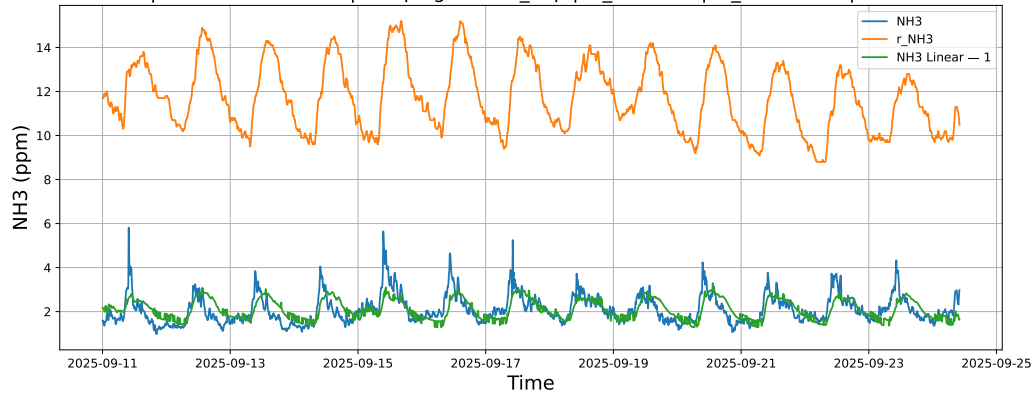


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

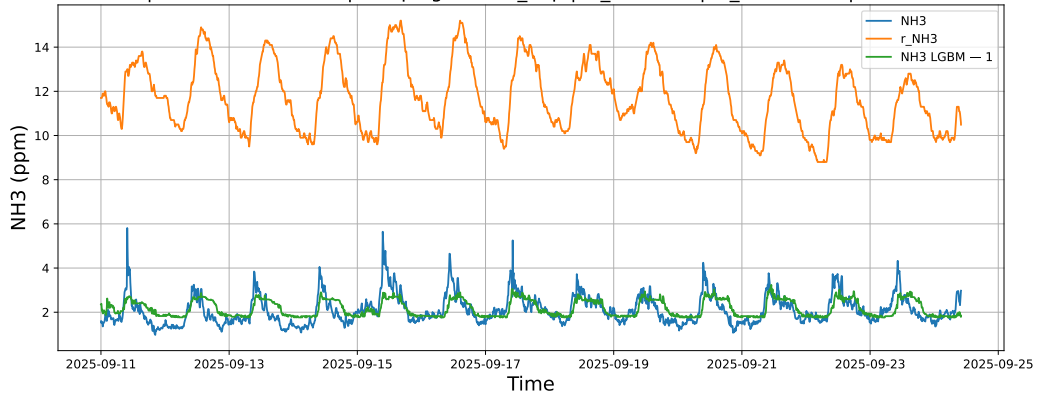


3.1.7.3 Time window : ALL

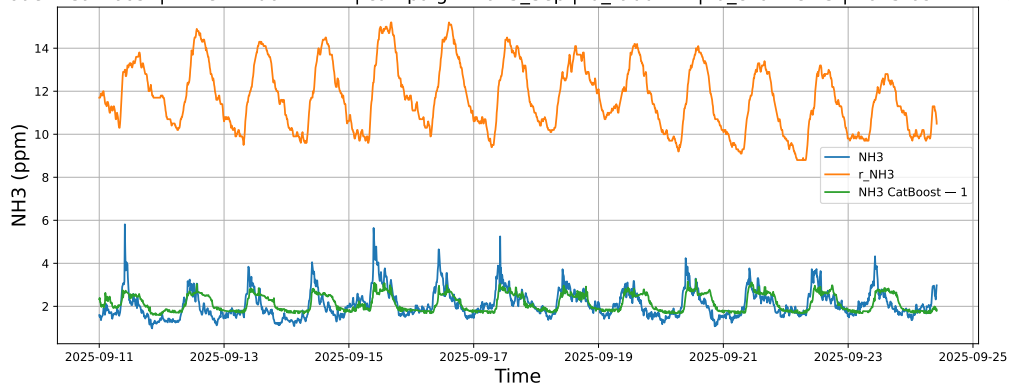
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



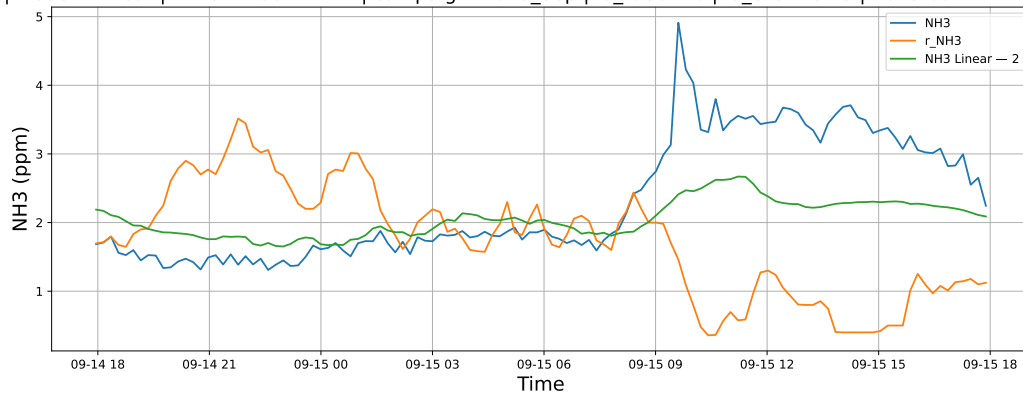
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



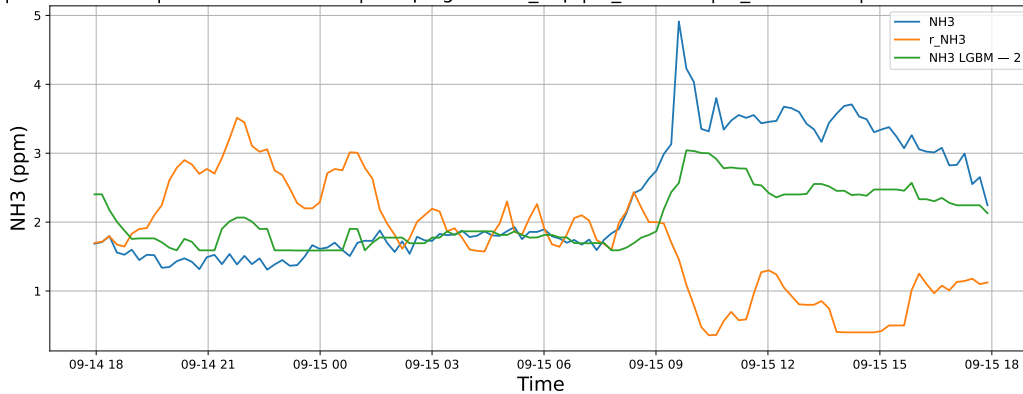
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

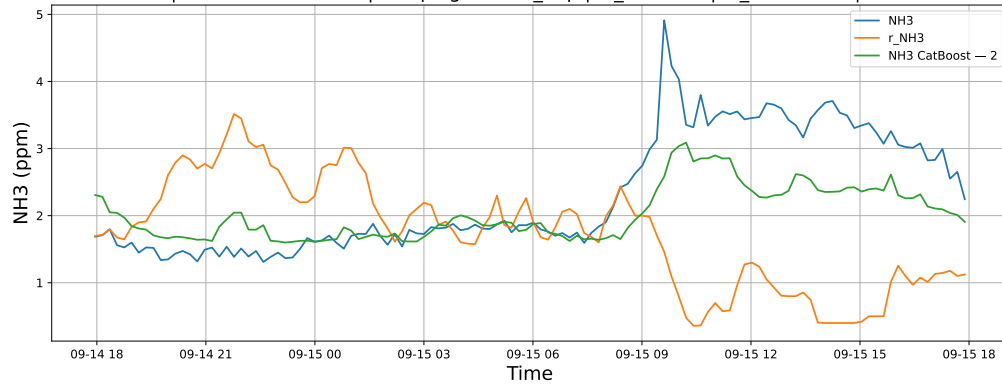
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

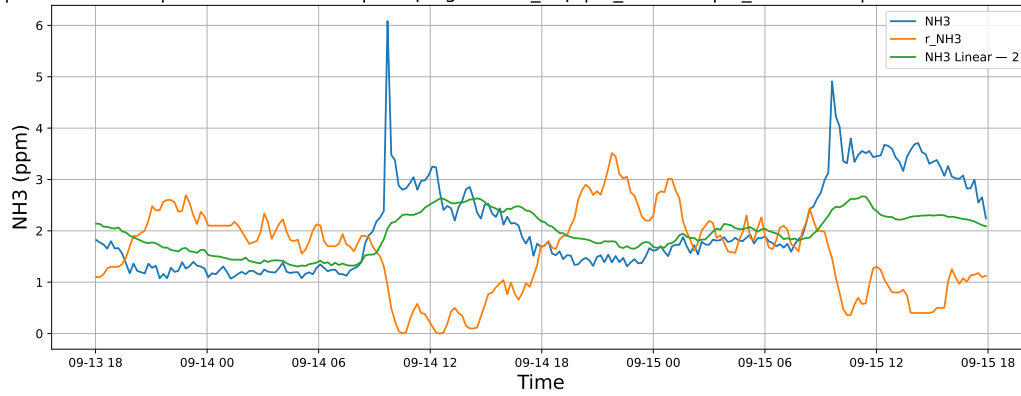


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

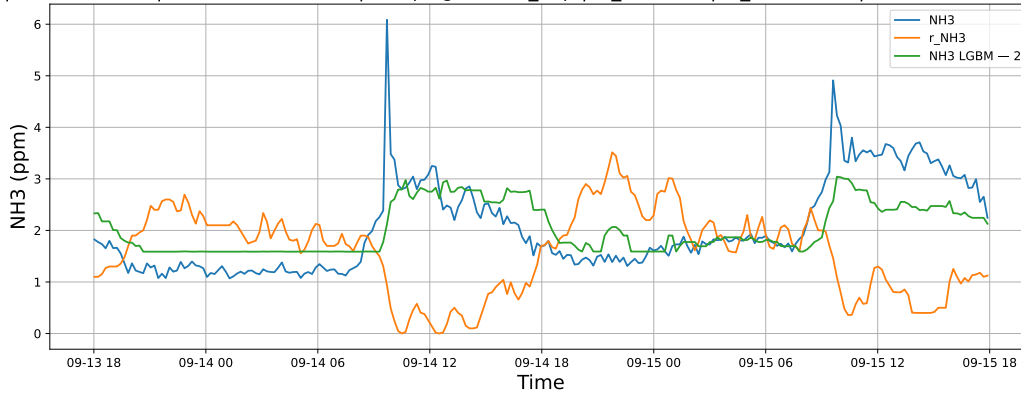


3.1.8.2 Time window : 48

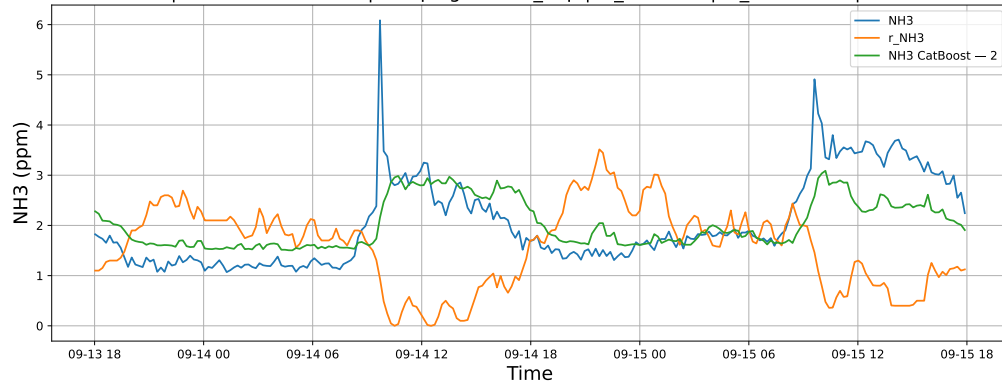
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

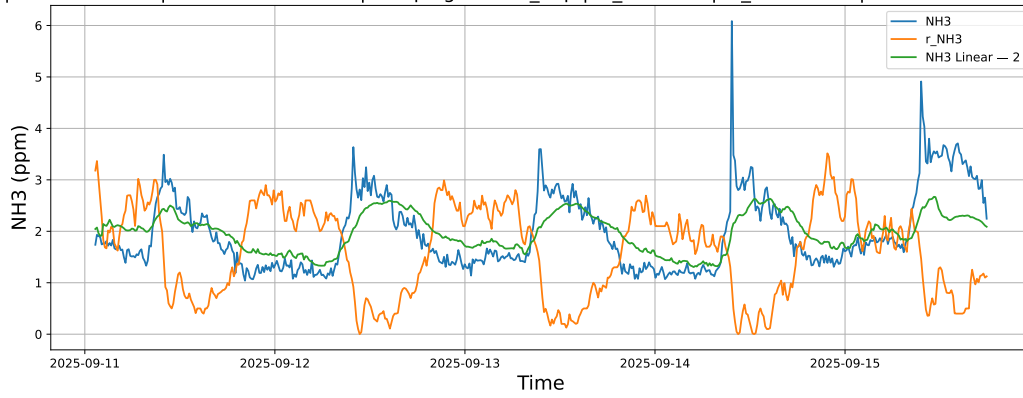


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

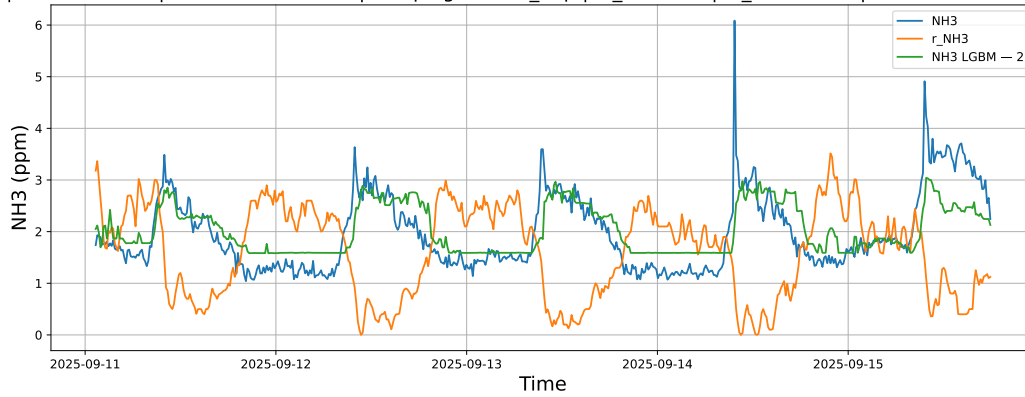


3.1.8.3 Time window : ALL

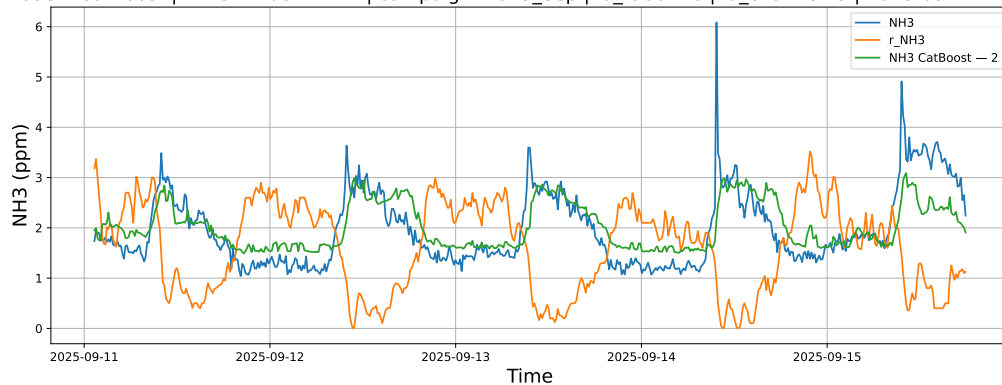
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

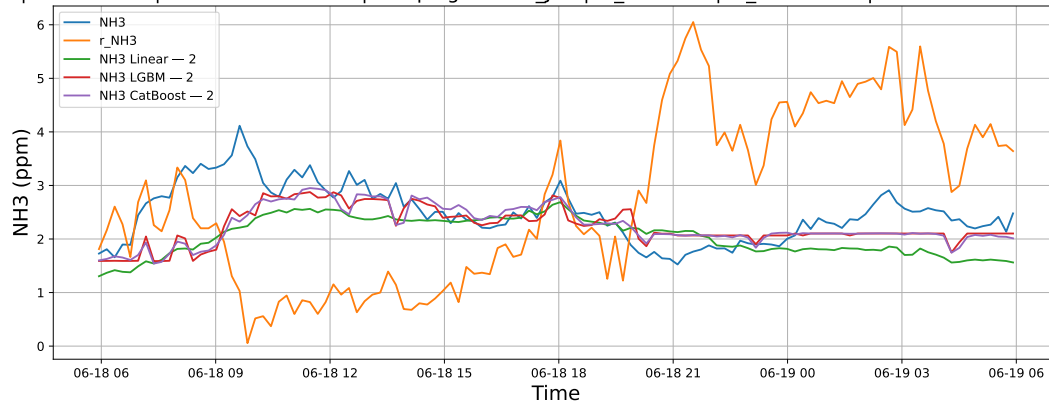


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

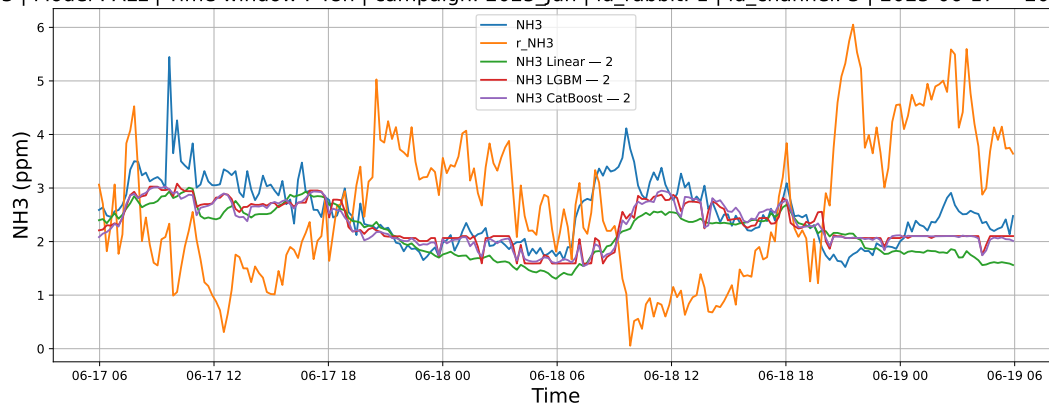
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



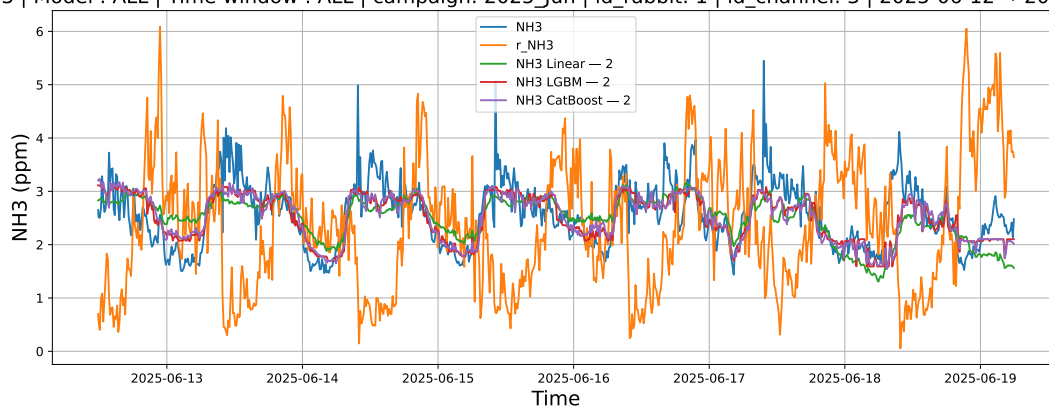
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

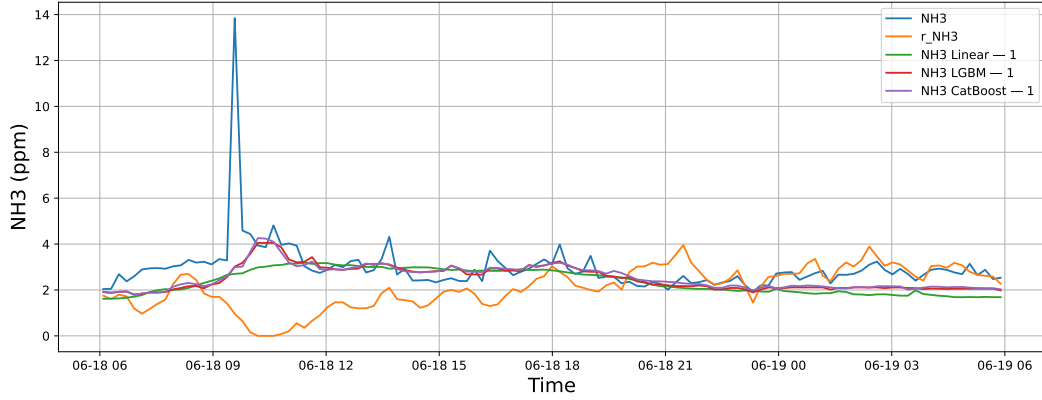
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

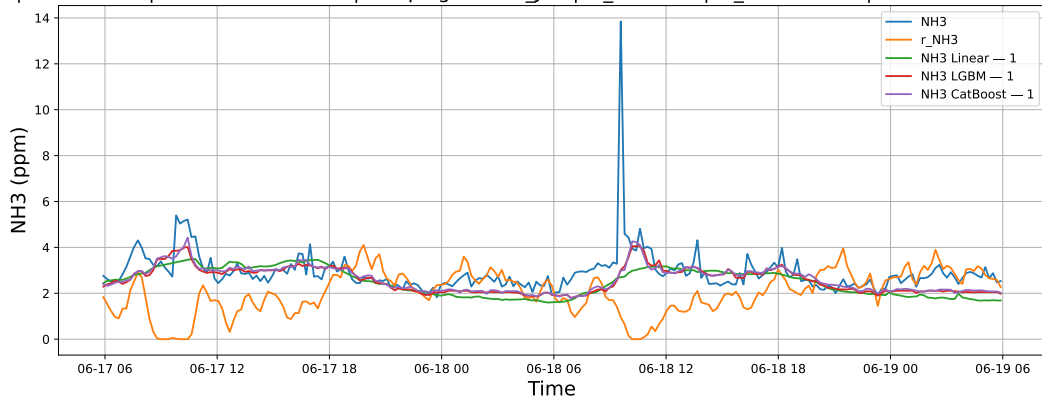
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



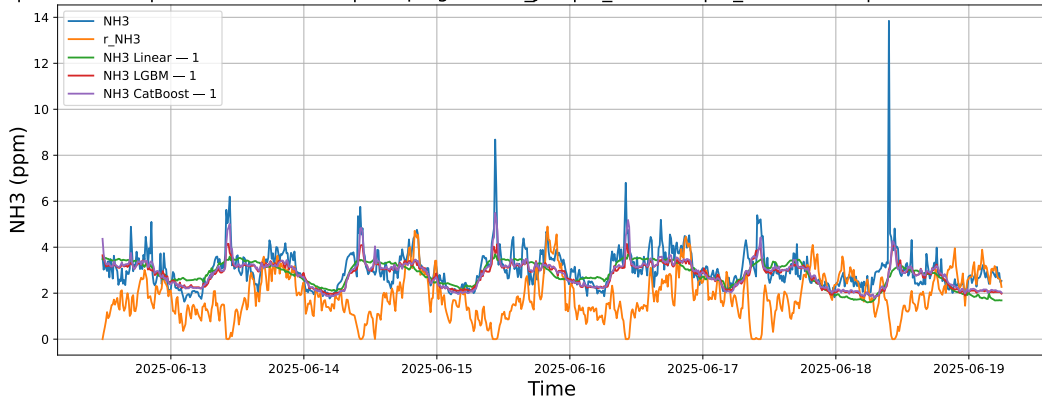
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

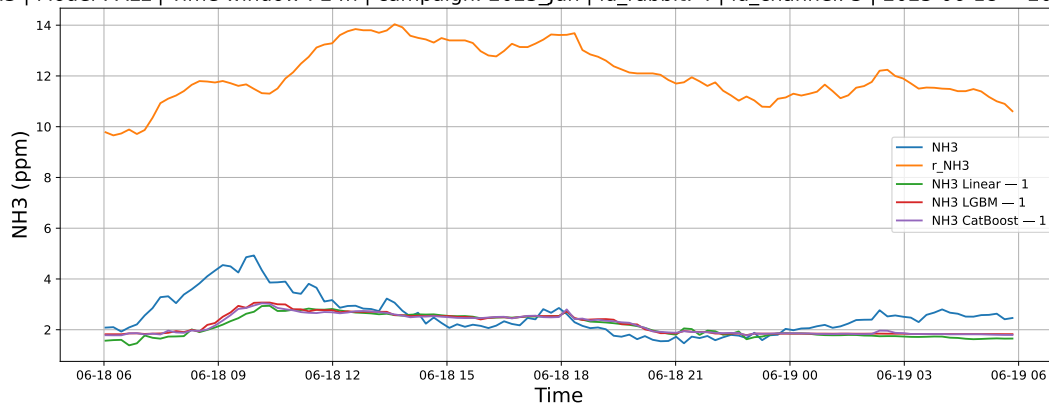
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

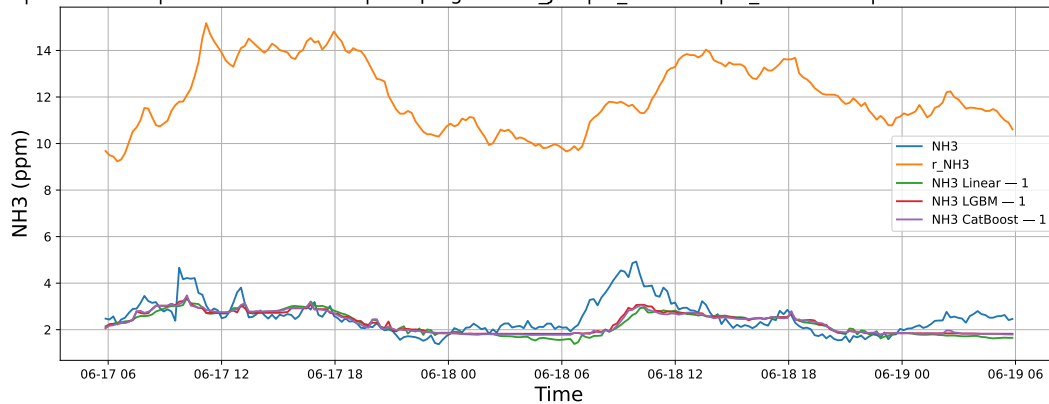
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



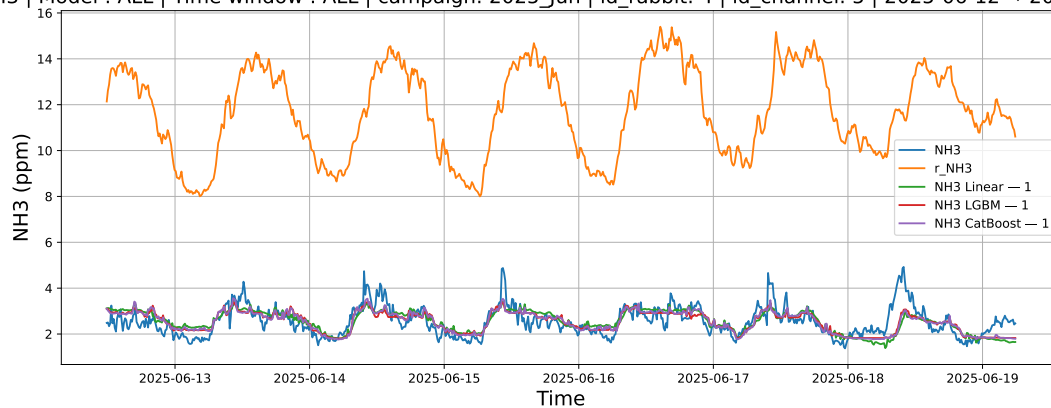
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

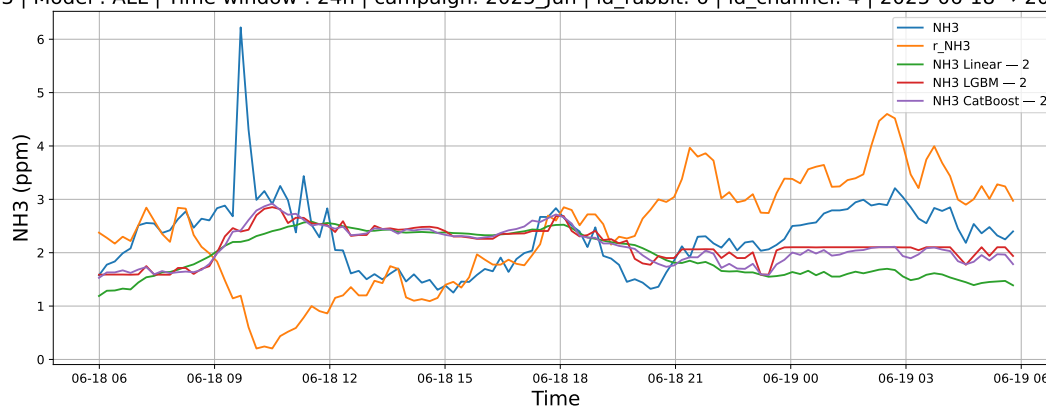
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

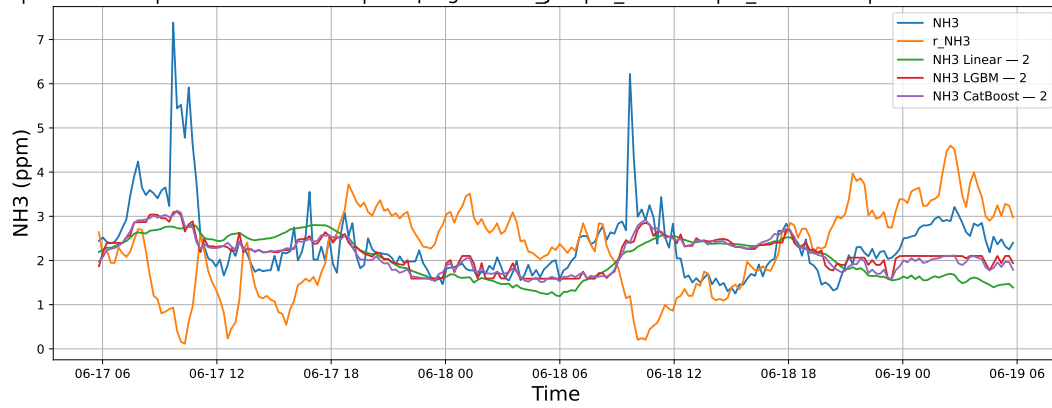
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



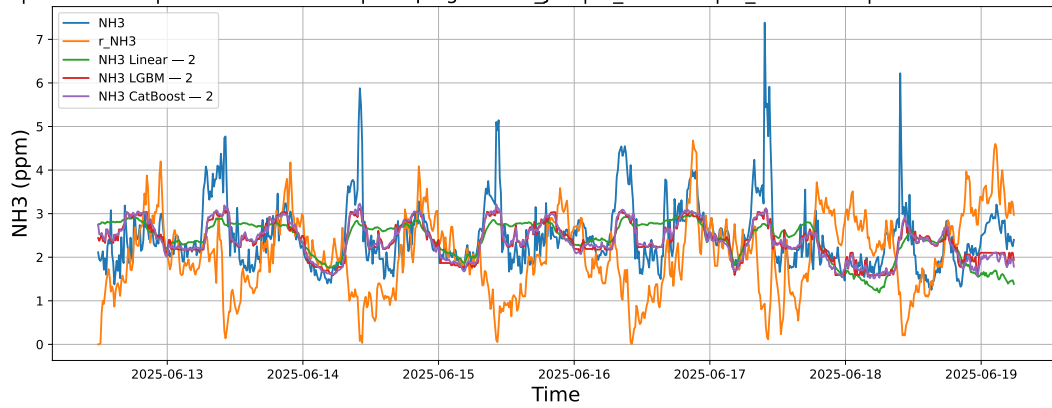
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

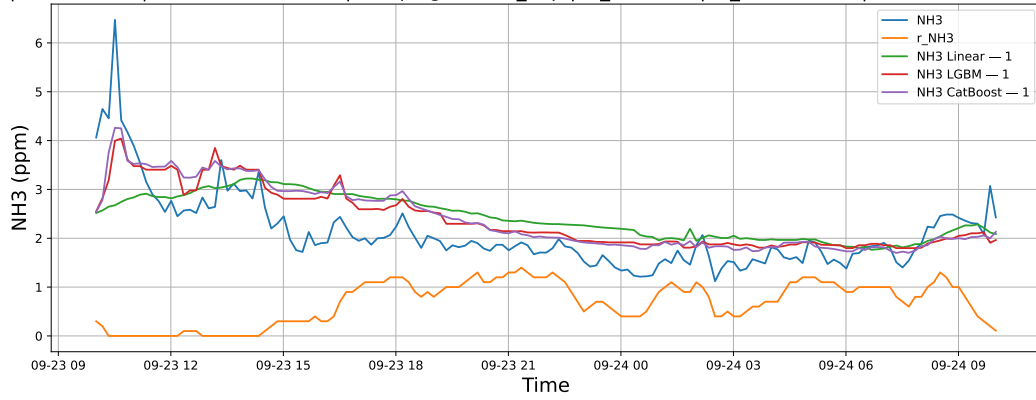
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

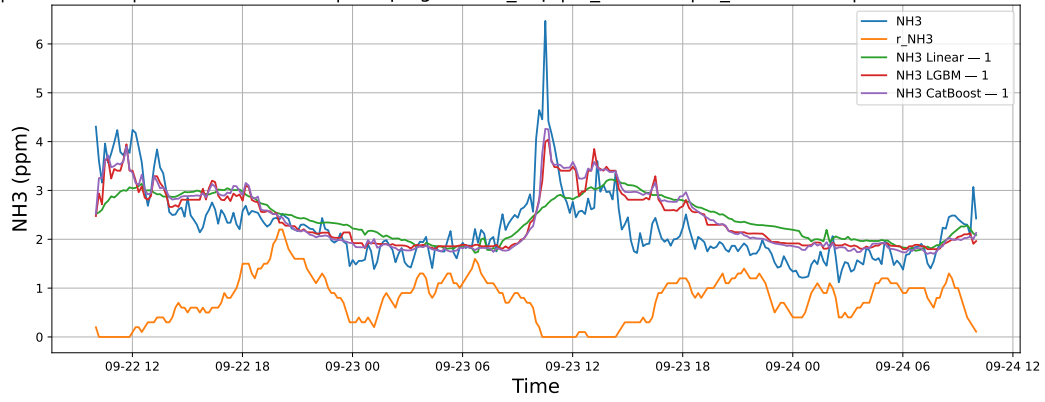
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



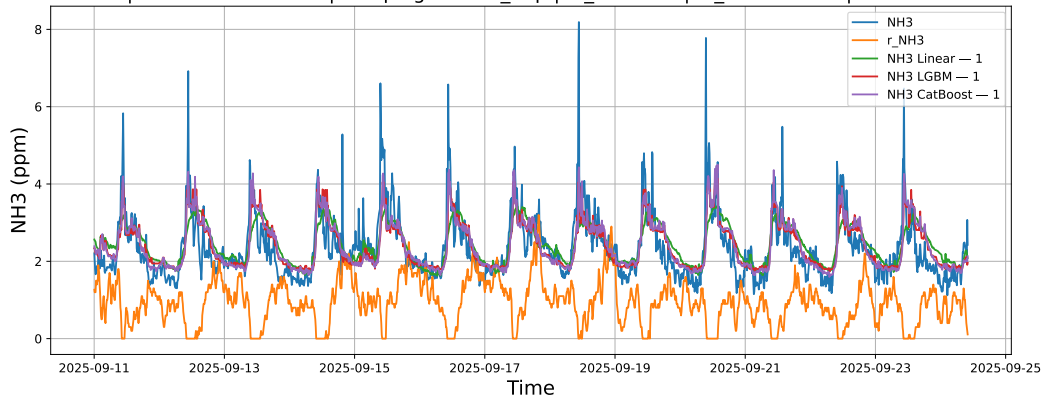
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

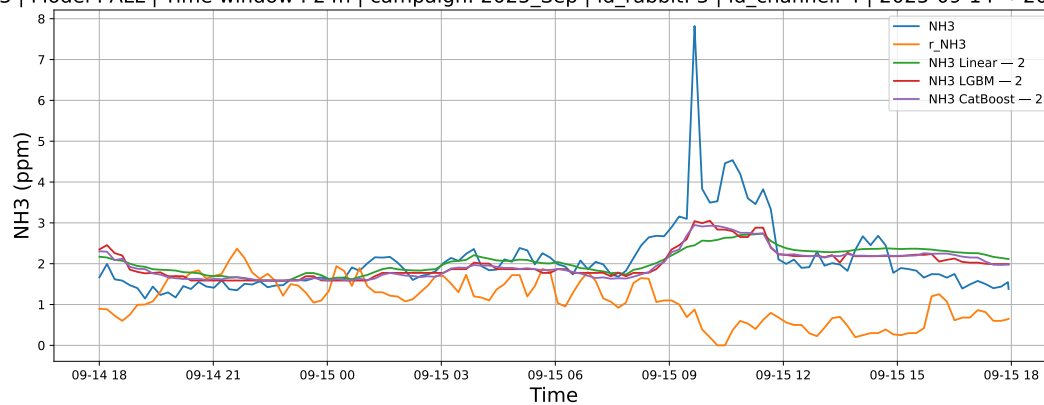
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

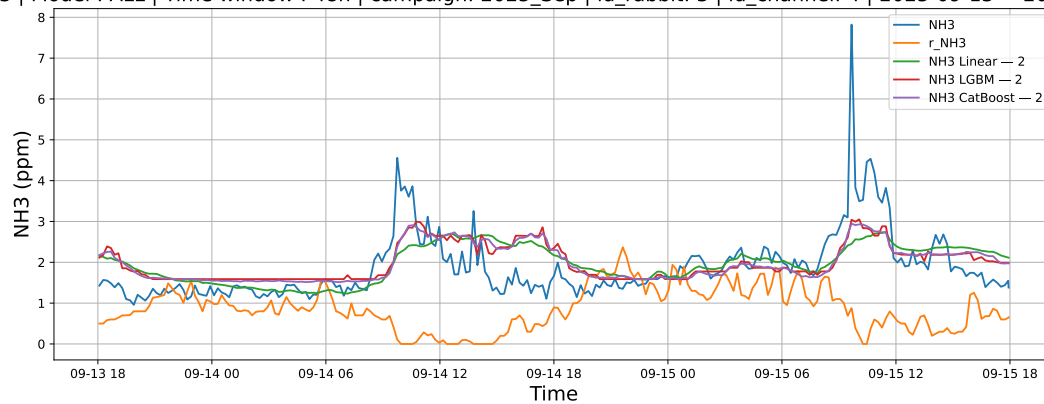
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



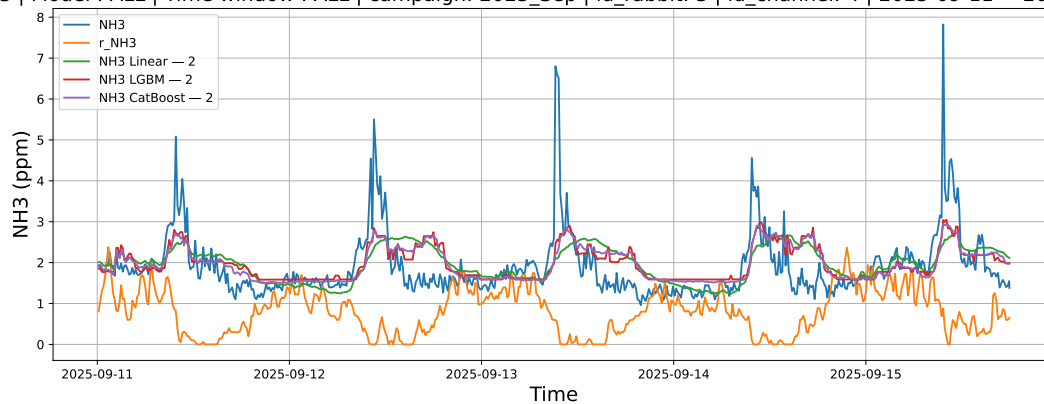
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

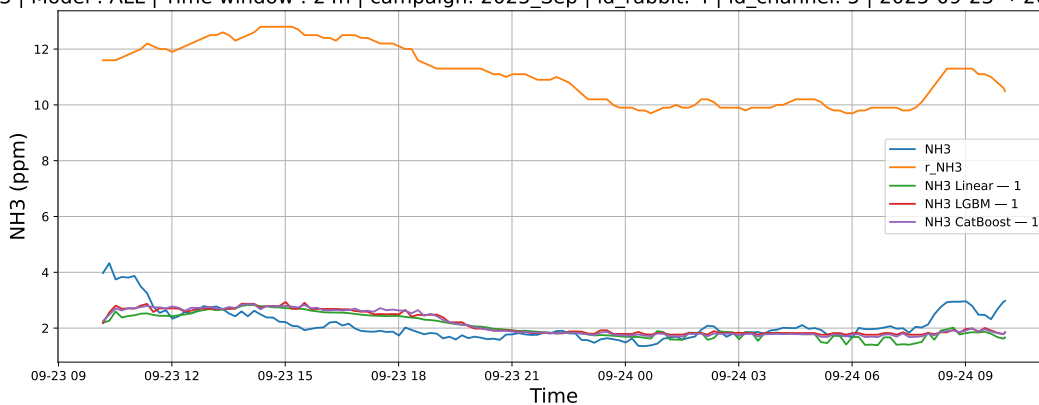
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

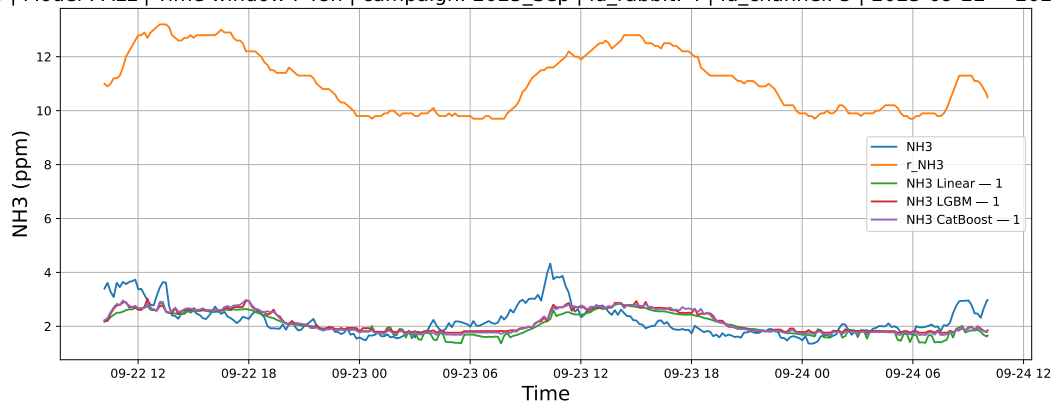
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



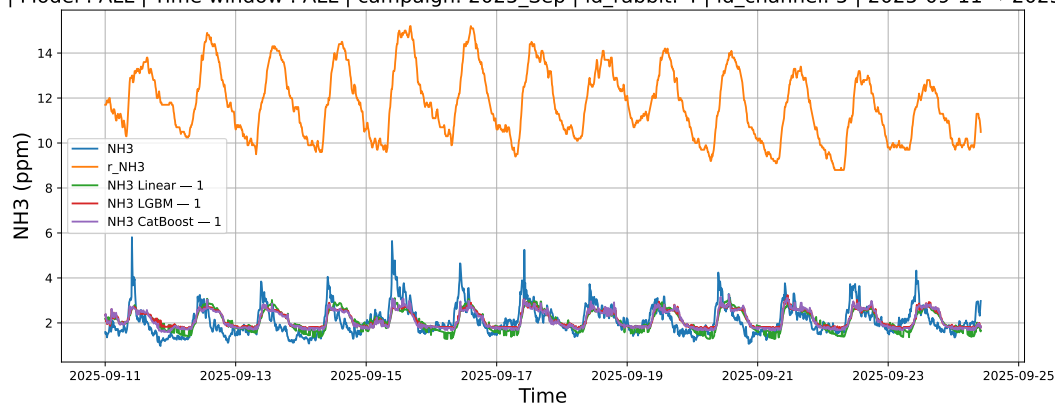
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

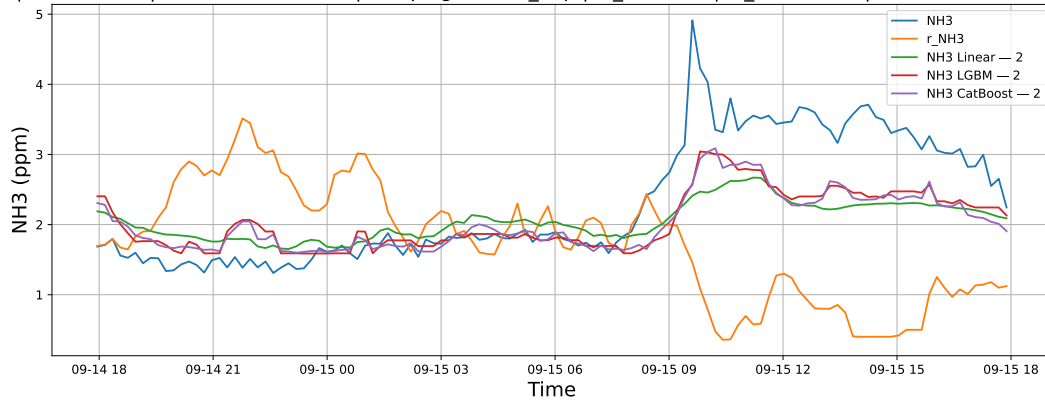
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

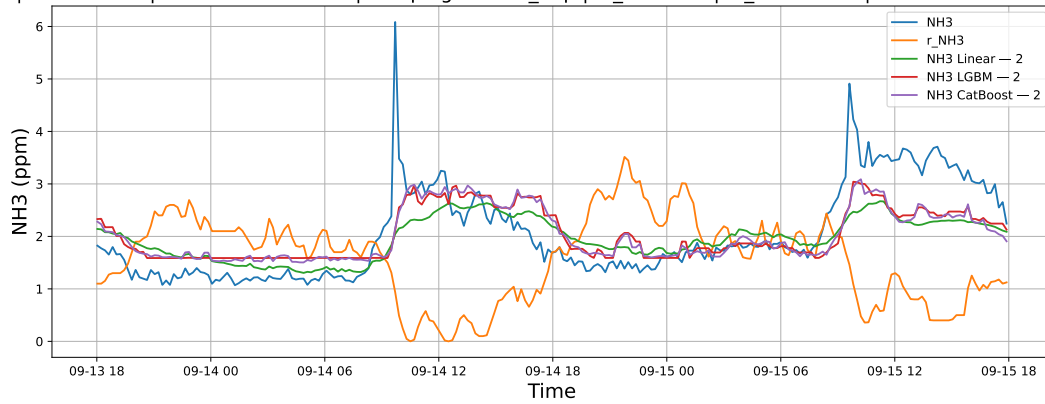
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

